INFINITARY REAL-VALUED LOGIC AND A REFINEMENT OF THE KOCHEN–SPECKER THEOREM

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Abstract. For nearly sixty years, refinements of the Bell–Kochen–Specker theorem have concentrated on finite configurations, combinatorics and robustness, while the logical form of the obstruction has remained essentially propositional: once a configuration is fixed, contextuality is witnessed by the inconsistency of a finite family of Boolean constraints. We replace this truth-functional analysis by infinitary real-valued model theory and model-theoretic forcing.

We axiomatize, in real-valued $L_{\omega_1\omega}$, the lattice of closed subspaces of a separable infinite-dimensional complex Hilbert space together with a quantum probability measure. In the standard models the theory recovers Gleason's theorem; after adjoining the axiom that the measure is $\{0,1\}$-valued, the resulting theory has no models, standard or nonstandard. The inconsistency is genuinely infinitary: every finite set of axioms has a model. A model existence game nevertheless converts this global failure into finite inconsistent sets of interval commitments on the values of sentences.

The same framework yields an exact definability theorem of independent descriptive-set-theoretic interest. Over the standard models, the subsets of $[0,1]$ definable from the range of the quantum probability predicate are exactly the σ-projective sets, in the sense of the least class containing the Borel sets and closed under complements, countable unions and projections along $\mathbb{N}^{\mathbb{N}}$. This classification is proved in ZFC.

For the Bell–Kochen–Specker theory, the finite obstructions refine the usual configurations by recording information that the propositional encoding suppresses. Contextual occurrences of a ray remain distinct until lattice-theoretic axioms identify them; probabilities are tracked only up to rational tolerances; and σ-additivity selects the finite-dimensional subspace in which the contradiction appears. For the eighteen-ray parity configuration of Cabello, Estebaranz and García-Alcaine, every identification required by the parity argument becomes visible. Thus the refinement is not a smaller Kochen–Specker set but a finer logical analysis of why contextuality is forced, what infinitary structure is responsible for the contradiction, and how that structure is compressed into a finite obstruction.

Contents

Introduction 2
Foundational implications for quantum theory 4
Main results of the paper 5
Organization of the paper 7
1. Lattices 8
2. Hermitian spaces 10
3. Projective geometry, Solèr lattices and pre-Solèr lattices 12

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3.1. Points and lines 13
3.2. From lattices to geometries 14
3.3. Harmonic conjugates and Solèr's condition 16
4. What a pre-Solèr lattice is missing (optional) 19
5. The logic and the vocabulary 21
5.1. Real-valued $L_{\omega_1\omega}$ 21
5.2. Henkin types and model existence 23
5.3. The vocabulary 24
6. The axioms 27
7. Standard models of T_0: pre-Solèr lattices 32
8. Standard models of T_1, proofs of Theorems A and B, and definable sets of values 35
8.1. Definable sets of values 38
9. The proof of Theorem C 43
10. The model existence game and finite obstructions 45
10.1. The setting 46
10.2. The model existence theorem as a game 49
10.3. Finite obstructions 52
10.4. The Bell–Kochen–Specker obstruction 55
11. Remarks 59
12. Finite obstructions and Kochen–Specker configurations 62
Appendix A. The space of Henkin types and the model existence game 65
The space of Henkin types 65
Positions and plays 66
Closed conditions, strict conditions and compactness 68
Baire category 68
References 70

Introduction

The Bell–Kochen–Specker theorem is one of the basic impossibility results of quantum foundations. It asserts, in particular, that a quantum probability measure on the lattice of closed subspaces of a Hilbert space cannot be dispersion-free: there is no map $\mu\colon L(H) \to \{0,1\}$ that is additive on every finite or countable orthogonal decomposition of the identity. In the usual physical reading, this rules out noncontextual hidden-variable assignments, in which every proposition is given a definite truth value independent of the compatible propositions measured together with it [BCG+22].

The theorem was announced by Specker in dimension three [Spe60], proved independently by J. S. Bell [Bel66] and by Kochen and Specker [KS67], and has since generated a large literature of increasingly efficient finite configurations and combinatorial proofs. Bell observed that the theorem follows from Gleason's theorem [Gle57] and also gave a direct continuity argument. Kochen and Specker instead exhibited a finite configuration of rays on which no noncontextual $\{0,1\}$-valuation can exist. Later derivations of finite obstructions from analytic forms of the theorem use compactness, explicitly or implicitly; see J. L. Bell [Bel96], Pitowsky [Pit98], and Hrushovski–Pitowsky [HP04].

From the standpoint of logic, however, these finite proofs use very little machinery. Once a Kochen–Specker configuration has been chosen, the contradiction is expressed by a finite family of Boolean constraints and verified by propositional inconsistency. The geometry

and combinatorics of the configurations are subtle, but the logical framework in which the obstruction is certified is essentially truth-functional propositional logic. For nearly sixty years, refinements of the theorem have changed the configurations, dimensions, robustness estimates and combinatorics, while leaving this underlying logical level largely unchanged.

The purpose of the present paper is to change that level. We replace the propositional analysis by real-valued infinitary model theory and model-theoretic forcing. More precisely, we work in the $L_{\omega_1\omega}$ extension of Keisler's model theory for real-valued structures [Kei23, ADI26]. The logic allows countable suprema and infima, several interacting sorts, and quantitative predicates taking values in real intervals. These features let us place the Hilbert space, its lattice of closed subspaces, countable families of vectors and a quantum probability measure inside one formal theory, and then analyze the failure of a $\{0,1\}$-valued measure by a model existence game rather than by a fixed truth table.

This shift in logic reveals structure that the usual propositional encoding collapses. A classical Kochen–Specker graph identifies a ray across all contexts in which it appears: there is one vertex, and hence one truth value, for that ray. In our finite obstructions, contextual occurrences remain separate until the lattice-theoretic axioms identify them. The equality of their probabilities is then not built into the graph but forced by the interaction between geometry and the measure. Likewise, the contradiction is reached from approximate interval commitments rather than from exact Boolean assignments, and the role of σ-additivity becomes visible as an infinitary mechanism that selects the finite-dimensional subspace in which the contradiction occurs.

The starting point is a countable theory T whose standard models are the closed unit balls of separable infinite-dimensional complex Hilbert spaces together with their lattices of closed subspaces and quantum probability measures. Solèr's representation theorem [Sol95], together with the coordinatization theorem of projective geometry, provides the lattice-theoretic part of the axiomatization. In the standard models, T recovers the setting of Gleason's theorem; this is stated precisely in Theorem A. The measure is named by a predicate μ taking values in $[0,1]$.

We then adjoin the single axiom saying that μ takes only the values 0 and 1. The resulting theory T^{01} has no models, standard or nonstandard; this is Theorem C. Its proof uses neither Gleason's theorem nor standardness. At the same time, every finite set of axioms of T^{01} has a model. Thus the inconsistency is genuinely infinitary: unlike an ordinary Kochen–Specker contradiction, it cannot be located in a finite subset of the axioms themselves.

The finite contradictions reappear at a different level, through a model existence game. One player, Olivia, maintains that the theory has no model; the other, Josh, attempts to build one by making finitely many interval commitments about the values of sentences and by answering challenges concerning witnesses, function values and countable suprema or infima. If Olivia follows a winning strategy, every play terminates after finitely many rounds at a finite set of commitments that no structure satisfies. These terminal positions are the finite obstructions of the present paper. The forcing information carried by a play therefore records how an infinitary model-theoretic inconsistency is compressed into a finite contradiction.

For the Bell–Kochen–Specker theory, these obstructions can be pictured using a refinement of the familiar Kochen–Specker graph. We use the parity configuration of Cabello, Estebaranz and García-Alcaine [CEGA96]. A ray occurring in two bases appears first as two contextual vertices; the lattice axioms identify the corresponding projections, and the measure axioms transport probability across that identification. Every identification used by the parity count is therefore displayed rather than silently assumed. If any one of the required ray identifications is removed, the odd-count contradiction disappears. The obstruction also

records rational tolerances on the relevant probabilities and basis sums, so that the parity argument survives small errors in the commitments.

The infinite-dimensional setting contributes one further phenomenon that is invisible in the finite propositional formulation. In models of finite fragments of T^{01}, a $\{0,1\}$-valued measure can survive by vanishing on every finite-dimensional subspace. Countable additivity is what prevents this in the full theory. Along a play, σ-additivity forces almost all of the probability onto finitely many rays of a fixed orthonormal basis; one of these rays then has probability close to 1, and the Cabello–Estebaranz–García-Alcaine contradiction is realized in a four-dimensional block containing it. The hypothetical measure therefore selects the finite-dimensional place in which its own failure becomes visible.

The refinement proved here is consequently not a new record for the size of a Kochen–Specker set. It is a refinement of the logical form of the theorem. The classical finite proof presents contextuality as a Boolean inconsistency on a fixed combinatorial configuration. Our analysis separates the geometric identifications, the quantitative probability commitments, the countable-additivity step and the finite contradiction, and organizes them inside a well-founded forcing tree. In this sense, the paper replaces a propositional certificate of impossibility by a model-theoretic analysis of how the impossibility is forced.

There is a second reason that the stronger logic matters. Because the probability predicate takes every value in $[0,1]$ in the standard models, formulas can genuinely probe the full continuum of quantum probabilities. This raises a natural expressive-power question: which subsets of $[0,1]$ can be singled out uniformly by formulas of the theory? We prove in Theorem 8.6 that the answer is exact: they are precisely the σ-projective subsets of $[0,1]$, where σ-projective denotes the least class containing the Borel sets and closed under complements, countable unions and projections along $\mathbb{N}^{\mathbb{N}}$. The class strictly contains the projective sets. The classification is proved in ZFC; no determinacy or large-cardinal hypothesis is used. The novelty here is the exact correspondence between syntactic expressive power and a natural descriptive-set-theoretic pointclass.

Foundational implications for quantum theory. The usual foundational conclusion of Bell–Kochen–Specker is negative: a noncontextual dispersion-free assignment does not exist. The present analysis does not strengthen that physical impossibility statement by ruling out a larger class of hidden-variable theories. Rather, it refines the mathematical anatomy of the impossibility. In the ordinary finite formulation, geometry, noncontextual identification and Boolean valuation are compressed into one static combinatorial object. In the model-theoretic formulation these components become logically distinct and can be followed separately through a forcing construction.

This separation clarifies where noncontextuality enters. A ray occurring in two compatible measurement contexts is not initially represented by one vertex carrying one preassigned value. The two contextual occurrences are first treated separately; the lattice axioms identify the corresponding projections, and only then does the probability predicate force equality of their values. Noncontextuality is therefore not merely an informal interpretation attached to the final graph. The formalism records the exact commitments by which context-independent identity is imposed and shows that the parity contradiction requires each of the relevant identifications.

The analysis also distinguishes finite experimental content from infinitary probabilistic structure. Every finite fragment of the $\{0,1\}$-valued theory has a model, so no finite list of the axioms is itself responsible for the contradiction. The obstruction appears only when the full countable-additivity requirement is present. Yet the model existence game turns

this infinitary inconsistency into a finite terminal position: a finite collection of quantitative commitments that no structure can realize. In this sense, the forcing analysis exhibits how an infinitary principle of quantum probability can generate a finite contextuality witness.

Finally, the contradiction is not tied in advance to a fixed four-dimensional sector of the infinite-dimensional Hilbert space. A hypothetical dispersion-free measure can survive on finite fragments by assigning value 0 to every finite-dimensional subspace. Countable additivity eventually forces probability into finitely many rays of a named orthonormal basis, and the hypothetical measure itself thereby determines a four-dimensional block in which the parity contradiction is activated. The resulting picture is dynamic rather than static: the forbidden valuation does not merely fail on a preselected Kochen–Specker configuration; under the rules imposed by quantum probability, it is forced to reveal a finite configuration on which it fails.

For quantum foundations, the point is therefore methodological as well as formal. The Bell–Kochen–Specker phenomenon is usually encoded at the level of finite propositional consistency. Real-valued infinitary logic and model-theoretic forcing make it possible to ask finer questions: which identifications are indispensable, which approximations still force contradiction, which infinitary probabilistic principles are doing the work, and how a global impossibility is converted into a finite certificate. These questions are invisible in the truth-table formulation but become mathematical objects in the present framework.

Main results of the paper. The theory T speaks of three kinds of objects, which the logic keeps apart as sorts. The elements of the *Hilbert sort* are vectors; those of the *Solèr sort* carry the operations of an ortholattice. In the intended interpretation the elements of the Solèr sort are closed subspaces, identified, as above for $L(H)$, with the orthogonal projections onto them. Those of the *sequence sort* are sequences of vectors, through which the completeness of the lattice becomes expressible. The vocabulary has a function symbol for the orthogonal projection of a vector onto an element of the Solèr sort. A vector belongs to an element of the Solèr sort when the projection onto that element leaves the vector fixed. The predicate μ assigns a probability to every element of the Solèr sort. Like every countable theory that has a model, the theory T has models whose sorts are countable. The theory also has models whose Hilbert sort and sequence sort are the full closed unit balls of Hilbert spaces. The theorems below distinguish the conclusions that hold in all models from those that require models of this second kind, which we now name. We call a model *standard* when its Hilbert and sequence sorts are the full closed unit balls of complex Hilbert spaces with their own operations, inner products and norms (see Definition 5.2 below). In a nonstandard model, either the Hilbert sort lacks some of the vectors of such a ball or the sequence sort is not a ball of this kind. This use of the term "nonstandard" is unrelated to nonstandard analysis. Gleason's theorem [Gle57] concerns the quantum probability measures on the lattice of closed subspaces of a separable Hilbert space of dimension at least three. It asserts that these measures are exactly the maps that send each closed subspace to $\mathrm{Tr}(\rho P)$, where P is the orthogonal projection onto the subspace and where ρ is a density operator. A density operator, that is, a positive operator of trace 1, represents a state of the quantum system. In the standard models, the theory T captures the setting of Gleason's theorem for separable infinite-dimensional complex Hilbert spaces. The point of the first theorem is therefore semantic as well as representational: it shows that the new formal language does not merely imitate the Hilbert-space picture, but recovers exactly the intended quantum-probability structures in its standard models.

Theorem A (Gleason's theorem in real-valued $L_{\omega_1\omega}$). *There is a countable theory T of real-valued $L_{\omega_1\omega}$ satisfying the following two properties.*

(1) In every standard model of T, the Hilbert sort is the closed unit ball of a separable infinite-dimensional complex Hilbert space H, and the map $a \mapsto S_a$, where S_a is the closed subspace of H spanned by the vectors of the Hilbert sort that belong to a, is an isomorphism of ortholattices from the Solèr sort onto the lattice $L(H)$ of closed subspaces of H; moreover, there is a unique density operator ρ on H such that $\mu(a) = \mathrm{Tr}(\rho P_a)$ for every element a of the Solèr sort, where P_a is the orthogonal projection onto S_a.

(2) Conversely, for every separable infinite-dimensional complex Hilbert space H and every density operator ρ on H, there is a standard model of T whose Hilbert sort is the closed unit ball of H, whose Solèr sort is $L(H)$, and in which $\mu(a) = \mathrm{Tr}(\rho P_a)$ for every $a \in L(H)$, where P_a is the orthogonal projection onto a.

In every separable infinite-dimensional Hilbert space, the probabilities that a quantum state assigns to the closed subspaces fill the whole interval $[0, 1]$, even when the state is pure (see Lemma 8.4 below). Thus the probability predicate is not merely a Boolean remnant inside the new formalism: in the intended models it ranges over the entire continuum. This makes the range theorem the starting point for the expressive-power question answered by Theorem 8.6: which subsets of that continuum can formulas of the theory distinguish uniformly?

Theorem B (The range of the measure). *In every standard model of T, the range of μ is the whole interval $[0, 1]$. Consequently, for every Borel set $B \subseteq \mathbb{R}$ that meets $[0, 1]$, the theory T, together with an axiom saying that no value of μ lies in B (see Lemma 5.1), has no standard models.*

The conclusion of Theorem B reaches well beyond the Borel sets. A subset of $[0, 1]$ is *definable over T* when some formula of the logic, with one free variable a of the Solèr sort, singles it out as follows (see Definition 8.5 below). In every standard model of T and at every element a of its Solèr sort, the formula takes the value 1 if $\mu(a)$ lies in the set and the value -1 otherwise. Because formulas may quantify over the Solèr sort, and because the measure takes every value in $[0, 1]$, the definable sets form a much larger class than the Borel sets. In Theorem 8.6 below we show that the definable sets are exactly the σ-projective subsets of $[0, 1]$. The σ-projective sets are the members of the least class of subsets of Polish spaces that contains the Borel sets and is closed under complements, countable unions and projections along $\mathbb{N}^{\mathbb{N}}$. This class contains every projective set, and the example following Theorem 8.6 shows that the containment is strict. Consequently, the hypothesis that the probabilities avoid a given nonempty σ-projective subset of $[0, 1]$ can be written as an axiom of the logic. It follows as well that the theory T together with that axiom has no standard models. The exact σ-projective classification, including the strict extension beyond the projective sets, is proved in ZFC.

The hypothesis that every probability is 0 or 1 can be written as a single axiom of the logic, displayed in Theorem C below. In that axiom, the symbol $\mathbf{1}$ denotes the sentence whose value is 1 in every structure, and for formulas φ and ψ the expression $\varphi \wedge \psi$ denotes the minimum of their values. Our next theorem shows that the axiom fails not only in the standard models but in every model of T.

Theorem C (The Bell–Kochen–Specker theorem in real-valued $L_{\omega_1\omega}$). *The theory T of Theorem A, together with the axiom $\mu(a) \wedge (\mathbf{1} - \mu(a)) \leq 0$, which, since the axioms of T confine μ to $[0,1]$, says that μ takes only the values 0 and 1, has no models, standard or not.*

The proof of Theorem C, given in Section 9, uses neither Gleason's theorem nor the standardness of the model. The inconsistency it establishes is genuinely infinitary, for every finite set of axioms of the theory of Theorem C has a model. The finite obstructions of this inconsistency are therefore not finite sets of axioms but finite sets of commitments on the values of sentences. These obstructions stand at the leaves of the well-founded trees of the game described above. The branching of these trees shows where the countable suprema and infima of the axioms enter. The rules of the game apply to every theory whose axioms are *closed conditions*, that is, assertions $\varphi \leq r$ or $\varphi \geq r$ about sentences φ, with r rational. An axiom with free variables takes this form once its variables are bound by suprema or infima, and the axioms of T are all of this kind. The next theorem isolates the general game-theoretic mechanism used later in the quantum application: from model-theoretic inconsistency one obtains a well-founded tree whose leaves are finite inconsistent interval commitments, and whose infinite branching records precisely the infinitary operations of the logic.

Theorem D (Finite obstructions). *Let T' be a countable theory of real-valued $L_{\omega_1\omega}$ that consists of closed conditions and has no models. In the model existence game of Section 10, the player who claims that T' has no models has a winning strategy. For every such strategy, the tree of plays in which that player follows it is well-founded; at each leaf, the commitments made along the play, each of which says that the value of a sentence lies in a closed interval with rational endpoints, form a finite set that is satisfied in no structure. The tree branches infinitely only at the moves that concern countable suprema and infima; once the other player's answers to those moves are fixed, the plays have bounded length. For the theory of Theorem C, every finite set of axioms has a model; consequently, the tree of every winning strategy for that theory branches infinitely.*

Organization of the paper. The background occupies Sections 1–4 and is drawn from lattice theory, the theory of hermitian forms and projective geometry. Its principal sources are MacNeille completions [Mac37], the coordinatization of projective geometries [Bae52, MM70, Hol95], and the theorems on orthomodular spaces due to Amemiya and Araki [AA66] and to Solèr [Sol95]. From these sources we obtain the Solèr lattices, which up to isomorphism are the lattices of closed subspaces of infinite-dimensional real, complex and quaternionic Hilbert spaces. The same sources yield the pre-Solèr lattices, in which every defining condition of a Solèr lattice holds except possibly completeness. In the standard models of T, the Solèr sort is a Solèr lattice. By contrast, in the standard models of the part of T that mentions

neither the sequence sort nor the measure, the Solèr sort need only be a pre-Solèr lattice. The logic and the theory T are presented in Sections 5 and 6. The proofs of Theorems A, B and C occupy Sections 7–9, together with that of Theorem 8.6, which identifies the sets of values that formulas define in the standard models. The model existence game, whose analysis rests on the determinacy theorem of Gale and Stewart [GS53], is the subject of Section 10, which also contains the proof of Theorem D. The reader may omit Section 4 and the remarks of Section 11 on a first reading. In Section 12 we compare the finite obstructions with the configurations of Kochen and Specker. The comparison bears on what a vertex is, the identifications and the axioms from which they follow, the tolerances of the intervals and the location of the configuration. The comparison also bears on the role of σ-additivity for configurations of rays in infinite dimension and on the topological property of the logic behind each kind of certificate. In Section A we relate the game to Keisler's model-theoretic forcing [Kei73] and compare it with the treatment of the Henkin construction in the space of Henkin types [ADI26].

1. Lattices

Birkhoff and von Neumann [BvN36] observed that the experimentally verifiable propositions about a quantum system correspond to the closed subspaces of a Hilbert space. These subspaces, they further observed, form a lattice in which the distributive law fails; the logic of quantum mechanics is therefore not that of a Boolean algebra. In place of the distributive law the two authors proposed the weaker modular law, which holds in finite dimensions but fails for the closed subspaces of an infinite-dimensional Hilbert space. The law that holds in every dimension is the still weaker orthomodular law. In this section we collect the notions from lattice theory that we need: orthocomplementation and orthomodularity; atoms together with the covering property and irreducibility; and the MacNeille completion. This completion supplies the joins and meets that an incomplete lattice lacks.

Definition 1.1. A *partially ordered set* is a set with a reflexive, antisymmetric and transitive relation $\leq$. For a subset S of a partially ordered set, an *upper bound* of S is an element b with $a \leq b$ for all $a \in S$, and the *join* $\bigvee S$ of S is its least upper bound, when it exists; lower bounds and the *meet* $\bigwedge S$ are defined dually. A *lattice* is a partially ordered set in which every pair of elements a, b has a join $a \vee b$ and a meet $a \wedge b$. A lattice is *bounded* when it has a least element 0 and a greatest element 1, and *complete* when every subset has a join.

In a lattice, the order can be recovered from the meet: for all elements a and b, the relation $a \leq b$ holds if and only if $a \wedge b = a$. It is in this form that the axioms of Section 6 express the order. Completeness, likewise, can be stated through meets: a lattice is complete if and only if every subset has a meet. Indeed, when every subset has a join, the meet of a subset is the join of its lower bounds; the converse follows dually.

Definition 1.2. An *orthocomplementation* on a bounded lattice Λ is a map $a \mapsto a^\perp$ from Λ to Λ such that, for all $a, b \in \Lambda$, the equations $a^{\perp\perp} = a$, $a \wedge a^\perp = 0$ and $a \vee a^\perp = 1$ hold, and the inequality $a \leq b$ implies $b^\perp \leq a^\perp$. A bounded lattice with an orthocomplementation is an *ortholattice*. Two elements a, b of an ortholattice are *orthogonal*, written $a \perp b$, when $a \leq b^\perp$. An ortholattice is *orthomodular* when, for all elements a and b, the inequality $a \leq b$ implies $b = a \vee (b \wedge a^\perp)$.

The lattice $L(H)$ of closed subspaces of a Hilbert space H is ordered by inclusion and orthocomplemented by $M \mapsto M^\perp$. In this lattice, orthomodularity says that every closed subspace N containing a closed subspace M is the orthogonal sum of M and $N \cap M^\perp$. Orthomodular or not, every ortholattice satisfies the De Morgan law $(\bigvee S)^\perp = \bigwedge\{a^\perp : a \in S\}$

for every subset S whose join exists, since the map $a \mapsto a^\perp$ is an order-reversing bijection that is its own inverse. It is through this law that orthocomplements of joins are computed below, in the MacNeille completion and in the coordinatization theorem of Section 3.

Definition 1.3. An *atom* of a bounded lattice is a nonzero element p such that every element b with $b \leq p$ is equal to 0 or to p. A bounded lattice is *atomistic* when every element a is the join of the atoms below it (this join being required to exist). It has the *covering property* when, for every atom p and every element a with $p \not\leq a$, no element lies strictly between a and $a \vee p$. It is *irreducible* when, for any two distinct atoms p and q, some atom r distinct from p and q lies below $p \vee q$.

For complete orthomodular lattices that are atomistic and have the covering property, irreducibility in this sense has an equivalent formulation through direct products (see Maeda and Maeda [MM70] and Holland [Hol95, §4]). Such a lattice is irreducible in this sense exactly when it admits no decomposition as a direct product of two lattices, each with more than one element. We adopt the definition above because it is the form in which irreducibility enters the coordinatization theorem (see Theorem 3.3).

Definition 1.4. A *sub-ortholattice* of an ortholattice Λ is a subset that contains 0 and 1 and is closed under the operations $\vee$, $\wedge$ and $^\perp$ of Λ. An *embedding* of ortholattices is an injective map that preserves $\vee$, $\wedge$, $^\perp$, 0 and 1.

Among the pre-Solèr lattices of Section 3 are the Solèr sorts of the standard models of the part of T that mentions neither the sequence sort nor the measure. Pre-Solèr lattices, these Solèr sorts included, need not be complete (see Example 7.6). Suppose that a pre-Solèr lattice embeds in the lattice of orthogonally closed subspaces of a hermitian space, the atoms of the pre-Solèr lattice being carried onto the rays of the space. In Theorems 4.1 and 7.5 below we show that the lattice of orthogonally closed subspaces is then the MacNeille completion of the embedded lattice. This completion, which we construct presently, is characterized by a notion of density that concerns the order alone; no topology is involved. Let Γ be a complete lattice, and let Λ be a subset of Γ. We say that Λ is *join-dense* in Γ when every $g \in \Gamma$ satisfies $g = \bigvee\{a \in \Lambda : a \leq g\}$, the join being computed in Γ, and that Λ is *meet-dense* in Γ when every $g \in \Gamma$ satisfies $g = \bigwedge\{a \in \Lambda : a \geq g\}$. For instance, the set $\mathbb{Q}$ of rational numbers is join-dense and meet-dense in the extended real line $[-\infty, \infty]$.

A join-dense and meet-dense subset determines, up to isomorphism, the complete lattice that contains it (see Proposition 1.5 below), and that lattice can therefore be built from the subset alone. The construction, due to MacNeille [Mac37], is as follows. For a partially ordered set Λ and a subset X of Λ, let X^u and X^l be the sets of upper and lower bounds of X in Λ. We call X a *cut* when $X = (X^u)^l$. Ordered by inclusion, the cuts form a complete lattice $\mathrm{DM}(\Lambda)$, the *MacNeille completion* of Λ. Into this lattice the map $a \mapsto \{b \in \Lambda : b \leq a\}$ embeds Λ, preserving all joins and meets that exist in Λ. The letters DM recall the other name of the completion, the Dedekind–MacNeille completion.

When Λ is an ortholattice, its orthocomplementation induces one on the cuts. For a cut X, let $X^\perp$ be the set $\{b \in \Lambda : b \leq a^\perp \text{ for every } a \in X\}$ of lower bounds of $\{a^\perp : a \in X\}$. The set $X^\perp$, as a set of lower bounds, is a cut, and the map $X \mapsto X^\perp$ reverses inclusion. An element b lies in $X^\perp$ exactly when $b^\perp$ is an upper bound of X, whence $X^{\perp\perp} = (X^u)^l = X$. Moreover, the intersection $X \cap X^\perp$ is $\{0\}$, because an element b with $b \leq b^\perp$ satisfies $b = b \wedge b^\perp = 0$. Since $X \mapsto X^\perp$ is an order-reversing bijection that is its own inverse, it carries the meet $X \cap X^\perp$ to the join of $X^\perp$ and X, which is therefore $\{0\}^\perp = \Lambda$. Thus $X \mapsto X^\perp$ is an orthocomplementation of $\mathrm{DM}(\Lambda)$; it extends that of Λ, since it sends the cut $\{b \in \Lambda : b \leq c\}$ to the cut $\{b \in \Lambda : b \leq c^\perp\}$, and with it we regard $\mathrm{DM}(\Lambda)$ as an ortholattice.

Proposition 1.5 (Banaschewski [Ban56], Schmidt [Sch56]). *Let Λ be a partially ordered set, and let Γ be a complete lattice that contains Λ as a join-dense and meet-dense subset, the order of Γ inducing on Λ the order of Λ. Then the map that sends each $g \in \Gamma$ to the set $\{a \in \Lambda : a \leq g\}$ is an isomorphism of Γ onto $\mathrm{DM}(\Lambda)$ that restricts on Λ to the canonical embedding $a \mapsto \{b \in \Lambda : b \leq a\}$. If Λ is an ortholattice, then the lattice Γ has at most one orthocomplementation that extends that of Λ; any such orthocomplementation is given by $g^\perp = \bigwedge\{a^\perp : a \in \Lambda,\ a \leq g\}$, and the isomorphism carries it to the orthocomplementation $X \mapsto X^\perp$ of $\mathrm{DM}(\Lambda)$.*

Proof. For $g \in \Gamma$, write $\Lambda_{\leq g} = \{a \in \Lambda : a \leq g\}$; we show that each $\Lambda_{\leq g}$ is a cut and that the map $g \mapsto \Lambda_{\leq g}$ is an order embedding whose image contains every cut. The assertions on orthocomplementations then follow from the De Morgan law. By join-density, we have $g = \bigvee \Lambda_{\leq g}$; hence the upper bounds of $\Lambda_{\leq g}$ in Λ are the elements $b \in \Lambda$ with $b \geq g$. By meet-density, the common lower bounds of these elements are the elements $a \in \Lambda$ with $a \leq g$; thus $\Lambda_{\leq g}$ is a cut. For $g, g' \in \Gamma$, join-density also shows that $g \leq g'$ if and only if $\Lambda_{\leq g} \subseteq \Lambda_{\leq g'}$; hence the map is an order embedding. Because the order of Γ induces that of Λ, the map restricts on Λ to the canonical embedding. If X is a cut and $g = \bigvee X$ in Γ, then the cut X is contained in $\Lambda_{\leq g}$. Conversely, an element $a \in \Lambda$ with $a \leq g$ lies below every $b \in X^u$ because each such b is an upper bound of X in Γ, whence $a \in X = (X^u)^l$. Thus $X = \Lambda_{\leq g}$, and the map is onto $\mathrm{DM}(\Lambda)$. The formula for $g^\perp$ follows from $g = \bigvee \Lambda_{\leq g}$ and the De Morgan law. By that formula, an element $b \in \Lambda$ lies below $g^\perp$ exactly when $b \leq a^\perp$ for every $a \in \Lambda_{\leq g}$; hence $\Lambda_{\leq g^\perp} = (\Lambda_{\leq g})^\perp$. ☻

⧖ **Historical note.** Birkhoff and von Neumann rejected the lattice of closed subspaces of an infinite-dimensional Hilbert space as the logic of quantum mechanics. Letters that von Neumann wrote to Birkhoff in 1935, while the two were preparing their paper of 1936, show why: that lattice is not modular. Von Neumann, however, wanted the probabilities of quantum propositions to behave like a dimension function (that is, like a strictly increasing valuation). Only a modular lattice carries such a function [Réd07]. In the paper itself, Birkhoff and von Neumann postulated instead an abstract projective geometry, whose lattice of subspaces is modular [BvN36]. A year later Husimi observed that the closed subspaces of a Hilbert space satisfy the orthomodular law [Hus37], which was rediscovered several times in the 1950s and 1960s. In his thesis of 1964, Piron took completeness, orthomodularity, atoms and the covering property as axioms for the propositions about a quantum system. He showed that the irreducible lattices with these properties are represented by hermitian forms over a division ring when their dimension is at least four [Pir64].

2. Hermitian spaces

The coordinatization theorem, which we state in Section 3, turns the atoms of a suitable ortholattice into the one-dimensional subspaces of a vector space over a division ring. Under the same theorem, the orthocomplementation of the ortholattice becomes orthogonality with respect to a hermitian form on that vector space. In this section we collect the facts about such spaces that we need. The most important of these facts, on which Sections 3 and 4 depend, is that orthomodularity of the lattice of orthogonally closed subspaces is equivalent to a property of the space itself. This property asks that every orthogonally closed subspace and its orthogonal complement have the whole space as their sum (see Proposition 2.8).

Definition 2.1. A *division ring* is a ring with $1 \neq 0$ in which every nonzero element is invertible; multiplication need not be commutative. An *involution* of a division ring Δ is a

map $\alpha \mapsto \alpha^*$ such that $(\alpha+\beta)^* = \alpha^* + \beta^*$, $(\alpha\beta)^* = \beta^*\alpha^*$ and $\alpha^{**} = \alpha$ for all $\alpha, \beta \in \Delta$. Let E be a left vector space over Δ, that is, a left Δ-module.[1] A *hermitian form* on E is a map $\langle \cdot \mid \cdot \rangle \colon E \times E \to \Delta$ such that $\langle \alpha x + \beta y \mid z \rangle = \alpha\langle x \mid z\rangle + \beta\langle y \mid z\rangle$ and $\langle y \mid x\rangle = \langle x \mid y\rangle^*$ for all $x, y, z \in E$ and $\alpha, \beta \in \Delta$; it is *anisotropic* when $\langle x \mid x\rangle \neq 0$ for every nonzero vector $x \in E$. A *hermitian space* is a left vector space over a division ring with an involution, together with an anisotropic hermitian form.

It follows from these identities that $\langle x \mid \alpha y\rangle = \langle x \mid y\rangle\alpha^*$ for all $x, y \in E$ and $\alpha \in \Delta$. The bracket is linear in its first argument, as is customary in mathematics; in the Dirac notation used by physicists it is linear in the second. Nothing in the definition forces the scalars to be real or complex: over $\mathbb{Q}$ with the identity involution, the form $\sum_n x_n y_n$ on finitely supported rational sequences $x = (x_n)_{n\in\mathbb{N}}$ and $y = (y_n)_{n\in\mathbb{N}}$ is anisotropic. The anisotropy of this form rests on the fact that a sum of rational squares vanishes only when every square does.

Definition 2.2. In a hermitian space E, two vectors x and y are *orthogonal*, written $x \perp y$, when $\langle x \mid y\rangle = 0$. For $S \subseteq E$, let $S^\perp$ be the set $\{x \in E : \langle x \mid y\rangle = 0 \text{ for all } y \in S\}$. A subspace M is *orthogonally closed* when $M^{\perp\perp} = M$.

Orthogonality is symmetric, since $\langle y \mid x\rangle = \langle x \mid y\rangle^*$ and the involution fixes 0. In particular, every subset S of E satisfies $S \subseteq S^{\perp\perp}$.

Definition 2.3 (The lattice $\mathcal{C}(E)$). For a hermitian space E, we write $\mathcal{C}(E)$ for the set of orthogonally closed subspaces of E, ordered by inclusion.

Every set of the form $S^\perp$ is an orthogonally closed subspace: applying the inclusion-reversing operation $^\perp$ to $S \subseteq S^{\perp\perp}$ gives $S^{\perp\perp\perp} \subseteq S^\perp$, and the reverse inclusion is the same fact for $S^\perp$. If $(M_j)_{j\in J}$ is a family of orthogonally closed subspaces indexed by an arbitrary set J, then their intersection, which equals $\bigcap_j M_j^{\perp\perp} = \big(\bigcup_j M_j^\perp\big)^\perp$, is orthogonally closed. Hence $\mathcal{C}(E)$ is a complete lattice with least element $\{0\}$ and greatest element E. In this lattice the family has meet $\bigcap_j M_j$ and join $\big(\sum_j M_j\big)^{\perp\perp}$, which may be strictly larger than $\sum_j M_j$. For $M \in \mathcal{C}(E)$, anisotropy gives $M \cap M^\perp = \{0\}$, whence $M \vee M^\perp = (M^\perp \cap M)^\perp = E$. Since the map $M \mapsto M^\perp$ also reverses inclusion and is its own inverse on $\mathcal{C}(E)$, the lattice $\mathcal{C}(E)$ is, with this map, a complete ortholattice.

Definition 2.4. A hermitian space E is *orthomodular* when $M + M^\perp = E$ for every $M \in \mathcal{C}(E)$, where $M + M^\perp$ is the set of sums $v + y$ with $v \in M$ and $y \in M^\perp$.

The lattice join $M \vee M^\perp$ is always E; orthomodularity asks that the algebraic sum be E as well.

Definition 2.5. A *Hilbert space* is a hermitian space H over the real numbers $\mathbb{R}$, the complex numbers $\mathbb{C}$ or the quaternions $\mathbb{H}$, with the usual involution (the identity, complex conjugation or quaternionic conjugation), such that $\langle x \mid x\rangle > 0$ for every nonzero vector x and the space is complete in the norm $\|x\| = \langle x \mid x\rangle^{1/2}$.

Let H be a Hilbert space. By the projection theorem, every closed subspace M of H satisfies $M + M^\perp = H$, and every subspace M satisfies $M^{\perp\perp} = \overline{M}$. Hence the orthogonally closed subspaces of H are exactly its closed subspaces, which form the lattice $L(H) = \mathcal{C}(H)$. The space H is orthomodular, and so, by Proposition 2.8 below, is the lattice $L(H)$.

Lemma 2.6. *In a hermitian space E, every finite-dimensional subspace F is orthogonally closed and satisfies $F + F^\perp = E$.*

[1] Physicists usually let quaternionic scalars act on the right; their quaternionic Hilbert spaces are therefore right modules. The choice of side is immaterial: a left vector space over Δ is a right vector space over the opposite division ring, and an involution is an isomorphism of Δ onto its opposite.

Proof. The idea is to write down the orthogonal projection onto F. Let n be the dimension of F. By anisotropy, the Gram–Schmidt process gives F an orthogonal basis $w_1, \dots, w_n$ with $\langle w_i \mid w_i \rangle \neq 0$ for $i = 1, \dots, n$. For $z \in E$, the vector $Pz = \sum_i \langle z \mid w_i \rangle \langle w_i \mid w_i \rangle^{-1} w_i$ lies in F, and the difference $z - Pz$ lies in $F^\perp$; hence $F + F^\perp = E$. If $y \in F^{\perp\perp}$, the vector $y - Py$ lies in $F^\perp \cap F^{\perp\perp}$ and is therefore orthogonal to itself; by anisotropy it is 0, whence $y \in F$. ☺

Lemma 2.7. *In a hermitian space E over a division ring Δ, the subspace $M + \Delta x$ belongs to $\mathcal{C}(E)$ for every $M \in \mathcal{C}(E)$ and every $x \in E$.*

Proof. If $x \in M$, there is nothing to prove; we therefore assume that $x \notin M$. Since $M + \Delta x$ is contained in $(M + \Delta x)^{\perp\perp}$, it suffices to show that every vector of $(M + \Delta x)^{\perp\perp}$ differs from a multiple of x by a vector of M. The idea is that such a vector induces on $M^\perp$, through the form, a functional obtained from the one induced by x by multiplication on the right by a scalar. Put $W = M^\perp$, and let ξ_x be the functional $z \mapsto \langle z \mid x \rangle$ on W. This functional is nonzero, since $x \notin W^\perp = M$, and we may choose $z_0 \in W$ with $\xi_x(z_0) = 1$. Let y be an element of $(M + \Delta x)^{\perp\perp}$, and let ξ_y be the functional $z \mapsto \langle z \mid y \rangle$ on W. Since $W \cap x^\perp = (M + \Delta x)^\perp$, the functional ξ_y vanishes on the kernel of ξ_x; as $z - \xi_x(z) z_0$ lies in that kernel for every $z \in W$, we obtain $\xi_y(z) = \xi_x(z)\alpha$ with $\alpha = \xi_y(z_0)$. Because $\langle z \mid \alpha^* x \rangle = \langle z \mid x \rangle \alpha$, the vector $y - \alpha^* x$ is orthogonal to W and therefore lies in $W^\perp = M$, whence $y \in M + \Delta x$. ☺

Proposition 2.8. *For a hermitian space E, the lattice $\mathcal{C}(E)$ is orthomodular if and only if E is an orthomodular space.*

Proof. Suppose first that E is orthomodular, and let $M \subseteq M'$ be elements of $\mathcal{C}(E)$. Every vector $z \in M'$ can be written as $v + y$ with $v \in M$ and $y \in M^\perp$; since $y = z - v$ lies in M', we have $M' = M + (M' \cap M^\perp) \subseteq M \vee (M' \wedge M^\perp) \subseteq M'$, which is the orthomodular law. Conversely, suppose that $\mathcal{C}(E)$ is orthomodular, and let Δ be the division ring of scalars of E. Let M be an element of $\mathcal{C}(E)$, and let x be a vector of $E \setminus M$. By Lemma 2.7, the subspace $M' = M + \Delta x$ belongs to $\mathcal{C}(E)$, and orthomodularity gives $M' = M \vee (M' \cap M^\perp)$. Since $M' \neq M$, the subspace $M' \cap M^\perp$ is nonzero. As this subspace meets M only in 0, and as M has codimension one in M', the subspace $M' \cap M^\perp$ is one-dimensional, spanned by a vector u. Then $M \subsetneq M + \Delta u \subseteq M'$, and the codimension count gives $M' = M + \Delta u$; hence $x \in M + M^\perp$. ☺

⌛ **Historical note.** In 1964 Piron published a theorem asserting that every orthomodular complex inner product space is complete, but his proof contained an error [Pir64]. Amemiya and Araki gave a correct proof in 1966, and their method also covers real and quaternionic scalars [AA66]. It was then speculated that, among infinite-dimensional hermitian spaces over arbitrary division rings, orthomodularity alone characterizes the Hilbert spaces over $\mathbb{R}$, $\mathbb{C}$ and $\mathbb{H}$. Keller ended the speculation in 1980 with a "non-classical Hilbert space", an infinite-dimensional orthomodular hermitian space over a division ring other than these three [Kel80]. Keller was a student of Herbert Gross, whose papers and book on forms in infinite-dimensional vector spaces shaped the theory; the story is told in Holland's survey of 1995 [Hol95].

3. Projective geometry, Solèr lattices and pre-Solèr lattices

In this section we explain how an ortholattice with a few combinatorial properties can be represented by orthogonally closed subspaces of a hermitian space. A hermitian space, we recall, is a vector space over a division ring with an anisotropic hermitian form. We also explain how Solèr's condition, together with completeness, forces the scalars to be real, complex

or quaternionic. The ortholattices that have the combinatorial properties in question, Solèr's condition and completeness are the Solèr lattices; those that have them all except possibly completeness are the pre-Solèr lattices. In the standard models of our theory, completeness comes from the sequences of vectors placed beside the lattice rather than from axioms about the lattice alone (see Section 5 and Theorem 8.3). Pre-Solèr lattices therefore recur throughout the paper. We assume familiarity with Hilbert spaces, but none with projective geometry.

3.1. **Points and lines.** In a complex Hilbert space H, the atoms of the lattice $L(H)$ of closed subspaces are the *rays*, the one-dimensional subspaces $\mathbb{C}x$ spanned by nonzero vectors $x \in H$. The join of two distinct rays is the two-dimensional subspace, or *plane*, that they span. More generally, we call the one-dimensional subspaces of any vector space its *rays*. Which rays lie in which planes is information about the lattice alone (see Figure 1). For a vector space of dimension at least four, this information determines the space and its division ring of scalars, in the sense made precise by Theorem 3.2 below.

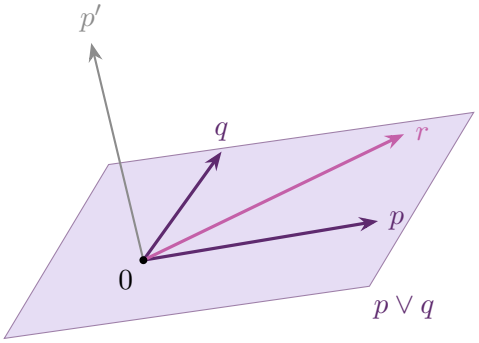


FIGURE 1. In the lattice $L(H)$ of closed subspaces of a complex Hilbert space H, the atoms are the rays. The join $p \vee q$ of two rays p and q is the plane they span; the ray r in that plane is an atom below $p \vee q$, while the ray p' is not.

Definition 3.1. A *projective geometry* is a set of *points* together with a family of subsets, called *lines*, such that:

(PG1) any two distinct points lie on exactly one line;
(PG2) every line contains at least three points;
(PG3) (*the Veblen–Young axiom*) if a, b, a', b' are distinct points and the line through a and b meets the line through a' and b', then the line through a and a' meets the line through b and b'.

A set of points is a *projective subspace* when it contains the whole line through any two of its points. The projective subspace *generated* by a set of points is the intersection of all projective subspaces containing it. A *projective plane* is the projective subspace generated by three points not on one line. Axiom (PG3) says that two lines lying in a common projective plane always meet (see Figure 2); a projective geometry has no parallel lines. The word *projective* comes from perspective drawing, in which parallel rails, projected onto the canvas, meet at the horizon.

The basic example comes from linear algebra. Let V be a left vector space of dimension at least two over a division ring Δ. Take as points the rays of V, and as lines the sets of rays contained in a two-dimensional subspace. Axiom (PG1) holds because two distinct rays

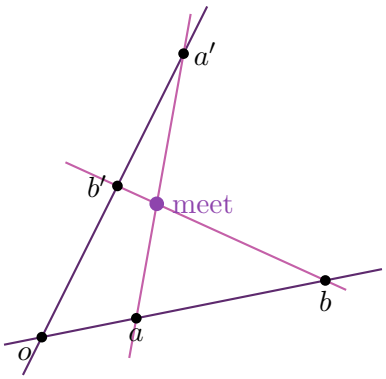


FIGURE 2. The Veblen–Young axiom. The lines ab and $a'b'$ meet at o; then the lines aa' and bb' meet as well.

span a unique two-dimensional subspace. Axiom (PG2) holds because the two-dimensional subspace spanned by linearly independent vectors x and y contains the three distinct rays Δx, Δy and $\Delta(x+y)$. For (PG3), suppose that the two-dimensional subspaces spanned by a, b and by a', b' have a ray in common. In that case the rays a, b, a', b' lie in a subspace of dimension at most three. By counting dimensions, any two two-dimensional subspaces of that subspace have a ray in common, in particular those spanned by a, a' and by b, b'.

Points are *independent* when none of them lies in the projective subspace generated by the others. In the basic example, rays are independent exactly when nonzero vectors spanning them, one in each ray, are linearly independent. Two projective geometries are *isomorphic* when there is a bijection between their sets of points under which the lines of the one correspond exactly to the lines of the other. Let E and E' be vector spaces over division rings Δ and Δ', respectively. A *semilinear isomorphism* from E onto E' is an additive bijection $f\colon E \to E'$ for which there is an isomorphism $\sigma\colon \Delta \to \Delta'$ with $f(\alpha x) = \sigma(\alpha) f(x)$ for all $\alpha \in \Delta$ and $x \in E$. The following theorem says that the basic example is, up to isomorphism, the only one as soon as there are four independent points.

Theorem 3.2 (Coordinatization of projective geometries; see Baer [Bae52]). *Every projective geometry that contains four independent points is isomorphic to the geometry of rays of a left vector space E over a division ring Δ. The pair (Δ, E) is unique up to a semilinear isomorphism.*

The division ring Δ is thus recovered from the geometry alone, whatever it may be: every division ring occurs. Among the division rings so recovered are the rational numbers, the non-archimedean ordered fields (fields with infinitesimals, such as fields of formal Laurent series) and the quaternions. The hypothesis of four independent points, on the other hand, is indispensable: a projective plane regarded as a projective geometry in its own right need not come from a division ring. The planes that do are exactly those in which Desargues's theorem holds (if two triangles are in perspective from a point, the three intersection points of corresponding sides lie on one line). Planes in which Desargues's theorem fails, the *non-Desarguesian* planes, exist. In a geometry with four independent points, by contrast, that theorem follows from the axioms.

3.2. **From lattices to geometries.** To apply Theorem 3.2 to an orthomodular lattice that is atomistic and has the covering property, we take the atoms of the lattice as the points of a geometry. The lines of this geometry are the sets $\{r : r \text{ an atom with } r \leq p \vee q\}$ for distinct atoms p and q; in $L(H)$, these sets consist of the rays in the plane spanned by p

and q. The covering property gives this geometry (PG1) and, with orthomodularity, (PG3); irreducibility in the sense of Definition 1.3 is exactly (PG2) (see Maeda and Maeda [MM70, §§33–34] and Holland [Hol95, §4]).

Orthocomplementation adds a relation between points: by the definition in Section 1, atoms p and q are orthogonal when $p \le q^\perp$. In $L(H)$ this is orthogonality of rays, and it determines the inner product up to the ambiguity described at the end of Theorem 3.3 below. The following properties of an ortholattice Λ enter the theorems of this section:

(C) the lattice is complete;
(OM) it is orthomodular;
(A) it is atomistic;
(CP) it has the covering property;
(I) it is irreducible;
(D) it contains four mutually orthogonal atoms.

Condition (D) supplies the four independent points of Theorem 3.2, since mutually orthogonal atoms are independent. The coordinatization theorem below goes back, in finite dimensions, to Birkhoff and von Neumann [BvN36]. The theorem shows that every ortholattice with (OM), (A), (CP), (I) and (D) is represented in the manner of $L(H)$. Under this representation, the atoms of such an ortholattice correspond to the rays of a vector space, and orthogonality of atoms to orthogonality of rays with respect to an anisotropic hermitian form.

Theorem 3.3 (Coordinatization; Baer [Bae52], Maeda and Maeda [MM70, §34], Holland [Hol95, §3]). *Let Λ be an ortholattice that satisfies (OM), (A), (CP), (I) and (D), but not necessarily (C). Then there are a division ring Δ with an involution, a hermitian space E over Δ, and a bijection $p \mapsto \hat{p}$ from the atoms of Λ onto the rays of E, such that two atoms p and q are orthogonal exactly when $\hat{p} \perp \hat{q}$. The map*

$$\gamma\colon \Lambda \to \mathcal{C}(E), \qquad \gamma(a) = \{0\} \cup \bigcup \{\hat{p} : p \text{ an atom with } p \le a\},$$

where $\mathcal{C}(E)$ is the lattice of orthogonally closed subspaces of E, is an embedding of ortholattices whose image contains every finite-dimensional subspace of E and the orthogonal complement of each. The triple formed by Δ, E and the form is unique up to a semilinear isomorphism followed by a rescaling $\langle x \mid y\rangle' = \langle x \mid y\rangle\zeta$, where $\zeta^* = \zeta \ne 0$, with the new involution $\alpha \mapsto \zeta^{-1}\alpha^*\zeta$.

Proof of the statements about γ. The orthocomplementation of Λ induces a *polarity* on the geometry of its atoms, in the sense of Holland [Hol95, Definition 3.2]. This polarity is the map that sends each projective subspace to the set of points orthogonal to all its points. The polarity reverses inclusion and sends the set of all points to the empty set. Under the polarity, each point p goes to the set of atoms below $p^\perp$, a hyperplane (that is, a maximal proper projective subspace) whose image is again $\{p\}$ (see Holland [Hol95, §4]). By Theorem 3.2, the atoms become the rays of a vector space E. Baer's theorem [Bae52] (see Holland [Hol95, Theorem 3.6]) shows that every polarity of this kind under which no point is orthogonal to itself is orthogonality for an anisotropic hermitian form on E. If two atoms lie below a, every atom on the line they determine lies below a; hence $\gamma(a)$ is a subspace. For an atom r the following are equivalent: $r \le a^\perp$; $a \le r^\perp$; $p \le r^\perp$ for every atom $p \le a$ (because (A) makes a the join of these atoms); $\hat{r} \perp \hat{p}$ for every atom $p \le a$. Thus $\gamma(a^\perp) = \gamma(a)^\perp$, and applying this twice gives $\gamma(a) = \gamma(a)^{\perp\perp}$. By (A), the relation $a \le b$ holds if and only if $\gamma(a) \subseteq \gamma(b)$; hence γ is injective. Meets are preserved because an atom lies below $a \wedge b$ exactly when it lies below a and b, and joins are preserved by the De Morgan law. Finally, let F be a finite-dimensional subspace of E with basis $x_1, \dots, x_k$, and let p_i be the atom with $\hat{p}_i = \Delta x_i$, for $i = 1, \dots, k$.

The join $p_1 \vee \cdots \vee p_k$ exists, as every finite join does in a lattice. Because γ preserves joins, the image of this join is $(\Delta x_1 + \cdots + \Delta x_k)^{\perp\perp} = F^{\perp\perp}$, which is F by Lemma 2.6. The orthogonal complement $F^\perp$ is therefore $\gamma\big((p_1 \vee \cdots \vee p_k)^\perp\big)$. ☻

Piron [Pir64] proved this representation for complete lattices; completeness is not needed for the embedding, and in Theorem 3.6 below it serves only to make γ surjective.

3.3. **Harmonic conjugates and Solèr's condition.** The hypotheses of Theorem 3.3 do not force the division ring Δ to be $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{H}$. Indeed, the lattice of all subspaces of $\mathbb{Q}^4$, orthocomplemented by orthogonality for the form $\sum_{i<4} x_i y_i$, satisfies these hypotheses with $\Delta = \mathbb{Q}$. Solèr's condition, together with completeness, singles out the real numbers, the complex numbers and the quaternions. This condition is phrased through the harmonic conjugate, a classical construction of projective geometry that uses joins and meets alone. Consider, in the hermitian space of Theorem 3.3, a ray of a plane spanned by two orthogonal rays and the harmonic conjugate of that ray with respect to these two. This harmonic conjugate is the orthocomplement of the ray in the plane exactly when the ray is spanned by the sum of two vectors of equal "length", one from each orthogonal ray. We verify this equivalence below (see Figure 4). Agreement along an infinite orthogonal sequence of atoms therefore yields, after a rescaling of the form, an infinite orthonormal sequence. This orthonormal sequence, together with orthomodularity, is what Solèr's theorem requires (see Lemma 3.4 and Theorem 3.6).

Let p and q be distinct points of a projective geometry, and write $p+q$ for the line through them. In the basic example, this line consists of the rays of the plane spanned by p and q. Fix a point c of $p+q$ other than p and q. By (PG1) and (PG3), two distinct lines in a common projective plane meet in exactly one point. With a slight abuse of notation, we denote that point by the intersection of the two lines. Choose a point r not on $p+q$ and a point r' on $r+p$ other than r and p; put $c' = (c+r') \cap (q+r)$ and $c'' = (p+c') \cap (q+r')$. The point $h = (r+c'') \cap (p+q)$ is the *harmonic conjugate* of c with respect to p and q (see Figure 3). In the basic example the point h does not depend on the choices of r and r', as the computation below shows. By Theorem 3.2, the same holds in every projective geometry with four independent points. Consider now an ortholattice that satisfies (OM), (A), (CP), (I) and (D). Its atoms form a projective geometry with four independent points, as in the preceding subsection. The same construction applies to these atoms, with $\vee$ in place of $+$ and $\wedge$ in place of $\cap$. The relation "h is the harmonic conjugate of c with respect to p and q" is then expressed by a first-order formula in the language of lattices.

In the basic example, the construction is simple in coordinates. Let u and v be vectors spanning p and q, respectively, scaled so that $c = \Delta(u+v)$; this is possible because c lies on $p+q$ and differs from p and q. Likewise, let w be a vector spanning r, scaled so that $r' = \Delta(u+w)$, which is possible because r' lies on $r+p$ and differs from r and p. A direct computation gives $c' = \Delta(v-w)$, $c'' = \Delta(u-v+w)$ and $h = \Delta(u-v)$. All coefficients are ± 1, so the computation is valid over every division ring:

$$\text{the harmonic conjugate of } \Delta(u+v) \text{ with respect to } \Delta u \text{ and } \Delta v \text{ is } \Delta(u-v). \tag{1}$$

Suppose now that the vector space carries an anisotropic hermitian form, as the space E of Theorem 3.3 does, and that $p \perp q$; then $u \perp v$. The orthocomplement of c within the plane $p \vee q$, the ray $c^\perp \wedge (p \vee q)$, consists of the vectors $\alpha u + \beta v$ with $\langle \alpha u + \beta v \mid u+v \rangle = \alpha \langle u \mid u \rangle + \beta \langle v \mid v \rangle = 0$. It coincides with the harmonic conjugate $\Delta(u-v)$ exactly when $\langle u \mid u \rangle = \langle v \mid v \rangle$, that is, when u and v have the same "length" (see Figure 4). In characteristic 2, that is, when $1+1=0$ in Δ, the vectors $u+v$ and $u-v$ coincide. In that case the equality $\langle u \mid u \rangle = \langle v \mid v \rangle$

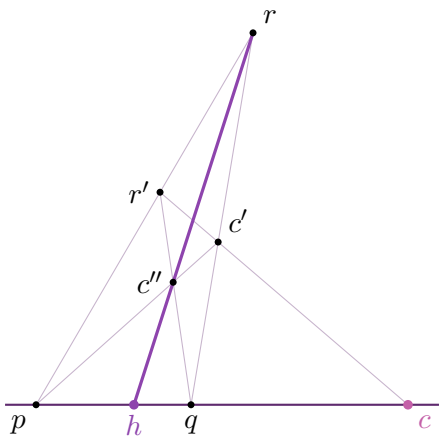


FIGURE 3. The construction of the harmonic conjugate h of a point c of the line through p and q, with respect to p and q. The construction uses the auxiliary points r, r', c' and c''.

would make $u+v$ isotropic, because it would give $\langle u+v \mid u+v\rangle = (1+1)\langle u \mid u\rangle = 0$. Hence, when the harmonic conjugate of c coincides with its orthocomplement in the plane $p \vee q$, the characteristic of Δ is not 2.

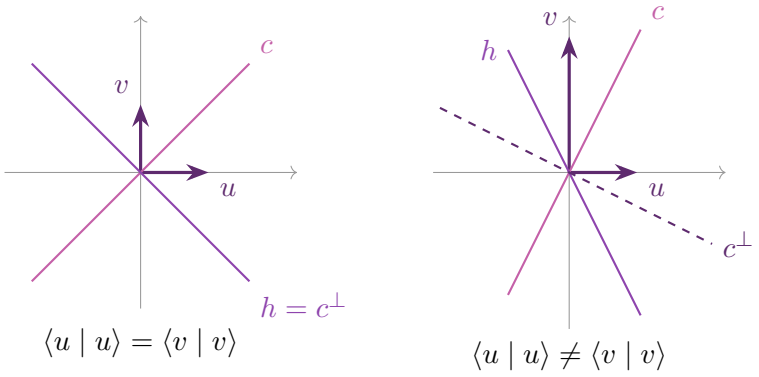


FIGURE 4. The ray $c = \Delta(u+v)$ spanned by the sum of orthogonal vectors u and v, and its harmonic conjugate $h = \Delta(u-v)$ with respect to Δu and Δv. The orthocomplement $c^\perp$ of c in the plane of u and v is drawn as well. The rays h and $c^\perp$ coincide when u and v have equal length (left), and differ otherwise (right).

Solèr's condition, in the form we use, asks for an infinite orthogonal sequence of atoms along which the orthocomplement and the harmonic conjugate agree at every step:

(S) There are mutually orthogonal atoms p_n, for $n \in \mathbb{N}$, and atoms $c_n \leq p_n \vee p_{n+1}$ with $c_n \neq p_n, p_{n+1}$, such that, for every $n \in \mathbb{N}$, the harmonic conjugate of c_n with respect to p_n and p_{n+1} is $c_n^\perp \wedge (p_n \vee p_{n+1})$.

This formulation of Solèr's condition through harmonic conjugates is due to Holland [Hol95, Theorems 3.7 and 4.1]. Morash's angle-bisecting axiom [Mor73], which asks for bisectors as in the left half of Figure 4, may be used instead (see Holland [Hol95, §4]).

Lemma 3.4. *Let Λ be an ortholattice that satisfies (OM), (A), (CP), (I), (D) and (S), and let E be a hermitian space over a division ring Δ and $p \mapsto \hat{p}$ a bijection from the atoms of Λ onto the rays of E that Theorem 3.3 provides for Λ. After a rescaling of the form, the space E contains an infinite orthonormal sequence.*

Proof. Let p_n and c_n, for $n \in \mathbb{N}$, be atoms as in (S). Choose e_0 with $\hat{p}_0 = \Delta e_0$, and inductively choose e_{n+1} with $\hat{p}_{n+1} = \Delta e_{n+1}$ and $\hat{c}_n = \Delta(e_n + e_{n+1})$. The choice of e_{n+1} is possible because $\hat{c}_n$ is a ray of the plane spanned by e_n and e_{n+1}, distinct from Δe_n and Δe_{n+1}. By (1) and the computation that follows it, condition (S) gives $\langle e_n \mid e_n \rangle = \langle e_{n+1} \mid e_{n+1} \rangle$ for every $n \in \mathbb{N}$; call this common value ζ, which satisfies $\zeta^* = \zeta \neq 0$. The form $\langle x \mid y \rangle' = \langle x \mid y \rangle \zeta^{-1}$ is hermitian for the involution $\alpha \mapsto \zeta \alpha^* \zeta^{-1}$, has the same orthogonality relation, and satisfies $\langle e_n \mid e_n \rangle' = 1$ for every n. ☺

Definition 3.5. A *Solèr lattice* is an ortholattice that satisfies (C), (OM), (A), (CP), (I), (D) and (S). A *pre-Solèr lattice* is an ortholattice that satisfies (OM), (A), (CP), (I), (D) and (S), but not necessarily (C).

Theorem 3.6 (Piron, Amemiya and Araki, Solèr; in this form Holland [Hol95, Theorem 4.1]). *An ortholattice is a Solèr lattice if and only if it is isomorphic to $L(H)$ for some infinite-dimensional Hilbert space H over $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{H}$.*

Sketch. Let Λ be a Solèr lattice, with γ and E as in Theorem 3.3. Given $M \in \mathcal{C}(E)$, completeness provides the join a of the atoms p with $\hat{p} \subseteq M$, and this join satisfies $\gamma(a)^\perp = \gamma(a^\perp) = M^\perp$, whence $\gamma(a) = M$. Thus the embedding γ is an isomorphism onto $\mathcal{C}(E)$; the lattice $\mathcal{C}(E)$ is therefore orthomodular, and by Proposition 2.8 so is the space E. By Lemma 3.4, the space E contains, after a rescaling of the form that changes neither $\mathcal{C}(E)$ nor its orthocomplementation, an infinite orthonormal sequence. Solèr's theorem [Sol95] (see also Prestel [Pre95]) asserts that an orthomodular space containing an infinite orthonormal sequence is a Hilbert space over $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{H}$. The space E, with its form so rescaled, is therefore such a Hilbert space. Conversely, the lattice $L(H)$ is complete and satisfies the other conditions. For (S), take an orthonormal sequence $(e_n)_{n \in \mathbb{N}}$ in H, and let p_n be the ray of e_n and c_n the ray of $e_n + e_{n+1}$. ☺

Condition (S) cannot be omitted from Theorem 3.6. Keller [Kel80] constructed infinite-dimensional orthomodular hermitian spaces over division rings other than $\mathbb{R}$, $\mathbb{C}$ and $\mathbb{H}$. For such a space E the lattice $\mathcal{C}(E)$ satisfies (C), (OM), (A), (CP), (I) and (D) (see Proposition 2.8 and Lemmas 2.7 and 2.6). Nevertheless, by the uniqueness clause of Theorem 3.3, the lattice $\mathcal{C}(E)$ is isomorphic to no lattice $L(H)$ of a Hilbert space H over $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{H}$.

⌛ **Historical note.** Von Staudt's *Geometrie der Lage* of 1847 was the first treatment of projective geometry free of any metric. His later calculus of "throws" outlined the algebraization of an axiomatic geometry. Hilbert showed in his *Grundlagen der Geometrie* of 1899 that Desargues's theorem follows from the axioms of space but cannot be proved in the plane without the axioms of congruence. On that theorem he founded an algebra of segments that does not use those axioms [Hil99]. Veblen and Young gave axioms for projective geometry in 1908 [VY08]. In their book of 1910 they introduced coordinates through an algebra of points that descends from von Staudt's algebra of throws [VY10]. Solèr, a student of Herbert Gross, had just begun work on Gross's orthomodular problem when Gross died suddenly in 1989. She completed the work under the informal guidance of Alexander Prestel, a friend of Gross's at Konstanz, and her theorem appeared in 1995 [Sol95]. In a survey of the same year, Holland restated her condition through harmonic conjugates, constructed as in the book of Veblen and Young, in the form used in this section [Hol95].

4. What a pre-Solèr lattice is missing (optional)

A pre-Solèr lattice satisfies every condition in the definition of a Solèr lattice except, possibly, completeness. We show that the MacNeille completion of such a lattice supplies the joins and meets that the lattice may lack. We show further that the completion is a Solèr lattice exactly when it is orthomodular, which need not be the case. Neither completeness nor the orthomodularity of the completion is expressed, among pre-Solèr lattices, by any sentence of two-valued $L_{\omega_1\omega}$ about the lattice alone (see Proposition 4.5 below). The theory of Section 6 secures both in its standard models, through the sorts for vectors and for sequences of vectors that the vocabulary of Section 5 places beside the Solèr sort.

Throughout this section we fix a pre-Solèr lattice Λ and a hermitian space E that Theorem 3.3 provides for Λ. Under the embedding γ of that theorem, we identify Λ with its image $\gamma(\Lambda) \subseteq \mathcal{C}(E)$.

Theorem 4.1. *Let Λ be a pre-Solèr lattice, let E be a hermitian space that Theorem 3.3 provides for Λ, and identify Λ with its image $\gamma(\Lambda)$ in $\mathcal{C}(E)$ under the embedding γ of that theorem. Then Λ is join-dense and meet-dense in $\mathcal{C}(E)$; consequently $\mathcal{C}(E)$, with $M \mapsto M^\perp$, is isomorphic as an ortholattice to the MacNeille completion of Λ.*

Proof. Every $M \in \mathcal{C}(E)$ is the join in $\mathcal{C}(E)$ of the rays it contains, which belong to Λ by Theorem 3.3. If $M^\perp = \{0\}$, then the subspace M is E, which belongs to Λ. Otherwise M equals $M^{\perp\perp} = \bigcap\{y^\perp : 0 \neq y \in M^\perp\}$, a meet of hyperplanes, each of which belongs to Λ. Both assertions now follow from Proposition 1.5. ☺

The elements of the MacNeille completion that do not come from Λ are therefore orthogonally closed subspaces M such that M and $M^\perp$ are both infinite-dimensional. Indeed, the lattice Λ contains every finite-dimensional subspace and the orthogonal complement of each (see Theorem 3.3). Adjoining these elements to Λ need not produce a lattice isomorphic to some $L(H)$, because the completion may fail to be orthomodular (see Example 4.3). As the next theorem shows, orthomodularity of the completion is exactly what is needed.

Theorem 4.2. *Let Λ be a pre-Solèr lattice, and let E be a hermitian space that Theorem 3.3 provides for Λ. Then the following are equivalent:*

(1) the MacNeille completion of Λ is orthomodular;
(2) the space E is orthomodular;
(3) the lattice Λ is isomorphic to a join-dense and meet-dense sub-ortholattice of $L(H)$ for some infinite-dimensional Hilbert space H over $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{H}$.

When these conditions hold, the space E, with its form suitably rescaled, is a Hilbert space as in (3). Moreover, the lattice Λ is isomorphic to $L(H)$ for such an H if and only if Λ is complete.

Proof. Conditions (1) and (2) are equivalent by Theorem 4.1 and Proposition 2.8. If (2) holds, then Lemma 3.4 and Solèr's theorem show that the space E, with its form rescaled, is a Hilbert space. Rescaling changes neither $\mathcal{C}(E)$ nor its orthocomplementation, and condition (3) follows by Theorem 4.1. If (3) holds, then by Proposition 1.5 the orthomodular lattice $L(H)$ is isomorphic as an ortholattice to the MacNeille completion of Λ; this is (1). For the last assertion, a complete pre-Solèr lattice is a Solèr lattice and by Theorem 3.6 is therefore isomorphic to $L(H)$ for an infinite-dimensional Hilbert space H over $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{H}$. Conversely, every lattice $L(H)$ is complete. ☺

Example 4.3. Let E_0 be the space of finitely supported complex sequences $x = (x_n)_{n\in\mathbb{N}}$, with the inner product $\langle x \mid y\rangle = \sum_n x_n\overline{y_n}$ of the space ℓ^2 of square-summable sequences.

Let Λ_0 be the set of finite-dimensional subspaces of E_0 and their orthogonal complements in E_0. Then Λ_0 is a sub-ortholattice of $\mathcal{C}(E_0)$: its elements are orthogonally closed (the finite-dimensional ones by Lemma 2.6). Moreover, for finite-dimensional F and G the joins $F \vee G = F + G$, $F^\perp \vee G^\perp = (F \cap G)^\perp$ and $F \vee G^\perp = (F^\perp \cap G)^\perp$ belong to Λ_0, whence the De Morgan law places the meets in Λ_0 as well. The lattice Λ_0 is orthomodular, because every $M \in \Lambda_0$ satisfies $M + M^\perp = E_0$ by Lemma 2.6, and the argument of Proposition 2.8 applies. As a sub-ortholattice of $\mathcal{C}(E_0)$ that contains every ray of E_0, the lattice Λ_0 inherits (A), (CP) and (I) from $\mathcal{C}(E_0)$, in which (CP) holds by Lemma 2.7. Conditions (D) and (S) hold with $p_n = \mathbb{C}e_n$ and $c_n = \mathbb{C}(e_n + e_{n+1})$, where $(e_n)_{n\in\mathbb{N}}$ is the sequence of standard unit vectors. Consequently Λ_0 is a pre-Solèr lattice, and the space E_0, with $\hat{p} = p$ for every atom p of Λ_0, is a hermitian space that Theorem 3.3 provides for Λ_0. The map γ of that theorem is then the inclusion of Λ_0 in $\mathcal{C}(E_0)$. The MacNeille completion of Λ_0, which is $\mathcal{C}(E_0)$ by Theorem 4.1, is not orthomodular. Indeed, by the Amemiya–Araki theorem [AA66], a complex inner product space whose lattice of orthogonally closed subspaces is orthomodular is complete, which E_0 is not. Hence Λ_0 is an orthomodular lattice whose MacNeille completion is not orthomodular; Harding [Har91] studies such lattices.

Orthomodularity of the completion can be stated through the atoms of Λ alone, without reference to the completion or to the space E. It is in this form that Proposition 4.5 below shows that orthomodularity of the completion cannot be expressed by a sentence of two-valued $L_{\omega_1\omega}$ about the lattice. For a set $\mathcal{A}$ of atoms, let $\mathcal{A}^\perp$ be the set of atoms orthogonal to every element of $\mathcal{A}$, and call $\mathcal{A}$ *orthoclosed* when $\mathcal{A} = \mathcal{A}^{\perp\perp}$. Consider the following condition, named (MOM) for MacNeille orthomodularity:

(MOM) for every orthoclosed set $\mathcal{A}$ of atoms and every atom $p \notin \mathcal{A}$, some atom $q \in \mathcal{A}^\perp$ lies in $(\mathcal{A} \cup \{p\})^{\perp\perp}$.

Theorem 4.4. *The MacNeille completion of a pre-Solèr lattice is orthomodular if and only if the lattice satisfies (MOM).*

Proof. Let $\hat{p}$ denote, for each atom p, the ray of E that corresponds to p under Theorem 3.3. The map $\mathcal{A} \mapsto \{0\} \cup \bigcup\{\hat{p} : p \in \mathcal{A}\}$ is a bijection from the orthoclosed sets of atoms onto $\mathcal{C}(E)$, carrying $\mathcal{A}^\perp$ to the orthogonal complement of the image of $\mathcal{A}$. If M is the image of $\mathcal{A}$ and $\hat{p} = \Delta x$, where Δ is the division ring of scalars of E, then by Lemma 2.7 the set $(\mathcal{A} \cup \{p\})^{\perp\perp}$ corresponds to $M + \Delta x$. Condition (MOM) therefore says that $(M + \Delta x) \cap M^\perp \neq \{0\}$ whenever $M \in \mathcal{C}(E)$ and $x \notin M$. If $y = v + \alpha x$ is nonzero, with $v \in M$ and $y \in M^\perp$, then the scalar α is nonzero because $y \notin M$, whence the vector $x = \alpha^{-1}(y - v)$ lies in $M + M^\perp$. Conversely, if $x = v + y$ with $v \in M$ and $y \in M^\perp$, then the vector $y = x - v$ is a nonzero element of $(M + \Delta x) \cap M^\perp$. Thus (MOM) is equivalent to the orthomodularity of E, and the theorem follows from Theorem 4.2. ☺

By Lemma 2.6, the requirement in (MOM) holds automatically when $\mathcal{A}$ or $\mathcal{A}^\perp$ corresponds to a finite-dimensional subspace. The content of (MOM) lies in the orthoclosed sets $\mathcal{A}$ for which both $\mathcal{A}$ and $\mathcal{A}^\perp$ correspond to infinite-dimensional subspaces.

Proposition 4.5. *Neither (C) nor (MOM) is equivalent, among pre-Solèr lattices, to a sentence of two-valued $L_{\omega_1\omega}$ in the vocabulary of ortholattices expanded by constants for atoms p_n and c_n, for $n \in \mathbb{N}$, that witness (S).*

Proof. Suppose that a sentence φ of $L_{\omega_1\omega}$ were equivalent to (MOM) among pre-Solèr lattices. The axioms of ortholattices and the conditions (OM), (A), (CP), (I), (D) and (S) are expressed by countably many sentences of $L_{\omega_1\omega}$ in this vocabulary; together with φ, they hold in $L(\ell^2)$. By the downward Löwenheim–Skolem theorem for a countable fragment, they

hold in a countable structure Λ', which is a pre-Solèr lattice satisfying (MOM). By Theorems 4.4 and 4.2, the lattice Λ' contains every ray of an infinite-dimensional Hilbert space, and is therefore uncountable, a contradiction. The same argument applies to (C), using Theorem 3.6.

Historical note. MacNeille introduced the completion by cuts in 1937 [Mac37], generalizing the cuts by which Dedekind had constructed the real numbers from the rationals in 1872 [Ded72]. Banaschewski and Schmidt characterized it in 1956 by join-density and meet-density [Ban56, Sch56]. In 1969 Adams solved negatively Problem 36 of Birkhoff's *Lattice Theory*, in its third edition of 1967, by giving an orthocomplemented modular lattice whose completion by cuts is not orthomodular [Ada69]. His lattice consists of the finite-dimensional subspaces of an incomplete complex inner product space and their orthogonal complements, as in Example 4.3, and his argument also rests on the Amemiya–Araki theorem. Harding gave in 1991, without recourse to the Amemiya–Araki theorem, a construction of orthomodular lattices whose completions by cuts are not orthomodular. Following a construction that Kalmbach introduced in 1977, his construction glues together the Boolean algebras generated by the chains of a lattice [Har91]. Harding noted that whether every orthomodular lattice can be embedded in a complete one was still open.

5. The logic and the vocabulary

Completeness is the one condition in the definition of a Solèr lattice that a pre-Solèr lattice need not satisfy. The downward Löwenheim–Skolem theorem puts this condition beyond the reach of the sentences of $L_{\omega_1\omega}$, two-valued or real-valued, that speak of the lattice alone. Consider any countable fragment that contains sentences expressing the axioms of ortholattices and the other conditions in the definition of a Solèr lattice. A Solèr lattice, expanded by constants for atoms that witness Solèr's condition (S), has a countable elementary substructure. Every sentence of the fragment takes in this substructure the value that it takes in the Solèr lattice. This substructure is a pre-Solèr lattice, yet it is not complete. Were it complete, it would be a Solèr lattice, which by Theorem 3.6 is isomorphic to the lattice of closed subspaces of an infinite-dimensional Hilbert space and is therefore uncountable. In the two-valued case, this argument is given in detail in the optional Proposition 4.5. Because completeness thus eludes the sentences that speak of the lattice alone, it enters our theory through a Hilbert space placed beside the lattice. The joins that a pre-Solèr lattice may lack are supplied by that Hilbert space, through sequences of vectors whose closed spans the lattice is required to contain. In this section we first recall the features of the logic that we use. We then present a vocabulary in which the lattice stands beside a Hilbert space and a space of sequences of vectors. The three are linked by function symbols for orthogonal projection, for the passage between vectors and atoms, for the passage between vectors and sequences, and for the closed span of a sequence.

5.1. **Real-valued $L_{\omega_1\omega}$.** We use the real-valued infinitary logic of Aubone, Dueñez and Iovino [ADI26], which extends both Eagle's infinitary $[0,1]$-valued logic [Eag14] and the logic of Keisler's model theory for real-valued structures [Kei23]. Of this logic we use only the following features:

- The vocabulary is countable and many-sorted, and the structures are Keisler's general structures, up to affine rescalings of the predicate symbols [Kei23, Section 2.1]. The interpretation of a function symbol may be any function from the product of its argument sorts to its value sort, and no topology is prescribed in advance with respect

to which the predicates must be continuous. Eagle's metric structures, by contrast, have a master metric, and their predicates take values in $[0,1]$ [Eag14, Definition 1.4]; in a general structure each predicate symbol has a bound of its own, as the next item describes.

- ♠ Each predicate symbol R comes with a nonnegative real number r_R, its *a priori bound*, and takes its values in $[-r_R, r_R]$. In this paper the a priori bound of every predicate symbol is a positive integer. The vocabulary contains a nullary predicate symbol $\mathbf{1}$, the *logical unity constant*, with a priori bound 1 and value 1 in every structure.
- ♠ Formulas are built from the atomic formulas by the connectives $s \cdot \varphi$, for $s \in \mathbb{Q}$, and $\varphi \vee \psi$, interpreted as multiplication by s and as the maximum. To these are added the connectives $\sup_x$, one for each variable x, interpreted as the supremum over the sort of x. Moreover, the formulas are closed under countable joins $\bigvee_n \varphi_n$, or countable suprema, subject to two requirements on the family $(\varphi_n)_{n\in\mathbb{N}}$. The free variables of the formulas φ_n must lie in one finite set, and, for a positive rational number r, their values must all lie in $[-r, r]$ in every structure. The number r is part of the syntax of the join and serves as its a priori bound. The join is interpreted as the supremum of the values of its members, which may themselves contain joins. Aubone, Dueñez and Iovino form a join only when the a priori bounds r_{φ_n} of its members have a finite supremum, which is then its a priori bound. The explicit number r admits joins, such as $\lceil\varphi\rceil$ in (2) below, whose members take their values in $[0,1]$ although their a priori bounds grow without bound. In this paper we also allow continuous connectives $f(\varphi_1, \ldots, \varphi_\ell)$, for continuous functions $f\colon \mathbb{R}^\ell \to \mathbb{R}$, which we need for sums and products. The continuous connectives are taken from a fixed countable set, which contains those occurring in the theories we consider. With all continuous functions as connectives, the first-order formulas over a countable vocabulary would already be uncountable, and no fragment would be countable. Every formula φ receives an a priori bound r_φ, by an induction given in Section 10, and takes its values in the closed interval $I_\varphi = [-r_\varphi, r_\varphi]$ in every structure.
- ♠ We write 0 for $0 \cdot \mathbf{1}$, interpreted as the constant value 0, and $-\varphi$ for $(-1) \cdot \varphi$, interpreted as the negative. The minimum and the infimum over the sort of x are expressed by $\varphi \wedge \psi$, which abbreviates $-(-\varphi \vee -\psi)$, and by $\inf_x \varphi$, which abbreviates $-\sup_x -\varphi$. Finally, the countable infimum $\bigwedge_n \varphi_n$, interpreted as the infimum of the values, abbreviates $-\bigvee_n -\varphi_n$.
- ♠ The logic has no equality symbol. As in Keisler's model theory for real-valued structures [Kei23], a structure is *reduced* when any two elements that agree in every atomic formula with parameters are equal.

The statements of the logic are not formulas but conditions, which compare the value of a sentence with a rational number. For a sentence φ and a rational number r, a *condition* is an expression $\varphi > r$ or $\varphi < r$. A structure $\mathfrak{M}$ satisfies $\varphi > r$ when the value $\varphi^{\mathfrak{M}}$ of φ in $\mathfrak{M}$ is greater than r, and it satisfies $\varphi < r$ when that value is less than r. A *theory* is a set of conditions; a *model* of a theory is a reduced structure that satisfies every condition in it. Aubone, Dueñez and Iovino do not require models to be reduced; the difference is immaterial (see the end of the proof of Theorem 10.3). Indeed, identifying the elements of a structure that agree in every atomic formula with parameters yields a reduced structure in which every sentence keeps its value. Non-strict inequalities are recovered as countable theories: the *closed condition* $\varphi \geq r$ denotes the theory $\{\varphi > r - \frac{1}{n} : n \geq 1\}$, the closed condition $\varphi \leq r$ denotes $\{\varphi < r + \frac{1}{n} : n \geq 1\}$, and the closed condition $\varphi = r$ denotes the union of the two.

An *axiom* is a closed condition on a formula $\varphi(\bar{x})$, possibly without free variables, required of all values of $\bar{x}$. The axiom $\varphi(\bar{x}) \leq r$ is the closed condition $\sup_{\bar{x}} \varphi(\bar{x}) \leq r$, and the axiom $\varphi(\bar{x}) \geq r$ is $\inf_{\bar{x}} \varphi(\bar{x}) \geq r$; the axiom $\varphi(\bar{x}) = r$ is the union of the two. For formulas φ and ψ, we write $\varphi \leq \psi$ and $\varphi = \psi$ for the axioms $\varphi - \psi \leq 0$ and $\varphi - \psi = 0$, which are closed conditions when φ and ψ are sentences.

As in the work of Aubone, Dueñez and Iovino [ADI26], we think of 1 as true and of -1 as false. A formula is *two-valued* in a structure if it takes only the values 1 and -1 there. For two-valued formulas, negation, disjunction and conjunction are expressed by $-\varphi$, $\varphi \vee \psi$ and $\varphi \wedge \psi$. Likewise, the existential and the universal quantifier are expressed by $\sup_x$ and $\inf_x$, and countable disjunction and conjunction by $\bigvee_n$ and $\bigwedge_n$. A two-valued formula θ is asserted by the axiom $\theta \geq 1$; the implication from θ to θ' is the two-valued formula $-\theta \vee \theta'$. Real quantities such as inner products, norms and the measure keep their numerical values; every statement about these quantities is a closed condition or an axiom with a rational threshold. A measure or an assignment that takes only the values 0 and 1 is always described as such, or as $\{0,1\}$-valued, never as two-valued. The measure μ, which we introduce in Section 5.3, is a predicate on the Solèr sort. The axioms concerning μ are written with μ itself, which they confine to $[0,1]$, whereas the corresponding truth value at an element a of the Solèr sort is $2\mu(a) - 1$.

An implication whose hypothesis is a closed condition on a quantity requires a formula that records whether the quantity is positive. For a formula φ, let

$$\lceil \varphi \rceil = \bigvee_{n \geq 1} \big(((n \cdot \varphi) \wedge \mathbf{1}) \vee 0 \big), \tag{2}$$

a countable join with a priori bound 1, whose members take their values in $[0,1]$ in every structure, as the definition of a join requires. The formula $\lceil \varphi \rceil$ takes the value 0 where $\varphi \leq 0$ and the value 1 where $\varphi > 0$. Its values are thus 0 and 1, not -1 and 1, a fact on which form (iii) below depends. Let θ be a formula that is two-valued in the structures under consideration, and let ψ be a formula with $|\psi| \leq c$ in each of them, for a positive rational number c. Three forms of implication then occur among the axioms:

(i) The axiom $\theta \vee \lceil \varphi \rceil \geq 1$ says that $\varphi \leq 0$ implies θ.
(ii) The axiom $\theta \wedge \psi \leq 0$ says that θ implies $\psi \leq 0$.
(iii) The axiom $\psi \leq c\lceil \varphi \rceil$ says that $\varphi \leq 0$ implies $\psi \leq 0$.

In each case, if the hypothesis fails, the axiom holds whatever the value of the conclusion.

5.2. **Henkin types and model existence.** The model existence argument proceeds by assigning values to sentences and requiring those assignments to respect the semantics of the logic. Henkin types express these requirements for structures whose elements are named by closed terms. We recall their construction here so that the logical framework remains self-contained; in Section 10 we express the model existence argument through challenges and answers, and in Section A we compare that game with the topological treatment of Aubone, Dueñez and Iovino.

Following Aubone, Dueñez and Iovino [ADI26], we call a *fragment* a set of formulas that contains the first-order formulas and is closed under subformulas and under substitution of terms. A fragment must, in addition, be closed under the connectives $s \cdot \varphi$, $\varphi \vee \psi$ and $\sup_x$. For a countable fragment $\mathcal{F}$, let C be a set of new constants, countably many of each sort. Let $\mathcal{F}_C$ be the fragment that $\mathcal{F}$ generates in the vocabulary enlarged by C, and let $S_C^{\mathcal{F}}$ be the set of sentences of $\mathcal{F}_C$. An $S_C^{\mathcal{F}}$-*Henkin type* is an element of the product of the intervals I_φ, over the sentences φ of $S_C^{\mathcal{F}}$, that satisfies the *Henkin conditions*, which are as follows.

The constant $\mathbf{1}$ receives the value 1, and the value of $s \cdot \varphi$ is s times that of φ. The value of $\varphi \vee \psi$ is the maximum of the values of φ and ψ, and the value of a countable join is the supremum of the values of its members. Finally, the value of $\sup_x \varphi(x)$ is the supremum of the values of the sentences $\varphi(t)$, for closed terms t of the sort of x. For formulas built with continuous connectives, we add the condition that the value of $f(\varphi_1, \dots, \varphi_\ell)$ is the value of f at the values of $\varphi_1, \dots, \varphi_\ell$. With the topology induced by the product topology, the $S_C^{\mathcal{F}}$-Henkin types form the *space of Henkin types* $\mathcal{H}_C^{\mathcal{F}}$, a G_δ subset of the product and therefore a Čech-complete space [ADI26, Proposition 5 and Corollary 6]. The Baire category theorem holds in this space, to which we return in Section A.

5.3. **The vocabulary.** The vocabulary $\mathcal{L}$ has three sorts. Two of them, the Hilbert sort and the sequence sort, are *vector sorts*, intended to be the closed unit balls of Hilbert spaces. Working in unit balls keeps every predicate bounded.

- ♠ The *Hilbert sort* $\mathbf{H}$ has a binary function $(x, y) \mapsto \lambda x + \lambda' y$ for each pair λ, λ' of elements of the field $\mathbb{Q}(i)$ of complex numbers with rational real and imaginary parts such that $|\lambda| + |\lambda'| \leq 1$. The sort $\mathbf{H}$ also has the predicates $\mathrm{Re}\langle x \mid y\rangle$, $\mathrm{Im}\langle x \mid y\rangle$ and $\|x\|$, with a priori bound 1, and constants e_n for $n \in \mathbb{N}$. We write 0 for the closed term $0e_0 + 0e_0$, and $\langle x \mid y\rangle$ for the pair of predicates $\mathrm{Re}\langle x \mid y\rangle$ and $\mathrm{Im}\langle x \mid y\rangle$.
- ♠ The *sequence sort* $\mathbf{K}$ has the same functions and predicates as $\mathbf{H}$, and one constant ε; in this sort, we write 0 for the closed term $0\varepsilon + 0\varepsilon$.
- ♠ The *Solèr sort* $\mathbf{L}$ has the constants 0 and 1, the binary functions $\wedge$ and $\vee$, the unary function $^\perp$ and a predicate $e(a, b)$ with a priori bound 1, the *equality predicate*.
- ♠ Function symbols connecting the sorts:

$$\pi\colon \mathbf{L} \times \mathbf{H} \to \mathbf{H}, \quad \mathrm{at}\colon \mathbf{H} \to \mathbf{L}, \quad \mathrm{rep}\colon \mathbf{L} \to \mathbf{H},$$
$$\iota_n\colon \mathbf{H} \to \mathbf{K}, \quad \kappa_n\colon \mathbf{K} \to \mathbf{H}\ (n \in \mathbb{N}), \quad \mathrm{span}\colon \mathbf{K} \to \mathbf{L}.$$

- ♠ A predicate μ on $\mathbf{L}$, the *measure*, with a priori bound 2. The bound is wider than 1 so that the inequality $\mu(a) \leq 1$, like $\mu(a) \geq 0$, is imposed by the axiom (M1) of Section 6 below, and not by the vocabulary.

The symbol 0 thus names a formula, a closed term of each vector sort and a constant of the Solèr sort. The symbols $\wedge$ and $\vee$, for their part, name both connectives and functions of the Solèr sort, and the arguments to which they are applied always show which is meant. For vector terms t_1, t_2 of the same sort, the equation $t_1 \doteq t_2$ abbreviates $\|\frac{1}{2}t_1 - \frac{1}{2}t_2\| \leq 0$; for terms t_1, t_2 of the Solèr sort, the equation $t_1 = t_2$ abbreviates $e(t_1, t_2) \geq 1$. An equation that stands as an axiom in Section 6 is an axiom in the sense of Section 5, required of all values of its free variables. A vector x *belongs* to an element a of the Solèr sort, or is a *member* of a, when $\pi(a, x) \doteq x$ (that is, when $\|\frac{1}{2}\pi(a, x) - \frac{1}{2}x\| = 0$).

The intended meanings are as follows. Each element a of the Solèr sort names a closed subspace of the Hilbert space whose closed unit ball is the Hilbert sort. The function $\pi(a, \cdot)$ is the orthogonal projection onto that subspace, and the members of a are therefore the vectors of the unit ball that lie in the subspace. The predicate e is equality, with the value 1 at equal elements and -1 at distinct ones. Vectors and elements of the Solèr sort are linked by at and rep. The function at sends a nonzero vector x to the element whose members are the multiples of x in the unit ball. In the opposite direction, each nonzero element a is sent by rep to a unit vector that belongs to a. The constants e_n, for $n \in \mathbb{N}$, form an orthonormal basis of the Hilbert space. The elements of the sequence sort are the square-summable sequences of vectors of that space whose ℓ^2 norm is at most 1. These sequences are linked to the Hilbert sort by ι_n and κ_n. The first sends a vector x to the sequence that has x in position n and

0 elsewhere, and the second reads the entry in position n. The constant ε is the sequence $(2^{-n-1}e_n)_{n\in\mathbb{N}}$. The function span sends a sequence s to the element whose members are the vectors of the unit ball in the closed span of the entries of s.

The axiom of Theorem B excludes the points of a Borel set from the range of the measure. This axiom needs a two-valued formula that says whether a value of the measure lies in that set. For a Borel set $B \subseteq \mathbb{R}$, we write $\beta_B(a)$ for any formula of real-valued $L_{\omega_1\omega}$, with the single free variable a of the Solèr sort, whose values are prescribed as follows. At every element a of every structure for $\mathcal{L}$, the formula takes the value 1 if $\mu(a) \in B$ and -1 otherwise.

Lemma 5.1. *For every Borel set $B \subseteq \mathbb{R}$, there is a formula $\beta_B(a)$ as just described.*

Proof. Let $\mathcal{S}$ be the class of sets $B \subseteq \mathbb{R}$ for which such a formula exists. For rational numbers $q \leq q'$, the distance from a real number t to the interval $[q, q']$ is $\operatorname{dist}(t, [q, q']) = \max(0, q - t, t - q')$, a continuous function of t and hence a continuous connective. The formulas $(\mathbf{1} - n \cdot \operatorname{dist}(\mu(a), [q, q'])) \vee -\mathbf{1}$, for $n \geq 1$, take their values in $[-1, 1]$, and their countable infimum

$$\bigwedge_{n\geq 1} \big((\mathbf{1} - n \cdot \operatorname{dist}(\mu(a), [q, q'])) \vee -\mathbf{1}\big)$$

takes the value 1 when $\mu(a) \in [q, q']$ and -1 otherwise. Thus the class $\mathcal{S}$ contains the closed intervals with rational endpoints. If $B \in \mathcal{S}$, then the negation $-\beta_B(a)$ is a formula for $\mathbb{R} \setminus B$. If $B_n \in \mathcal{S}$ for every $n \in \mathbb{N}$, then the formulas $\beta_{B_n}(a)$ take their values in $[-1, 1]$, and their countable join is a formula for $\bigcup_n B_n$. The class $\mathcal{S}$ is therefore a σ-algebra, and since every open subset of $\mathbb{R}$ is a countable union of closed intervals with rational endpoints, it contains every Borel set. ☺

In the formula $\beta_B(a)$ that the proof of Lemma 5.1 constructs, the countable joins and infima are nested as in the construction of B from closed intervals with rational endpoints. The depth of this nesting is a countable ordinal, unbounded below ω_1 as B varies. The axioms of Theorem B therefore involve formulas whose joins are nested to arbitrarily large countable depth, although none of these formulas contains a quantifier.

Definition 5.2. A structure $\mathfrak{M}$ for $\mathcal{L}$, or for a vocabulary obtained from $\mathcal{L}$ by removing the predicate μ, the sequence sort together with the symbols that involve it, or both, is *standard* when its Hilbert sort is the closed unit ball of a complex Hilbert space H, its sequence sort, if it has one, is the closed unit ball of a complex Hilbert space K, and the functions $\lambda x + \lambda' y$ and the predicates $\operatorname{Re}\langle x \mid y\rangle$, $\operatorname{Im}\langle x \mid y\rangle$ and $\|x\|$ are the restrictions of the operations, the inner products and the norms of these spaces.

Besides T, we define in Section 6 below two theories T_0 and T_1 contained in it. The theory T_0 is formulated in the vocabulary without the measure and the sequence sort, and the theory T_1 in the vocabulary without the measure alone. A model of one of the theories T_0, T_1 and T is called *nonstandard* when it is not standard. The word has nothing to do with the nonstandard models of arithmetic or of nonstandard analysis: every predicate takes real values, and no element is infinitely large or infinitesimal. In Lemma 6.3 below, we show that the Hilbert sort of every model of T_0 can be identified with a subset of the closed unit ball of a complex Hilbert space. Under this identification, the $\mathbb{Q}(i)$-linear span of the subset is dense in that space. The functions $\lambda x + \lambda' y$ and the predicates of the Hilbert sort are, moreover, the restrictions of the operations, the inner product and the norm of that space. Hence the same holds for the Hilbert sort of every model of T_1 or of T. A model of T_0 is standard if and only if that subset is the whole ball. A nonstandard model of T_0 therefore lacks some of the vectors of the ball and contains none that the ball lacks. The sequence sort, which lies outside the scope of Lemma 6.3, may fail to be the closed unit ball of a complex

Hilbert space in a nonstandard model of T_1 or of T. Such a failure can occur whether or not the Hilbert sort is the whole ball.

Standardness enters some of our theorems as a hypothesis on the model rather than as a consequence of axioms. Indeed, no countable theory in one of the vocabularies of Definition 5.2 that contains T_0 and has a model has only standard models. Every standard model of T_0 has an uncountable Hilbert sort, the closed unit ball of an infinite-dimensional Hilbert space (see Proposition 7.1 below). The downward Löwenheim–Skolem theorem, by contrast, gives every such theory a model whose sorts are countable. In particular, the theories T_0, T_1 and T (which have standard models by Theorem A(2) and by taking reducts) have nonstandard models whose sorts are countable. The conclusion of Theorem C holds in these countable models as well. The conclusions of Theorem A(1) and Theorem B, which concern the whole lattice of closed subspaces and the whole range of the measure, are by contrast proper to the standard models (see Remark 11.2).

Historical note. Carol Karp, a student of Henkin's, wrote her dissertation of 1959 on languages with expressions of infinite length and published a book on them in 1964 [Kar64]. In 1969 Makkai applied Smullyan's method of consistency properties to $L_{\omega_1\omega}$, building models in the manner of Henkin's completeness proof, and extended it to the infinitary logics on admissible sets [Mak69b, Mak69a]. In 1971 Keisler made the model existence theorem for consistency properties a central method of his book on $L_{\omega_1\omega}$ [Kei71, Chapter 3], which also contains the omitting types theorem for countable fragments [Kei71, Chapter 11]. In 1973 Keisler built countable models by forcing with finite sets of sentences satisfiable in a given class of models [Kei73, p. 100, Example 1.2]. In the same work he replaced principal types by $\forall\exists$ formulas over a set of formulas of a countable fragment [Kei73, p. 107]. His general omitting types theorem gives conditions under which a $\forall\exists$ class contains a countable model in which countably many given $\forall\exists$ sentences hold [Kei73, p. 108]. Keisler's finite conditions and $\forall\exists$ formulas have counterparts in the model existence game of Section 10 and in the refined Kochen–Specker graphs of Section 12 (see Section A). In 1974 Morley derived the omitting types theorem for countable fragments from the Baire category theorem, applied to a space of countable structures, which he showed to be a Baire space [Mor74]. In 2001 Keisler and Miller carried the theorem into first-order logic without equality, where its usual formulation does not hold. There the model is built from closed terms, and principal types give way to locally realized types [KM01, Section 4].

Model theory with truth values in the unit interval goes back to Chang's work of 1958 and 1961 on the infinite-valued logic of Łukasiewicz [Cha58, Cha61]. The continuous model theory of Chang and Keisler of 1966 took its truth values in a compact Hausdorff space [CK66]. Henson introduced the logic of positive bounded formulas in 1976 to study a Banach space through its nonstandard hull [Hen76]. In the passage from two-valued to real-valued structures, equality becomes a uniform structure, a metric for countable vocabularies, and the axioms of equality become axioms of continuity.[2] Uniform continuity with moduli prescribed by the signature was the setting of Iovino's thesis of 1994 and of his papers of 1999 on stable Banach spaces [Iov94, Iov99a, Iov99b]; in 2002 Henson and Iovino presented the model theory of normed space structures in it [HI02, Sections 2 and 3]. In 2008 Ben Yaacov, Berenstein, Henson and Usvyatsov set out the model theory of metric structures in continuous logic [BYBHU08]. In 2009 Ben Yaacov and Iovino introduced an infinitary continuous logic

[2] In Łukasiewicz predicate logic, the congruence axioms for a similarity, or graded equality, make every predicate 1-Lipschitz for the distance given by one minus the similarity (see [Háj98, Section 5.6] and the earlier work collected there).

for model-theoretic forcing [BYI09]. In 2010 Ben Yaacov and Usvyatsov built moduli of uniform continuity into the signatures of continuous logic [BYU10, Section 2]. In 2014 Eagle proved the omitting types theorem for infinitary $[0,1]$-valued logic, whose metric structures have a master metric with respect to which the predicates are uniformly continuous [Eag14, Definitions 1.4 and 1.5]. In Keisler's model theory for real-valued structures, of 2023, a general structure requires no metric signature, and its function and predicate symbols may be interpreted by arbitrary mappings, the predicates being bounded [Kei23, Sections 1 and 2.1]. General structures correspond to first-order structures without equality, as metric structures with a distance predicate correspond to first-order structures with equality [Kei23, Section 1]. For vocabularies with countably many function and predicate symbols, every general structure expands to one with a distance predicate that is a uniform limit of formulas [Kei23, Theorem 3.3.4]. The logic of this paper, whose conditions are strict inequalities, is the infinitary extension of the logic of Keisler's theory due to Aubone, Dueñez and Iovino, who build its models from closed terms and prove its omitting types theorem by Morley's method in the Čech-complete space of Henkin types [ADI26].

6. The axioms

In this section we state the axioms of the theory T of Theorem A and of two smaller theories contained in it. The theory T_0 governs the Hilbert sort and its relation to the Solèr sort. The theory T_1 adds the sequence sort and the spans of its sequences, and T adds the measure. Thus the three theories separate the representation of the lattice, its completeness in standard models and the quantum probability measure. Interpreted in a standard model, the axioms say first that the constants e_n form an orthonormal basis. The axioms say next that each element of the Solèr sort determines an orthogonal projection acting on the Hilbert sort. They say, moreover, that these projections are ordered and combined as the closed subspaces onto which they project. In the same reading, the axioms say that atoms correspond to unit vectors up to a phase, that is, up to a complex factor of modulus one. The axioms further assert that the function span sends each sequence of the sequence sort to an element of the Solèr sort. The members of that element, they stipulate, are the vectors of the unit ball lying in the closed span of the entries of that sequence. Finally, the axioms require the measure μ to be σ-additive, with values in $[0,1]$ and the value 1 at the greatest element of the Solèr sort. The axioms fall into eight groups: one for each of the three sorts, four for the remaining symbols that connect the sorts, and the group (M) for the measure. The groups for the sorts are (H), (OL) and (K), the last of which also governs the functions ι_n and κ_n. The groups (P) and (Sub) concern the projection π and its relation to the lattice operations, the group (At) governs at and rep, and the group (Sp) governs span.

Free variables are universally quantified: the variables x, y, z range over $\mathbf{H}$, the variables s, t over $\mathbf{K}$, and the variables a, b, c over $\mathbf{L}$. The letters λ, λ' range over the elements of $\mathbb{Q}(i)$ with $|\lambda| + |\lambda'| \leq 1$, and the letters n, m, k, N range over $\mathbb{N}$. These letters are not variables of the logic: an axiom in which they occur free stands for countably many axioms, one for each choice of their values. A bound occurrence of one of these letters, by contrast, indexes a finite sum, a countable join or a countable infimum. Each axiom is understood in the sense of Section 5, as a closed condition required of all values of its free variables. The equations written with $\doteq$ between vector terms and with $=$ between terms of the Solèr sort are the abbreviations introduced in Section 5.3. As in Section 5, an implication between two-valued statements is asserted by requiring the disjunction of its conclusion with the negation of its hypothesis to be at least 1. An implication that involves a real quantity takes one of the

forms (i)–(iii) that follow (2), of which (i) and (iii) use the formula $\lceil\varphi\rceil$. Some axioms are followed by a reading in words, in italics, which describes what the axiom says in standard models.

(H) The Hilbert sort.

(H1) *Inner product space.* For any two terms built from variables by the functions $\lambda x+\lambda' y$ that denote the same $\mathbb{Q}(i)$-linear combination of the variables, the equation between them; the closed conditions $\langle \lambda x+\lambda' y \mid z\rangle = \lambda\langle x \mid z\rangle + \lambda'\langle y \mid z\rangle$ and $\langle y \mid x\rangle = \overline{\langle x \mid y\rangle}$, each written as two real closed conditions; and $\|x\| = \sqrt{\max(0, \mathrm{Re}\langle x \mid x\rangle)}$.

(H2) *Separability.* $\langle e_n \mid e_m\rangle = \delta_{nm}$, where δ_{nm} is the Kronecker delta, and

$$\|x\|^2 = \bigvee_N \Big(\mathbf{1} \wedge \sum_{n<N} |\langle x \mid e_n\rangle|^2\Big).$$

The constants e_n are orthonormal and satisfy Parseval's identity.

(OL) The Solèr sort.

(OL1) The equations of ortholattices:

$$a\wedge b = b\wedge a, \quad a\wedge(b\wedge c) = (a\wedge b)\wedge c, \quad a\wedge(a\vee b) = a,$$
$$a\vee b = b\vee a, \quad a\vee(b\vee c) = (a\vee b)\vee c, \quad a\vee(a\wedge b) = a,$$
$$a\wedge 0 = 0, \quad a\vee 1 = 1, \quad a^{\perp\perp} = a, \quad (a\wedge b)^\perp = a^\perp \vee b^\perp, \quad a\wedge a^\perp = 0.$$

(OL2) *The equality predicate.*

$$e(a,b)\vee -e(a,b) \geq 1, \qquad e(a,a)\geq 1,$$
$$-e(a,b)\vee e(b,a)\geq 1, \qquad -e(a,b)\vee -e(b,c)\vee e(a,c)\geq 1;$$
$$-e(a,b)\vee e(a\wedge c, b\wedge c)\geq 1, \qquad -e(a,b)\vee e(a\vee c, b\vee c)\geq 1,$$
$$-e(a,b)\vee e(a^\perp, b^\perp)\geq 1;$$
$$e(a,b)\wedge \big\|\tfrac{1}{2}\pi(a,x) - \tfrac{1}{2}\pi(b,x)\big\| \leq 0, \qquad e(a,b)\wedge \big\|\tfrac{1}{2}\mathrm{rep}(a) - \tfrac{1}{2}\mathrm{rep}(b)\big\| \leq 0,$$
$$e(a,b)\wedge |\mu(a)-\mu(b)| \leq 0.$$

The predicate e is a two-valued equivalence relation, respected by the lattice operations, the projections, the function rep *and the measure.*

(P) Each element of the Solèr sort determines a projection.

(P1) $\pi(a, \lambda x+\lambda' y) \doteq \lambda\pi(a,x) + \lambda'\pi(a,y)$.

(P2) $\pi(a,\pi(a,x)) \doteq \pi(a,x)$.

(P3) $\langle \pi(a,x) \mid y\rangle = \langle x \mid \pi(a,y)\rangle$.

(P4) *Extensionality.* $e(a,b)\vee \big\lceil \sup_x \big\|\tfrac{1}{2}\pi(a,x) - \tfrac{1}{2}\pi(b,x)\big\|\big\rceil \geq 1$.

Two elements with the same projection are equal.

Axioms (Sub2) and (Sub3) express norm convergence through the infimum of the suprema over tails of a nonnegative sequence. The vanishing of this infimum says that the sequence converges to zero. Here the sequences are the errors in the alternating-projection approximations to the meet and the join; in Section 7 we identify their limits by von Neumann's alternating projection theorem.

(Sub) The order and the operations of the Solèr sort are those of subspaces.

(Sub0) $\pi(0,x) \doteq 0$ and $\pi(1,x) \doteq x$.
Only the zero vector belongs to 0*; every vector belongs to* 1.

(Sub1)
$$e(a \wedge b, a) \wedge \|\tfrac{1}{2}\pi(b,\pi(a,x)) - \tfrac{1}{2}\pi(a,x)\| \leq 0,$$
$$e(a \wedge b, a) \vee \lceil \sup_x \|\tfrac{1}{2}\pi(b,\pi(a,x)) - \tfrac{1}{2}\pi(a,x)\| \rceil \geq 1.$$
The relation $a \leq b$ holds if and only if every member of a is a member of b.

(Sub2) $\bigwedge_N \bigvee_{n\geq N} \|\tfrac{1}{2}\pi(a \wedge b, x) - \tfrac{1}{2}(P_a P_b)^n x\| \leq 0$, where $(P_a P_b)^n x$ is the term obtained by applying $\pi(b,\cdot)$ and then $\pi(a,\cdot)$, n times.
The members of $a \wedge b$ are the common members of a and b.

(Sub3) $\bigwedge_N \bigvee_{n\geq N} \|\tfrac{1}{4}\pi(a \vee b, x) - \tfrac{1}{4}x + \tfrac{1}{4}(P_{a^\perp} P_{b^\perp})^n x\| \leq 0$, where $(P_{a^\perp} P_{b^\perp})^n x$ is the term of (Sub2) for $a^\perp$ and $b^\perp$, and the vector inside the norm is denoted by a term built from the functions $\lambda x + \lambda' y$.
The closed subspace spanned by the members of $a \vee b$ is the closure of the sum of the closed subspaces spanned by the members of a and by those of b.

(Sub4) $\tfrac{1}{2}\pi(a^\perp, x) \doteq \tfrac{1}{2}x - \tfrac{1}{2}\pi(a,x)$.
The members of $a^\perp$ are the vectors orthogonal to all members of a.

Axiom (At3) below needs a formula that recognizes atoms. Let $\mathrm{nonatom}(a)$ be the formula $\sup_b\big(-e(b,0) \wedge -e(b,a) \wedge e(a \wedge b, b)\big)$. By (OL2), it is two-valued, with the value 1 if some b lies strictly between 0 and a, and -1 otherwise. Thus an element a with $e(a,0) = -1$ is an atom exactly when $\mathrm{nonatom}(a) = -1$.

(At) Atoms and unit vectors.

(At1) $\pi(\mathrm{at}(x),x) \doteq x$, and $\big|\langle \pi(\mathrm{at}(x),y) \mid \pi(\mathrm{at}(x),z)\rangle\big|^2 = \|\pi(\mathrm{at}(x),y)\|^2\,\|\pi(\mathrm{at}(x),z)\|^2$.
Every vector x belongs to $\mathrm{at}(x)$*, and any two members of* $\mathrm{at}(x)$ *are linearly dependent.*

(At2) $\pi(a,\mathrm{rep}(a)) \doteq \mathrm{rep}(a)$, and $-e(a,0) \wedge \big|\,\|\mathrm{rep}(a)\| - 1\,\big| \leq 0$.
The vector $\mathrm{rep}(a)$ *belongs to a, and it is a unit vector when $a \neq 0$.*

(At3) $e(a,0) \vee \mathrm{nonatom}(a) \vee e(\mathrm{at}(\mathrm{rep}(a)),a) \geq 1$.
Every atom a equals $\mathrm{at}(\mathrm{rep}(a))$.

(At4) $e(\mathrm{at}(x),\mathrm{at}(y)) \wedge \big(\|x\|\,\|y\| - |\langle x \mid y\rangle|\big) \leq 0$, and
$$e(\mathrm{at}(x),\mathrm{at}(y)) \vee \Big\lceil \|x\|\,\|y\| - |\langle x \mid y\rangle| + \big|1 - \|x\|\big| + \big|1 - \|y\|\big| \Big\rceil \geq 1.$$
Two unit vectors determine the same atom if and only if one is a multiple of the other by a complex number of modulus one.

(At5) $\mathrm{at}(0) = 0$, and $e(\mathrm{at}(x),\mathrm{at}(y)) \vee \lceil \|\tfrac{1}{2}x - \tfrac{1}{2}y\| \rceil \geq 1$.

(K) The sequence sort.

(K1) The axioms (H1), written for **K**.

(K2) $\iota_n(\lambda x+\lambda' y) \doteq \lambda\iota_n x+\lambda'\iota_n y$ and $\kappa_n(\lambda s+\lambda' t) \doteq \lambda\kappa_n s+\lambda'\kappa_n t$; $\langle \iota_n x \mid \iota_m y\rangle = \delta_{nm}\langle x \mid y\rangle$; and $\langle \kappa_n s \mid x\rangle = \langle s \mid \iota_n x\rangle$.

(K3) $\|s\|^2 = \bigvee_N \Big(\mathbf{1} \wedge \sum_{n<N} \|\kappa_n s\|^2\Big)$.

The norm of a sequence is the ℓ^2 norm of its entries, whence, by the linearity of the maps κ_n in (K2), a sequence is determined by its entries.

(K4) $\kappa_n\varepsilon \doteq 2^{-n-1}e_n$.

(Sp) Spans.

(Sp1) $\pi(\mathrm{span}(s), \kappa_n s) \doteq \kappa_n s$.

Every entry of s belongs to $\mathrm{span}(s)$.

(Sp2) $\|\pi(\mathrm{span}(s), x)\|^2 \leq \Big\lceil \bigvee_N \Big(\mathbf{1} \wedge \sum_{n<N} |\langle x \mid \kappa_n s\rangle|^2\Big)\Big\rceil$.

A vector orthogonal to every entry of s is orthogonal to every member of $\mathrm{span}(s)$.

(Sp3) $e(\mathrm{span}(s), \mathrm{span}(t)) \vee \lceil \|\frac{1}{2}s - \frac{1}{2}t\| \rceil \geq 1$.

The axiom of σ-additivity concerns sequences whose entries are mutually orthogonal; it therefore needs a quantity that measures how far the entries of a sequence are from being mutually orthogonal. Let $\mathrm{od}(s)$, the *orthogonality defect* of s, be the formula $\bigvee_N\big(\mathbf{1} \wedge \sum_{n,m<N,\, n\neq m} |\langle \kappa_n s \mid \kappa_m s\rangle|\big)$. It takes values in $[0,1]$, and it vanishes exactly when the entries $\kappa_n s$ of s are mutually orthogonal.

(M) The measure.

(M1) $\mu(a) \geq 0$ and $\mu(a) \leq 1$.

(M2) $\mu(1) = 1$.

(M3) *σ-additivity.*

$$\Big|\, \mu(\mathrm{span}(s)) - \bigvee_N \Big(\Big((2\cdot\mathbf{1}) \wedge \sum_{n<N} \mu(\mathrm{at}(\kappa_n s))\Big) \vee 0\Big)\Big| \leq 2\lceil \mathrm{od}(s)\rceil.$$

If the entries of s are mutually orthogonal, the value of μ at $\mathrm{span}(s)$ *is the sum of its values at the elements* $\mathrm{at}(\kappa_n s)$.

Since μ takes its values in $[0,1]$ in every structure that satisfies (M1), the left side of (M3) is at most 2 in such a structure. Within such a structure, axiom (M3) is an implication of the form (iii), with the left side of (M3) in place of ψ. The bound 2 takes the place of the constant of (iii), and the orthogonality defect $\mathrm{od}(s)$ that of φ.

Definition 6.1. The theory T_0 consists of the axioms (H), (OL), (P), (Sub) and (At), without the clause on μ in (OL2), in the vocabulary obtained from $\mathcal{L}$ by removing the sequence sort with its constant ε, the functions ι_n, κ_n and span, and the predicate μ. The theory T_1 consists of T_0 together with (K) and (Sp), in the vocabulary $\mathcal{L}$ without μ; the theory T consists of T_1 together with (M) and the clause on μ in (OL2).

Lemma 6.2. *In a model of T_0, of T_1 or of T, two elements a, b of the Solèr sort are equal if and only if $e(a,b) = 1$, and two vectors x, y are equal if and only if $\|\frac{1}{2}x - \frac{1}{2}y\| = 0$; the same holds in the sequence sort of a model of T_1 or of T. Consequently, the reduct of a model of T_1 or of T to the vocabulary of T_0, and the reduct of a model of T to the vocabulary of T_1, are reduced, and they are therefore models of T_0 and of T_1, respectively.*

Proof. Since models are reduced, it suffices to show that two elements that the predicate e, or the norm of their half-difference, does not distinguish agree in every atomic formula. The proof of this agreement rests on the Cauchy–Schwarz inequality, and hence on the positivity of the form, which we prove first. By (H1) and (H2), the form given by the pair of predicates $\langle x \mid y\rangle$ is positive semidefinite on the Hilbert sort, that is, $\mathrm{Re}\langle x \mid x\rangle \geq 0$ for every element x of that sort. Indeed, suppose that $\mathrm{Re}\langle x \mid x\rangle < 0$ for some x; then (H1) yields $\|x\| = 0$, and (H2) gives $\langle x \mid e_n\rangle = 0$ for every n. For the vector $\frac{1}{2}x + \frac{1}{2}e_0$, axiom (H2) then gives $\|\frac{1}{2}x + \frac{1}{2}e_0\|^2 = \frac{1}{4}$, whereas (H1) gives $\|\frac{1}{2}x + \frac{1}{2}e_0\|^2 = \max\bigl(0, \frac{1}{4}\,\mathrm{Re}\langle x \mid x\rangle + \frac{1}{4}\bigr) < \frac{1}{4}$, which is absurd. In a model of T_1 or of T, an argument of the same kind uses (K1)–(K3), the positivity just proved and the vector $\frac{1}{2}s + \frac{1}{2}\iota_0 e_0$. This argument shows that the form is positive semidefinite on the sequence sort as well, whence the Cauchy–Schwarz inequality holds for the predicates of each vector sort. We combine the Cauchy–Schwarz inequality with axioms (H1) and (OL2) as well as (P1)–(P3) and (At5), and in a model of T_1 or of T also with (K1), (K2) and (Sp3). This combination shows that every atomic formula takes the same values at a and b when $e(a,b) = 1$, and at x and y when $\|\frac{1}{2}x - \frac{1}{2}y\| = 0$. Since models are reduced, the elements in question are equal. The converse implications hold by (OL2) and (H1). Finally, consider the reduct of a model of T_1 or of T to a smaller one of these vocabularies. In that reduct, two elements of the Solèr sort that agree in every atomic formula with parameters satisfy $e(a,b) = e(b,b) = 1$, and two elements of a vector sort that do so satisfy $\|\frac{1}{2}x - \frac{1}{2}y\| = \|\frac{1}{2}y - \frac{1}{2}y\| = 0$. By what has been shown, the two elements of each such pair are equal, and the reduct is reduced. ☺

Lemma 6.3. *Let $\mathfrak{M}$ be a model of T_0. There is a complex Hilbert space $\bar{E}$ such that the Hilbert sort of $\mathfrak{M}$ can be identified with a subset Y of the closed unit ball of $\bar{E}$ whose $\mathbb{Q}(i)$-linear span E is dense, the functions $\lambda x + \lambda' y$ and the predicates being the restrictions of the operations, the inner product and the norm of $\bar{E}$. For every element a of the Solèr sort, the map $\pi(a,\cdot)$ extends uniquely to an orthogonal projection $\bar{P}_a$ of $\bar{E}$. If $\mathfrak{M}$ is standard, then Y is the whole closed unit ball of $\bar{E}$.*

Proof. The idea is to recover a vector space from the Hilbert sort by formal multiples. Each pair (k,x), with $k \geq 1$ an integer and x an element of the Hilbert sort, stands for kx. We write $(k,x) \sim (l,y)$ when $\|\frac{1}{2l}x - \frac{1}{2k}y\| = 0$, and we define the sum of two pairs by $(k,x) + (l,y) = (2kl, \frac{1}{2l}x + \frac{1}{2k}y)$. The product by an element λ of $\mathbb{Q}(i)$ is defined by $\lambda(k,x) = (km, \frac{\lambda}{m}x)$, where m is a positive integer with $m \geq |\lambda|$, and the form by $\langle (k,x) \mid (l,y)\rangle = kl\langle x \mid y\rangle$. Axiom (H1), together with the positivity of the form established in the proof of Lemma 6.2, makes the quotient a $\mathbb{Q}(i)$-vector space E with a positive definite hermitian form. The map that sends x to the class of $(1,x)$ is injective by Lemma 6.2. We let $\bar{E}$ be the completion of E and let Y be the image of the Hilbert sort, whose $\mathbb{Q}(i)$-linear span is E. By (P1)–(P3), the map $\pi(a,\cdot)$ extends to a $\mathbb{Q}(i)$-linear idempotent map Q_a on E that is symmetric with respect to the form. Hence we have $\|Q_a x\|^2 = \langle x \mid Q_a x\rangle \leq \|x\|\|Q_a x\|$; the map Q_a is thus contractive, and it extends by continuity to a self-adjoint idempotent $\bar{P}_a$ of $\bar{E}$. In a standard model, the Hilbert sort is the closed unit ball of a Hilbert space H. In that case, the map sending the class of (k,x) to kx identifies E with H, since every vector of H is a positive integer multiple of a vector of that ball. As H is complete, it follows that $\bar{E} = H$, and that Y is the whole closed unit ball of $\bar{E}$. ☺

⌛ **Historical note.** In the continuous logic set out by Ben Yaacov, Berenstein, Henson and Usvyatsov in 2008, the unit ball of a Banach space carries the maps $(x,y) \mapsto \alpha x + \beta y$ with $|\alpha| + |\beta| \leq 1$ as functions. In that logic, a Hilbert space carries in addition its inner product

as a predicate [BYBHU08], and the Hilbert sort follows both conventions. Von Neumann's alternating projection theorem, on which axioms (Sub2) and (Sub3) rest, comes from lecture notes that he distributed in mimeographed form in 1933. The notes were printed in 1950 as the second volume of his *Functional Operators* [vN50].

7. Standard models of T_0: pre-Solèr lattices

In this section we study the standard models of T_0, the theory whose vocabulary has neither the sequence sort nor the measure. We show that the Solèr sort of such a model is a pre-Solèr lattice, an ortholattice that satisfies every condition in the definition of a Solèr lattice except, possibly, completeness. We then show that the map sending each element of the Solèr sort to the closed subspace spanned by its members is an embedding of ortholattices. The embedding takes its values in the lattice of closed subspaces of the Hilbert space whose closed unit ball is the Hilbert sort. In that lattice, as we also show, the image of the embedding is join-dense and meet-dense in the order-theoretic sense of Section 1. Gleason's theorem already holds at this level, in the sense that the σ-additive states on the Solèr sort, defined below, are given by density operators. The Solèr sort of a standard model of T_0 need not be complete (see Example 7.6); completeness is obtained in Section 8 from the sequence sort.

Throughout this section, we fix a standard model $\mathfrak{M}$ of T_0. We write H for the Hilbert space whose closed unit ball is the Hilbert sort of $\mathfrak{M}$ (see Definition 5.2), and $\mathbf{L}^{\mathfrak{M}}$ for the Solèr sort of $\mathfrak{M}$. For each element a of $\mathbf{L}^{\mathfrak{M}}$, we write P_a for the projection $\bar{P}_a$ of Lemma 6.3, which extends $\pi(a,\cdot)$, and S_a for the range of P_a. This range is the closed subspace of H spanned by the members of a, which are the vectors x of the unit ball with $\pi(a,x) = x$.

Proposition 7.1. *The space H is a separable infinite-dimensional complex Hilbert space, and the constants e_n of the Hilbert sort, for $n \in \mathbb{N}$, form an orthonormal basis of H.*

Proof. By (H2), the vectors e_n are orthonormal, and Parseval's identity places every vector of the unit ball in their closed span, hence every vector of H as well. Thus $(e_n)_{n\in\mathbb{N}}$ is an orthonormal basis of H, which is therefore separable and infinite-dimensional; it is complex by Definition 5.2. ☺

We next show that the map $a \mapsto S_a$ respects the order and the operations of the Solèr sort. Axioms (Sub2) and (Sub3) describe meets and joins through von Neumann's alternating projection theorem, which reaches the projection onto the intersection of two closed subspaces by projecting alternately onto each.

Theorem 7.2 (von Neumann's alternating projection theorem [vN49, Lemma 22]**).** *Let U and V be closed subspaces of a Hilbert space, and let P_U, P_V and $P_{U\cap V}$ be the orthogonal projections onto U, V and $U \cap V$. Then, for every vector x of the space, the sequence $\big((P_U P_V)^n x\big)_{n\in\mathbb{N}}$ converges in norm to $P_{U\cap V}x$.*

Proposition 7.3. *The map $\tau\colon a \mapsto S_a$ is an embedding of ortholattices from $\mathbf{L}^{\mathfrak{M}}$ into the lattice $L(H)$ of closed subspaces of H; that is, it is injective and satisfies*

$$S_0 = \{0\}, \quad S_1 = H, \quad S_{a\wedge b} = S_a \cap S_b, \quad S_{a\vee b} = \overline{S_a + S_b}, \quad S_{a^\perp} = S_a^\perp.$$

Moreover, the relation $a \le b$ holds if and only if $S_a \subseteq S_b$.

Proof. By (OL1) and Lemma 6.2, the Solèr sort $\mathbf{L}^{\mathfrak{M}}$ is an ortholattice. Axiom (Sub0) gives the first two equalities, and axiom (Sub4) gives $P_{a^\perp} = \mathrm{Id} - P_a$, whose range is $S_a^\perp$. For meets and joins we combine the axioms with Theorem 7.2: axiom (Sub2) shows that $P_{a\wedge b} = P_{S_a\cap S_b}$, and axiom (Sub3) that $P_{a\vee b} = \mathrm{Id} - P_{S_a^\perp \cap S_b^\perp}$, whose range is $(S_a^\perp \cap S_b^\perp)^\perp = \overline{S_a + S_b}$. Injectivity follows from extensionality: if $S_a = S_b$, then the projections P_a and P_b coincide, and axiom

(P4) gives $a = b$. Finally, the order is read off from axiom (Sub1), which shows that $a \wedge b = a$ exactly when $P_b P_a = P_a$, or equivalently when $S_a \subseteq S_b$. ☺

Proposition 7.4. *For every nonzero x in the unit ball, the subspace $S_{\mathrm{at}(x)}$ is $\mathbb{C}x$. The atoms of $\mathbf{L}^{\mathfrak{M}}$ are exactly the elements $\mathrm{at}(x)$ with $x \neq 0$, and the map $x \mapsto \mathrm{at}(x)$ induces a bijection from the unit sphere of H, modulo multiplication by complex numbers of modulus one, onto the set of atoms of $\mathbf{L}^{\mathfrak{M}}$, whose inverse sends an atom a to the class of $\mathrm{rep}(a)$.*

Proof. By (At1), the vector x lies in $S_{\mathrm{at}(x)}$, and any two vectors of $S_{\mathrm{at}(x)}$ attain equality in the Cauchy–Schwarz inequality. Any two such vectors are therefore linearly dependent, whence $S_{\mathrm{at}(x)} = \mathbb{C}x$. Since this ray is an atom of $L(H)$ and the map τ is an order embedding, the element $\mathrm{at}(x)$ is an atom of $\mathbf{L}^{\mathfrak{M}}$. If a is an atom of $\mathbf{L}^{\mathfrak{M}}$, then axiom (At2) shows that $\mathrm{rep}(a)$ is a unit vector, and axiom (At3) gives $a = \mathrm{at}(\mathrm{rep}(a))$. For unit vectors x and y, axiom (At4) gives $\mathrm{at}(x) = \mathrm{at}(y)$ if and only if $|\langle x \mid y\rangle| = 1$. For unit vectors, the equality $|\langle x \mid y\rangle| = 1$ holds exactly when y is a multiple of x by a complex number of modulus one. In particular, for a unit vector x, the unit vector $\mathrm{rep}(\mathrm{at}(x))$ is such a multiple of x, because it lies in $S_{\mathrm{at}(x)} = \mathbb{C}x$ by (At2). The inverse of the bijection is therefore as stated. ☺

Theorem 7.5. *Let $\mathfrak{M}$ be a standard model of T_0, let H be the Hilbert space whose closed unit ball is its Hilbert sort, and, for each element a of its Solèr sort $\mathbf{L}^{\mathfrak{M}}$, let S_a be the closed subspace of H spanned by the members of a. Then $\mathbf{L}^{\mathfrak{M}}$ is a pre-Solèr lattice, and the image of the map $\tau\colon a \mapsto S_a$ is join-dense and meet-dense in the lattice $L(H)$ of closed subspaces of H, which is therefore isomorphic to the MacNeille completion of $\mathbf{L}^{\mathfrak{M}}$. The image of τ contains every finite-dimensional subspace of H and the orthogonal complement of each, and the map τ sends every join that exists in $\mathbf{L}^{\mathfrak{M}}$ to the corresponding join in $L(H)$.*

Proof. We begin with the finite-dimensional subspaces and then verify, condition by condition, that $\mathbf{L}^{\mathfrak{M}}$ is a pre-Solèr lattice. The zero subspace is S_0; now let F be a nonzero finite-dimensional subspace of H with a basis $x_1, \ldots, x_k$ of unit vectors. By Propositions 7.3 and 7.4, the map τ sends $\mathrm{at}(x_1) \vee \cdots \vee \mathrm{at}(x_k)$ to the closure of F, which is F itself, and $(\mathrm{at}(x_1) \vee \cdots \vee \mathrm{at}(x_k))^{\perp}$ to $F^{\perp}$. Condition (OM) holds because $\tau(\mathbf{L}^{\mathfrak{M}})$ is a sub-ortholattice of the orthomodular lattice $L(H)$. For (A), let a be a nonzero element of $\mathbf{L}^{\mathfrak{M}}$; the atoms below a are the elements $\mathrm{at}(x)$ with $x \in S_a$ and $0 < \|x\| \leq 1$. Any upper bound b of these atoms in $\mathbf{L}^{\mathfrak{M}}$ satisfies $S_b \supseteq S_a$, hence $b \geq a$, which makes a their join. For (CP), let a be an element of $\mathbf{L}^{\mathfrak{M}}$, and let $\mathrm{at}(x)$ be an atom not below a; then the subspace S_a does not contain x. By Proposition 7.3 and Lemma 2.7, the subspace $\tau(a \vee \mathrm{at}(x))$ is $S_a + \mathbb{C}x$, in which S_a has codimension one. Since τ is an order embedding, no element lies strictly between a and $a \vee \mathrm{at}(x)$. Condition (I) holds because the join of two distinct atoms corresponds to a plane, which contains infinitely many rays, all in $\tau(\mathbf{L}^{\mathfrak{M}})$. The atoms $\mathrm{at}(e_n)$, for $n \in \mathbb{N}$, give (D). For (S), take $p_n = \mathrm{at}(e_n)$ and $c_n = \mathrm{at}(\frac{1}{2}(e_n + e_{n+1}))$; since τ is injective and preserves the lattice operations and the orthocomplementation, it suffices to verify the condition for the images of these atoms. These images are the rays of e_n and of $e_n + e_{n+1}$, where the condition follows from (1) and the computation following it, because $\|e_n\| = \|e_{n+1}\|$.

Each closed subspace S of H is the join in $L(H)$ of its rays, and it equals $\bigcap\{y^{\perp} : y \in S^{\perp},\ 0 < \|y\| \leq 1\}$, where each $y^{\perp} = S_{\mathrm{at}(y)^{\perp}}$ lies in the image of τ. These two facts make the image of τ join-dense and meet-dense in $L(H)$. By Proposition 1.5, the lattice $L(H)$ is therefore isomorphic to the MacNeille completion of $\mathbf{L}^{\mathfrak{M}}$. Finally, let $A \subseteq \mathbf{L}^{\mathfrak{M}}$ have a join c in $\mathbf{L}^{\mathfrak{M}}$, and let W be the closed span of $\bigcup_{a \in A} S_a$; clearly $W \subseteq S_c$. If y is a vector of $S_c \cap W^{\perp}$ with $0 < \|y\| \leq 1$, then every $a \in A$ lies below $\mathrm{at}(y)^{\perp}$, and hence so does c. The vector y,

which lies in S_c, is then orthogonal to itself, which is absurd. Since $S_c = W + (S_c \cap W^\perp)$ by the projection theorem, it follows that $S_c = W$. ☺

Example 7.6. Let L_{fc} (for "finite dimension or finite codimension") be the set of finite-dimensional subspaces of the space ℓ^2 of square-summable complex sequences and their orthogonal complements. By the argument of Example 4.3, the set L_{fc} is a sub-ortholattice of $L(\ell^2)$. Take the closed unit ball of ℓ^2 as Hilbert sort and the standard unit vectors as the constants e_n, and take L_{fc}, with its lattice operations and orthocomplementation, as Solèr sort. Interpret e as equality, with $e(S, S') = 1$ when $S = S'$ and $e(S, S') = -1$ otherwise. Put $\pi(S, x) = P_S x$ and $\mathrm{at}(x) = \mathbb{C}x$, let $\mathrm{rep}(S)$ be a unit vector chosen in S for each $S \neq \{0\}$, and let $\mathrm{rep}(\{0\})$ be 0. Axioms (Sub2) and (Sub3) hold by Theorem 7.2, and the other axioms of T_0 by construction; the result is therefore a standard model of T_0. Its Solèr sort L_{fc} is not complete. Indeed, an upper bound in L_{fc} of the rays $\mathbb{C}e_{2n}$, for $n \in \mathbb{N}$, is infinite-dimensional and hence of the form $G^\perp$ for a finite-dimensional subspace G orthogonal to every e_{2n}. The closed span of the vectors e_{2n+1} is infinite-dimensional, whereas G is finite-dimensional; that span therefore contains a unit vector y orthogonal to G. The subspace $(G + \mathbb{C}y)^\perp$, which contains every e_{2n} but not y, is an upper bound of the same rays that lies strictly below $G^\perp$. The rays $\mathbb{C}e_{2n}$ therefore have no join in L_{fc}.

In the form we need, Gleason's theorem concerns two notions: density operators, by which quantum mechanics represents the states of a system, and σ-additive states on an ortholattice. A *density operator* on H is a positive operator ρ whose trace $\operatorname{Tr}\rho = \sum_k \langle \rho u_k \mid u_k \rangle$, computed in any orthonormal basis $(u_k)_{k\in\mathbb{N}}$ of H, equals 1. A *σ-additive state* on an ortholattice Λ is a map $\nu\colon \Lambda \to [0,1]$ with $\nu(1) = 1$ such that $\nu(\bigvee_{n\in J} a_n) = \sum_{n\in J} \nu(a_n)$ for every family $(a_n)_{n\in J}$ of mutually orthogonal elements of Λ, indexed by a set $J \subseteq \mathbb{N}$, whose join exists in Λ.

Theorem 7.7 (Gleason's theorem for T_0). *Let $\mathfrak{M}$ be a standard model of T_0, let H be the Hilbert space whose closed unit ball is its Hilbert sort, and, for each element a of its Solèr sort $\mathbf{L}^{\mathfrak{M}}$, let P_a be the orthogonal projection onto the closed subspace spanned by the members of a. The σ-additive states on $\mathbf{L}^{\mathfrak{M}}$ are exactly the maps $a \mapsto \operatorname{Tr}(\rho P_a)$, where ρ ranges over the density operators on H, and each such state determines its density operator.*

Proof. We first verify that density operators give σ-additive states, and then recover a density operator from a state by applying Gleason's theorem to its values on rays. Let ρ be a density operator on H. Suppose that the elements a_n, for n in a set $J \subseteq \mathbb{N}$, are mutually orthogonal and have a join c in $\mathbf{L}^{\mathfrak{M}}$. By Proposition 7.3 and Theorem 7.5, the subspace S_c is then the closure of the sum of the mutually orthogonal subspaces S_{a_n}, whence $P_c x = \sum_{n\in J} P_{a_n} x$ for every $x \in H$ and $\operatorname{Tr}(\rho P_c) = \sum_{n\in J} \operatorname{Tr}(\rho P_{a_n})$. Moreover, the map $a \mapsto \operatorname{Tr}(\rho P_a)$ takes the value $\operatorname{Tr}\rho = 1$ at 1, and its values lie in $[0,1]$. Conversely, let ν be a σ-additive state on $\mathbf{L}^{\mathfrak{M}}$. Additivity on the orthogonal pair $1, 0$, whose join is 1, gives $\nu(0) = 0$. For a unit vector x, put $f(x) = \nu(\mathrm{at}(x))$. For $a \neq 0$, let $\mathcal{U}_a$ be an orthonormal basis of S_a, which is countable because H is separable. The atoms $\mathrm{at}(u)$, for $u \in \mathcal{U}_a$, are mutually orthogonal, and their join in $\mathbf{L}^{\mathfrak{M}}$ is a. Indeed, the element a is an upper bound of these atoms, and every upper bound b of them satisfies $S_b \supseteq S_a$. Hence the value $\nu(a)$ is the sum $\sum_{u\in\mathcal{U}_a} f(u)$. Taking $a = 1$ shows that f is a nonnegative *frame function* of weight 1: its sum over every orthonormal basis of H is 1. Since H is separable of dimension at least three, Gleason's theorem [Gle57] gives a unique density operator ρ with $f(x) = \langle \rho x \mid x \rangle$ for every unit vector x. It follows that $\nu(a) = \sum_{u\in\mathcal{U}_a} \langle \rho u \mid u \rangle = \operatorname{Tr}(\rho P_a)$ for every $a \neq 0$; the same equality holds for $a = 0$, since $\nu(0) = 0$. ☺

Historical note. Gleason proved his theorem in 1957 to answer a question of Mackey's. Mackey had asked whether the Born rule is the only way to assign probabilities to the closed subspaces of a Hilbert space [Mac57, Gle57]. The Born rule gives a closed subspace, in the state described by a density operator ρ, the probability $\mathrm{Tr}(\rho P)$, where P is the orthogonal projection onto that subspace. Frame functions are the heart of Gleason's proof: he shows first, by means of spherical harmonics, that the continuous frame functions on the unit sphere of $\mathbb{R}^3$ come from quadratic forms. Gleason then shows, in the hardest step, that every nonnegative frame function on that sphere is continuous. Finally, he passes from $\mathbb{R}^3$ to real and complex separable Hilbert spaces of dimension at least three. Gleason noted that the restriction on the dimension is essential: in dimension two the theorem fails. Cooke, Keane and Moran gave in 1985 an elementary proof, longer than Gleason's but with fewer prerequisites [CKM85].

8. Standard models of T_1, proofs of Theorems A and B, and definable sets of values

The axioms of T that mention neither the sequence sort nor the measure, which form the theory T_0 of Section 7, govern the elements that the Solèr sort contains. These axioms allow the Solèr sort to consist of the subspaces of finite dimension or finite codimension alone, as the lattice L_{fc} of Example 7.6 shows. The sequence sort brings in every other closed subspace: every closed subspace of a separable Hilbert space is the closed span of a square-summable sequence of vectors. The function span sends such a sequence to an element of the Solèr sort whose members span the same closed subspace as the sequence. In this section we consider the Hilbert space whose closed unit ball is the Hilbert sort of a standard model of T_1, the part of T that does not mention the measure. We show accordingly that every closed subspace of this space is spanned by the members of an element of the Solèr sort of that model, and we prove Theorems A and B. In Section 8.1 we identify the sets of values that formulas define in the standard models.

Throughout this section we fix a standard model $\mathfrak{M}$ of T_1, whose Solèr sort we denote by $\mathbf{L}^{\mathfrak{M}}$. We write H and K for the Hilbert spaces whose closed unit balls are the Hilbert sort and the sequence sort of $\mathfrak{M}$, respectively. Since T_1 contains T_0, the reduct of $\mathfrak{M}$ to the vocabulary of T_0 is a standard model of T_0, to which the results of Section 7 apply. As there, for each element a of $\mathbf{L}^{\mathfrak{M}}$ we write P_a for the orthogonal projection of H that extends $\pi(a,\cdot)$. Its range, the closed subspace spanned by the members of a, is again denoted by S_a, and the map $a \mapsto S_a$ by τ.

Lemma 8.1. *For each $n \in \mathbb{N}$, the map κ_n from the sequence sort to the Hilbert sort, which in the intended interpretation reads the entry in position n of a sequence (see Section 5.3), extends to a bounded linear map from K to H, and the map $s \mapsto (\kappa_n s)_{n\in\mathbb{N}}$ is a linear isometry of K onto the space $\ell^2(\mathbb{N}, H)$ of sequences $(x_n)_{n\in\mathbb{N}}$ in H with $\sum_n \|x_n\|^2 < \infty$.*

Proof. By (K2), each map κ_n is linear on the unit ball of K, and by (K3) it satisfies $\|\kappa_n s\| \le \|s\|$; it therefore extends to a bounded linear map from K to H. By (K2), we also have $\kappa_m \iota_n x = \delta_{mn} x$ for the maps ι_n, which in the intended interpretation place a vector in position n of a sequence. By the same axiom, the maps ι_n extend to linear isometries from H into K with mutually orthogonal ranges. For s in the unit ball of K and $N \in \mathbb{N}$, the identity $\langle \kappa_n s \mid x\rangle = \langle s \mid \iota_n x\rangle$ of (K2) and the Cauchy–Schwarz inequality give $\sum_{n<N} \|\kappa_n s\|^2 = \big\langle s, \sum_{n<N} \iota_n \kappa_n s\big\rangle \le \|s\| \big(\sum_{n<N} \|\kappa_n s\|^2\big)^{1/2}$, whence Bessel's inequality $\sum_{n<N} \|\kappa_n s\|^2 \le \|s\|^2 \le 1$. The truncation by $\mathbf{1}$ in (K3) is therefore never active, and axiom (K3) shows that the map $s \mapsto (\kappa_n s)_{n\in\mathbb{N}}$ is isometric. Given $(x_n)_{n\in\mathbb{N}} \in \ell^2(\mathbb{N}, H)$, the partial

sums $\sum_{n<N} \iota_n x_n$ therefore form a Cauchy sequence in K. Since K is complete, these partial sums converge to some s, which satisfies $\kappa_n s = x_n$ by the continuity of κ_n. ☺

Lemma 8.2. *For every s in the unit ball of K, the subspace $S_{\mathrm{span}(s)}$ is the closed span of the entries $\kappa_n s$.*

Proof. Let W be that closed span. By (Sp1), the subspace $S_{\mathrm{span}(s)}$ contains every entry of s; hence $W \subseteq S_{\mathrm{span}(s)}$. If $x \perp W$, axiom (Sp2) gives $P_{\mathrm{span}(s)}x = 0$; hence $S_{\mathrm{span}(s)} \subseteq W^{\perp\perp} = W$. ☺

Theorem 8.3. *Let $\mathfrak{M}$ be a standard model of T_1, let H be the Hilbert space whose closed unit ball is its Hilbert sort, and, for each element a of its Solèr sort, let S_a be the closed subspace of H spanned by the members of a. Then H is a separable infinite-dimensional complex Hilbert space, and the map $\tau\colon a \mapsto S_a$ is an isomorphism of ortholattices from the Solèr sort of $\mathfrak{M}$ onto the lattice $L(H)$ of closed subspaces of H; in particular, that Solèr sort is a Solèr lattice. Conversely, for every separable infinite-dimensional complex Hilbert space H', there is a standard model of T_1 whose Hilbert sort is the closed unit ball of H' and whose Solèr sort is $L(H')$.*

Proof. By Proposition 7.1, the space H is separable and infinite-dimensional. By Proposition 7.3, the map τ is an embedding; it remains to be shown that it is onto. Let S be a closed subspace of H, and let $(u_n)_{n\in\mathbb{N}}$ enumerate an orthonormal basis of S, which is countable by Proposition 7.1, followed by zeros if the basis is finite. By Lemma 8.1 there is s in the unit ball of K with $\kappa_n s = 2^{-n-1}u_n$ for all n, since $\sum_n 4^{-n-1} < 1$, and the subspace $S_{\mathrm{span}(s)}$ is S by Lemma 8.2.

For the converse, we build the model sort by sort. The Hilbert sort is the closed unit ball of H', and the constants e_n are interpreted as an orthonormal basis $(e_n)_{n\in\mathbb{N}}$ of H'. The Solèr sort is $L(H')$, with e the equality predicate, with $\pi(S, x) = P_S x$ for the orthogonal projection P_S onto S, and with $\mathrm{at}(x) = \mathbb{C}x$. For each $S \neq \{0\}$ we let $\mathrm{rep}(S)$ be a unit vector chosen in S, and we let $\mathrm{rep}(\{0\})$ be 0. The sequence sort is the closed unit ball of $\ell^2(\mathbb{N}, H')$, with $\iota_n x$ the sequence with x in position n and 0 elsewhere, $\kappa_n s$ the entry of s in position n, and $\varepsilon = (2^{-n-1}e_n)_{n\in\mathbb{N}}$. Finally, the function span sends each element s of the sequence sort to the closed span of its entries. Axioms (Sub2) and (Sub3) hold by Theorem 7.2, and the others by construction. ☺

Proof of Theorem A. Let $\mathfrak{M}$ be a standard model of T. Its reduct to the vocabulary of T_1 is a standard model of T_1, to which the notation fixed at the beginning of this section applies. By Theorem 8.3, the space H is a separable infinite-dimensional complex Hilbert space, and the map τ is an isomorphism of $\mathbf{L}^{\mathfrak{M}}$ onto $L(H)$. The idea is to apply axiom (M3) to a sequence whose entries are multiples of the vectors of an orthonormal basis of a closed subspace. Axiom (M3) then expresses the measure of that subspace as the sum of the measures of the atoms of those vectors (Step 2). This identity makes μ a σ-additive state, to which Gleason's theorem applies (Step 3).

Step 1: $\mu(0) = 0$. Recall that the formula od of Section 6 vanishes exactly at the sequences whose entries are mutually orthogonal. It therefore vanishes at the zero element s of the sequence sort, whose entries, all 0, are mutually orthogonal. Moreover, the subspace $S_{\mathrm{span}(s)}$ is $\{0\}$ by Lemma 8.2, whence $\mathrm{span}(s) = 0$ by the injectivity of τ. Since the entries of s are 0, axiom (At5) gives $\mathrm{at}(\kappa_n s) = \mathrm{at}(0) = 0$; axiom (M3) then says that $\mu(0)$ is the supremum over

N of $\max(0, \min(2, N\mu(0)))$. If $\mu(0)$ were positive, this supremum would be 2, contradicting (M1).

Step 2: sums along orthonormal bases. Let a be an element of $\mathbf{L}^{\mathfrak{M}}$, and let $(u_k)_{k\in\mathbb{N}}$ enumerate an orthonormal basis of S_a, followed by zeros if it is finite. By Lemma 8.1 there is s in the unit ball of K with $\kappa_k s = 2^{-k-1}u_k$. Its entries are mutually orthogonal, whence $\mathrm{od}(s) = 0$; by Lemma 8.2 and the injectivity of τ, the element $\mathrm{span}(s)$ is a. For $u_k \neq 0$, the subspaces $S_{\mathrm{at}(2^{-k-1}u_k)}$ and $S_{\mathrm{at}(u_k)}$ are both $\mathbb{C}u_k$ by Proposition 7.4, and the injectivity of τ gives $\mathrm{at}(\kappa_k s) = \mathrm{at}(u_k)$. For $u_k = 0$, the element $\mathrm{at}(\kappa_k s)$ is 0, whose measure is 0 by Step 1. Axiom (M3) and $\mu(a) \leq 1$ give

$$\mu(a) = \sum_k \mu(\mathrm{at}(u_k)). \tag{$*$}$$

Step 3: application of Gleason's theorem. We show that μ is a σ-additive state on $\mathbf{L}^{\mathfrak{M}}$, in order to apply Theorem 7.7. Its values lie in $[0,1]$ by (M1), and $\mu(1) = 1$ by (M2). Let $(a_n)_{n\in J}$, for a set $J \subseteq \mathbb{N}$, be a family of mutually orthogonal elements of $\mathbf{L}^{\mathfrak{M}}$. This family has a join c, because the map τ is onto $L(H)$. By Proposition 7.3 and Theorem 7.5, the subspaces S_{a_n} are mutually orthogonal, and the closure of their sum is S_c; orthonormal bases of the subspaces S_{a_n} therefore form together an orthonormal basis of S_c. Applied to c, the identity $(*)$ expresses $\mu(c)$ as the sum of the values $\mu(\mathrm{at}(u))$ over the vectors u of that basis. Applied to each a_n, the identity expresses $\mu(a_n)$ as the sum of the same values over the vectors that come from S_{a_n}. Since these values are nonnegative by (M1), the first sum may be grouped by n, whence $\mu(c) = \sum_{n\in J} \mu(a_n)$. The reduct of $\mathfrak{M}$ to the vocabulary of T_0 is a standard model of T_0 with the same Hilbert sort and the same Solèr sort, to which Theorem 7.7 applies. By that theorem, there is a unique density operator ρ on H with $\mu(a) = \mathrm{Tr}(\rho P_a)$ for every $a \in \mathbf{L}^{\mathfrak{M}}$.

The converse. Let a separable infinite-dimensional complex Hilbert space H' and a density operator ρ on H' be given. We expand a standard model of T_1 built for H' as in the proof of Theorem 8.3 by $\mu(a) = \mathrm{Tr}(\rho P_a)$, where P_a is the orthogonal projection onto the closed subspace a of H'. The clause on μ in (OL2) holds because e is the equality predicate. Axioms (M1) and (M2) hold because $\mathrm{Tr}(\rho P) \in [0,1]$ for every orthogonal projection P and $\mathrm{Tr}\,\rho = 1$. For (M3), let s be an element of the sequence sort, and suppose first that its entries $x_n = \kappa_n s$ for $n \in \mathbb{N}$ are mutually orthogonal. The element $\mathrm{span}(s)$ is then the closed span of these entries. Writing $\hat{x}_n = x_n/\|x_n\|$ for $x_n \neq 0$, we obtain $\mathrm{Tr}(\rho P_{\mathrm{span}(s)}) = \sum_{x_n\neq 0} \langle \rho\hat{x}_n \mid \hat{x}_n\rangle = \sum_n \mu(\mathrm{at}(x_n)) \leq 1$, whence the left side of (M3) is 0. If the entries are not mutually orthogonal, the right side of (M3) is 2, while the left side is at most 2. ☺

The proof of Theorem B rests on a property of the density operators on a separable infinite-dimensional complex Hilbert space. For each such operator ρ, the map $S \mapsto \mathrm{Tr}(\rho P_S)$ on the closed subspaces of that space takes every value in $[0,1]$, unlike a measure with only the values 0 and 1.

Lemma 8.4. *For every separable infinite-dimensional complex Hilbert space H and every density operator ρ on H, the set $\{\mathrm{Tr}(\rho P_S) : S \in L(H)\}$, where P_S is the orthogonal projection onto S, is the interval $[0,1]$.*

Proof. The values 0 and 1 are attained at $\{0\}$ and H, respectively. Fix α with $0 < \alpha < 1$; to show that α is attained, we rotate one vector of a subspace spanned by eigenvectors of ρ continuously and apply the intermediate value theorem. The rotation leads from a subspace whose value under $S \mapsto \mathrm{Tr}(\rho P_S)$ is at least α to one whose value is smaller. Since ρ is a positive compact operator of trace 1, there are orthonormal eigenvectors $v_1, v_2, \ldots$

(finitely or infinitely many) whose eigenvalues $\eta_1 \geq \eta_2 \geq \cdots > 0$ are the nonzero eigenvalues of ρ, repeated according to multiplicity, and these eigenvalues satisfy $\sum_i \eta_i = 1$. Write $\eta_{\leq k} = \eta_1 + \cdots + \eta_k$, with $\eta_{\leq 0} = 0$, and choose an integer $k \geq 0$ with $\eta_{\leq k} < \alpha \leq \eta_{\leq k+1}$. On the unit vectors of $W = \{v_1, \ldots, v_{k+1}\}^\perp$, the quantity $\langle \rho x \mid x \rangle$ has infimum 0: either W meets the kernel of ρ in a nonzero vector, or the eigenvalues η_i with $i > k+1$ tend to 0. Choose a unit vector $u \in W$ with $\langle \rho u \mid u \rangle < \alpha - \eta_{\leq k}$. For $0 \leq \vartheta \leq 1$, let w_ϑ be the unit vector $\sqrt{1-\vartheta}\, v_{k+1} + \sqrt{\vartheta}\, u$, and let S_ϑ be the subspace spanned by $v_1, \ldots, v_k$ and w_ϑ. Since $\rho v_{k+1} = \eta_{k+1} v_{k+1}$ and the vector u is orthogonal to $v_1, \ldots, v_{k+1}$, the cross terms vanish. Consequently, we have

$$\mathrm{Tr}(\rho P_{S_\vartheta}) = \eta_{\leq k} + (1-\vartheta)\eta_{k+1} + \vartheta \langle \rho u \mid u \rangle,$$

a continuous function of ϑ that equals $\eta_{\leq k+1} \geq \alpha$ at $\vartheta = 0$ and is smaller than α at $\vartheta = 1$. By the intermediate value theorem, it takes the value α. ☺

Proof of Theorem B. Let $\mathfrak{M}$ be a standard model of T, and let H be the Hilbert space whose closed unit ball is its Hilbert sort. By Theorem A, there is a density operator ρ on H with $\mu(a) = \mathrm{Tr}(\rho P_a)$ for every element a of the Solèr sort, and the map $a \mapsto S_a$ is onto $L(H)$. By Lemma 8.4, the range of μ is $[0,1]$. Since B meets $[0,1]$, some element a of the Solèr sort satisfies $\mu(a) \in B$, and the value of $\beta_B(a)$ at that element is 1. Hence the axiom $\beta_B(a) \leq -1$ of Theorem B fails in $\mathfrak{M}$. ☺

⧖ **Historical note.** Sierpiński proved in 1922 that a measure without atoms takes, on the measurable subsets of a set, every value between 0 and the measure of that set [Sie22]. A $\{0,1\}$-valued measure, which is what an assignment of truth values amounts to, is the opposite extreme: the whole space is an atom. By contrast, the measures $S \mapsto \mathrm{Tr}(\rho P_S)$ that density operators ρ define on the closed subspaces of a separable infinite-dimensional Hilbert space take every value in $[0,1]$ (see Lemma 8.4). Such a measure takes all these values even when ρ is a pure state, the orthogonal projection onto a ray, whose classical counterpart would be a point mass.

8.1. **Definable sets of values.** The formulas $\beta_B(a)$ constructed in the proof of Lemma 5.1 use the connectives of the logic but none of its quantifiers. With quantifiers over the Solèr sort, formulas define more sets of values in the standard models, because the measure takes every value in $[0,1]$ there. In Theorem 8.6 below we show that the sets so defined are exactly the σ-projective subsets of $[0,1]$, and that the conclusion of Theorem B extends to all of them.

Aguilera, Müller and Schlicht [AMS21] study the σ-projective sets, under this name, as subsets of the finite powers of $\mathbb{N}^\mathbb{N}$. Let Γ assign to each Polish space X a class $\Gamma(X)$ of subsets of X that contains the Borel subsets of X and is closed under complements in X and countable unions. Suppose moreover that, whenever $A \in \Gamma(X \times \mathbb{N}^\mathbb{N})$, the set $\{x \in X : (x,z) \in A \text{ for some } z \in \mathbb{N}^\mathbb{N}\}$ belongs to $\Gamma(X)$, and take for Γ the least assignment with these properties. A subset of a Polish space X is *σ-projective* when it belongs to $\Gamma(X)$. On the finite powers of $\mathbb{N}^\mathbb{N}$, the σ-projective sets are the sets of Aguilera, Müller and Schlicht. Their class is the least that contains the open sets and is closed under complements, countable unions and projections along $\mathbb{N}^\mathbb{N}$. This class is therefore contained in Γ, which has these properties on the finite powers of $\mathbb{N}^\mathbb{N}$. Conversely, for a Polish space X, consider the sets $A \subseteq X$ such that $g^{-1}[A]$ belongs to the class of Aguilera, Müller and Schlicht for every k and every Borel map $g\colon (\mathbb{N}^\mathbb{N})^k \to X$. The class of Aguilera, Müller and Schlicht contains the Borel sets and is closed under the operations that define Γ. Hence these sets A include

the Borel sets and are closed under complements, countable unions and projections along $\mathbb{N}^{\mathbb{N}}$. These sets therefore include $\Gamma(X)$, as the identity maps show for $X = (\mathbb{N}^{\mathbb{N}})^k$. Every projective set [Kec95, Section 37] is σ-projective, and so is every countable union of projective sets. The σ-projective sets are closed under preimages by Borel maps between Polish spaces. Indeed, the sets $A \subseteq X$ such that $g^{-1}[A]$ is σ-projective for every Borel map g from a Polish space into X include the Borel sets and are closed under complements and countable unions. They are also closed under projections along $\mathbb{N}^{\mathbb{N}}$, since $(g \times \mathrm{id})^{-1}$ commutes with the projection; these sets therefore include every σ-projective set. The σ-projective sets are also closed under projections along every nonempty Polish space Y. Given a continuous surjection $f\colon \mathbb{N}^{\mathbb{N}} \to Y$ [Kec95, Theorem 7.9] and a set $A \subseteq X \times Y$, the projection of A to X is the projection along $\mathbb{N}^{\mathbb{N}}$ of the preimage of A under $\mathrm{id} \times f$. A function f from a Polish space to $\mathbb{R}$ is Γ-*measurable* when the set $\{f > r\}$ is σ-projective for every real number r. The set $\{f \geq r\}$, which is the intersection of the sets $\{f > r - \frac{1}{n}\}$, is then σ-projective as well.

Definition 8.5. A set $D \subseteq [0,1]$ is *definable over* T if there is a formula $\gamma(a)$ of real-valued $L_{\omega_1\omega}$, with the single free variable a of the Solèr sort, whose value, in every standard model of T and at every element a of its Solèr sort, is 1 if $\mu(a) \in D$ and -1 otherwise; we say that γ *defines* D. Since (M1) confines μ to $[0,1]$ in every model of T, a formula that, in every standard model of T and at every element a of its Solèr sort, takes the value 1 if $\mu(a)$ lies in a set $E \subseteq \mathbb{R}$ and -1 otherwise defines $E \cap [0,1]$; only subsets of $[0,1]$ need be considered.

Theorem 8.6. *A subset of* $[0,1]$ *is definable over* T *if and only if it is* σ*-projective. Consequently, if* $D \subseteq [0,1]$ *is a nonempty* σ*-projective set and* $\gamma(a)$ *defines* D*, then the theory* T *together with the axiom* $\gamma(a) \leq -1$*, which says that no value of* μ *lies in* D*, has no standard models.*

The class of σ-projective subsets of $[0,1]$ is strictly larger than that of projective ones. For each $n \geq 1$, the hierarchy theorem for projective sets [Kec95, Section 37] provides a $\mathbf{\Sigma}^1_{n+1}$ set $C_n \subseteq [0,1]$ that is not $\mathbf{\Sigma}^1_n$, and we let h_n be a homeomorphism of $[0,1]$ onto $[2^{-n}, 2^{-n+1}]$. The set $C = \bigcup_n h_n[C_n \cap (0,1)]$ is a countable union of projective sets, and it contains none of the points 2^{-m}. If C were $\mathbf{\Sigma}^1_k$, then for $n > k$ the set $h_n^{-1}[C]$, which equals $C_n \cap (0,1)$, would be $\mathbf{\Sigma}^1_k$. The set C_n, which differs from $C_n \cap (0,1)$ by at most two points, would then be $\mathbf{\Sigma}^1_k$ and hence $\mathbf{\Sigma}^1_n$, which is false. If C were $\mathbf{\Pi}^1_k$, it would be $\mathbf{\Sigma}^1_{k+1}$, which is excluded in the same way. Hence C is not projective, and by Theorem 8.6 it is definable over T.

The two directions use different features of the standard models. To define every σ-projective set, we use the surjectivity of μ: quantification over the Solèr sort then realizes projection along the interval of probabilities. To bound the expressive power of formulas, we choose one standard model in which every symbol is Borel on Polish spaces. In that model, a continuous choice of a projection of each probability transfers the descriptive complexity of a formula to the subset of $[0,1]$ that it defines.

Proof of Theorem 8.6. Every σ*-projective subset of* $[0,1]$ *is definable.* For $m \geq 1$, consider a formula $\gamma(a_1, \dots, a_m)$ with free variables $a_1, \dots, a_m$ of the Solèr sort. The formula γ *strongly defines* a set $A \subseteq [0,1]^m$ if it is two-valued in every structure for $\mathcal{L}$ and, in every standard model of T, equals 1 at $(a_1, \dots, a_m)$ exactly when $(\mu(a_1), \dots, \mu(a_m)) \in A$. Let $\mathcal{A}$ be the class of sets $A \subseteq [0,1]^m$, for $m \geq 1$, that some formula strongly defines. We show that $\mathcal{A}$ contains every σ-projective subset of every cube $[0,1]^m$; a formula that strongly defines a set $D \subseteq [0,1]$ defines it in the sense of Definition 8.5.

First, $\mathcal{A}$ contains the Borel sets. We repeat the proof of Lemma 5.1 with the closed boxes $[q_1, q_1'] \times \cdots \times [q_m, q_m']$, for rational numbers $q_i \leq q_i'$, in place of the closed intervals with rational endpoints. The distance to such a box is written, by means of the connectives

$\operatorname{dist}(\cdot,[q,q'])$ of that proof, as the maximum

$$\operatorname{dist}(\mu(a_1),[q_1,q_1'])\vee\cdots\vee\operatorname{dist}(\mu(a_m),[q_m,q_m'])$$

of the distances of the coordinates. For every Borel set $B\subseteq\mathbb{R}^m$, the modified proof gives a formula $\beta_B(a_1,\dots,a_m)$. In every structure for $\mathcal{L}$, its value at every tuple of elements of the Solèr sort is 1 if $(\mu(a_1),\dots,\mu(a_m))\in B$ and -1 otherwise. The formula $\beta_B(a_1,\dots,a_m)$ strongly defines $B\cap[0,1]^m$. Next, if γ strongly defines $A\subseteq[0,1]^m$, then the formula $-\gamma$ strongly defines $[0,1]^m\setminus A$. If γ_n strongly defines $A_n\subseteq[0,1]^m$ for every $n\in\mathbb{N}$, then the join $\bigvee_n\gamma_n$, whose members take their values in $[-1,1]$, strongly defines $\bigcup_n A_n$. Finally, if $\gamma(a_1,\dots,a_m,b)$ strongly defines $A\subseteq[0,1]^{m+1}$, then the formula $\sup_b\gamma(a_1,\dots,a_m,b)$ is two-valued in every structure. Indeed, the Solèr sort contains the constant 0, and a supremum of values in $\{-1,1\}$ over a nonempty set is attained. In a standard model of T, the formula $\sup_b\gamma(a_1,\dots,a_m,b)$ takes the value 1 at $(a_1,\dots,a_m)$ exactly when $(\mu(a_1),\dots,\mu(a_m),\mu(b))\in A$ for some element b. By Theorem B, such an element b exists exactly when $(\mu(a_1),\dots,\mu(a_m),t)\in A$ for some $t\in[0,1]$. Thus $\mathcal{A}$ contains the Borel subsets of the cubes $[0,1]^m$ and is closed under complements, countable unions and projections along $[0,1]$.

It remains to be shown that these operations produce every σ-projective subset of every cube $[0,1]^m$. Consider the least class Γ_0 of subsets of the cubes $[0,1]^m$, for $m\geq 1$, containing their Borel subsets and closed under complements in the cube, countable unions and projections from $[0,1]^{m+1}$ to $[0,1]^m$. We have already shown that the class Γ_0 is contained in $\mathcal{A}$. Fix a Borel isomorphism θ of $[0,1]$ onto $\mathbb{N}^\mathbb{N}$ [Kec95, Theorem 15.6]. For each Polish space X, let $\Gamma'(X)$ be the class of sets $A\subseteq X$ such that $g^{-1}[A]\in\Gamma_0$ for every $m\geq 1$ and every Borel map $g\colon[0,1]^m\to X$. Each class $\Gamma'(X)$ contains the Borel subsets of X and is closed under complements in X and countable unions. Let $A\in\Gamma'(X\times\mathbb{N}^\mathbb{N})$ be given, and let $g\colon[0,1]^m\to X$ be a Borel map. Since θ is onto $\mathbb{N}^\mathbb{N}$, we have

$$g^{-1}\big[\{x:\exists z\in\mathbb{N}^\mathbb{N}\ (x,z)\in A\}\big]=\{u\in[0,1]^m:\exists t\in[0,1]\ (u,t)\in(g\times\theta)^{-1}[A]\},$$

the projection along $[0,1]$ of a set of Γ_0, because $g\times\theta$ is a Borel map on $[0,1]^{m+1}$; this projection belongs to Γ_0. Hence the assignment Γ' is closed under projections along $\mathbb{N}^\mathbb{N}$, and we have $\Gamma(X)\subseteq\Gamma'(X)$ for every Polish space X. For a σ-projective set $A\subseteq[0,1]^m$, taking for g the identity shows that $A\in\Gamma_0$ and therefore that $A\in\mathcal{A}$.

Every definable subset of $[0,1]$ *is* σ*-projective.* Let $\gamma(a)$ define a set $D\subseteq[0,1]$. Let $H=\ell^2(\mathbb{N})$, let v and w be orthogonal unit vectors of H, and let ρ be the projection onto $\mathbb{C}v$, a density operator. By Theorem A(2) and its proof, there is a standard model $\mathfrak{M}$ of T that can be described as follows. Its Hilbert sort is the closed unit ball B_H of H, its sequence sort is the closed unit ball B_K of $K=\ell^2(\mathbb{N},H)$, and its Solèr sort is $L(H)$. The symbols are interpreted as in that proof, with the standard unit vectors of $\ell^2(\mathbb{N})$ as the constants e_n and with $\mu(S)=\operatorname{Tr}(\rho P_S)$. In that proof, the value $\operatorname{rep}(S)$ may be any unit vector of S when $S\neq\{0\}$, and here we let it be $P_Se_n/\|P_Se_n\|$ for the least n with $P_Se_n\neq 0$. We identify $L(H)$ with the set of orthogonal projections on H, with the strong operator topology. Fix a dense sequence $(x_k)_{k\in\mathbb{N}}$ in B_H, and give the set of operators of norm at most 1 the strong operator topology. For such operators, convergence at the vectors x_k implies convergence at every vector. The map $Q\mapsto(Qx_k)_{k\in\mathbb{N}}$ is therefore a homeomorphism of the set of operators of norm at most 1 onto its image in the Polish space $B_H^\mathbb{N}$. The image is closed: if $(y_k)_{k\in\mathbb{N}}$ is the limit of sequences $(Q_jx_k)_{k\in\mathbb{N}}$ in the image, then we have $\|\sum_k c_ky_k\|\leq\|\sum_k c_kx_k\|$ for every finitely supported family of complex numbers c_k. The map $x_k\mapsto y_k$ therefore extends to an operator of norm at most 1. The projections form a closed subset, since a strong limit Q of projections satisfies $\langle Qx\mid x\rangle=\|Qx\|^2=\langle Q^*Qx\mid x\rangle$ for every x, whence $Q=Q^*Q$ by

polarization, and so $Q = Q^* = Q^2$. Thus the space $L(H)$ is Polish, and so are B_H and B_K with their norm topologies.

Every symbol of $\mathcal{L}$ is interpreted in $\mathfrak{M}$ by a Borel function on a product of these spaces. The vector operations, the inner products and the norms are continuous, as are the maps ι_n and κ_n, the constants and the orthocomplement $Q \mapsto I - Q$. The map $(Q, x) \mapsto Qx$, which interprets π, is continuous because $\|Qx - Q_0x_0\| \leq \|x - x_0\| + \|(Q - Q_0)x_0\|$. The measure, given by $\mu(Q) = \mathrm{Tr}(\rho Q) = \langle Qv \mid v\rangle$, is continuous as well. The meet $Q \wedge Q'$ is the strong limit of $(QQ'Q)^n$ as $n \to \infty$, by Theorem 7.2 and the identity $(QQ'Q)^n = (QQ')^nQ$. As a pointwise limit of continuous maps, the meet is Borel, and the join is given by $Q \vee Q' = I - ((I - Q) \wedge (I - Q'))$. The predicate e is 1 on the diagonal, a closed set, and -1 off it. The map at sends $x \neq 0$ to the projection $y \mapsto \langle y \mid x\rangle x/\|x\|^2$, continuously, and 0 to 0. For rep, the set of projections Q with $Qe_k = 0$ for $k < n$ and $Qe_n \neq 0$ is Borel, and on it $\mathrm{rep}(Q) = Qe_n/\|Qe_n\|$ is continuous. For span, let $P_N(s)$ be the projection onto the span of the entries $\kappa_0 s, \ldots, \kappa_{N-1}s$ of s. The ball B_K splits into finitely many sets according to the set of indices $i < N$ such that $\kappa_i s$ is not in the span of $\kappa_0 s, \ldots, \kappa_{i-1}s$. On each of these finitely many sets, the map P_N is continuous, being given by Gram–Schmidt orthonormalization of the corresponding entries. These sets are Borel, since the rank of the Gram matrix of $(\kappa_0 s, \ldots, \kappa_i s)$ is a Borel function of s, the set where it is at least r being open. The projections $P_N(s)$ increase with N to the projection $\mathrm{span}(s)$ onto the closed span of the entries of s, strongly, and the map span is Borel. It follows by induction on terms that every term is interpreted by a Borel function.

We show by induction on formulas that, for every formula $\varphi(\bar{x})$, the function $\bar{x} \mapsto \varphi^{\mathfrak{M}}(\bar{x})$ is Γ-measurable on the product of the Polish spaces that interpret the sorts of $\bar{x}$. Atomic formulas are Borel functions of terms, hence Borel. For a continuous connective $f(\varphi_1, \ldots, \varphi_\ell)$, the set $\{f(\varphi_1, \ldots, \varphi_\ell) > r\}$ is the preimage under $(\varphi_1, \ldots, \varphi_\ell)$ of an open subset of $\mathbb{R}^\ell$, which is a countable union of open boxes with rational corners. The preimage of each of these boxes is a finite intersection of sets $\{\varphi_i > q\}$ and complements of sets $\{\varphi_i \geq q'\}$. The connectives $s \cdot \varphi$ and $\varphi \vee \psi$ are of this kind. For a countable join, the set $\{\bigvee_n \varphi_n > r\}$ is the union $\bigcup_n \{\varphi_n > r\}$. For a supremum over a variable y, the set $\{\bar{x} : \sup_y \varphi(\bar{x}, y) > r\}$ is the projection of $\{(\bar{x}, y) : \varphi(\bar{x}, y) > r\}$ along the Polish space that interprets the sort of y. In each case the set is σ-projective.

Now the function $\gamma^{\mathfrak{M}}$ is Γ-measurable, and the set $\{Q \in L(H) : \gamma^{\mathfrak{M}}(Q) > 0\}$ is σ-projective. For $t \in [0, 1]$, let $\xi(t)$ be the projection at$(\sqrt{t}\, v + \sqrt{1 - t}\, w)$ onto the line spanned by the unit vector $\sqrt{t}\, v + \sqrt{1 - t}\, w$. The map ξ is continuous and satisfies $\mu(\xi(t)) = |\langle v \mid \sqrt{t}\, v + \sqrt{1 - t}\, w\rangle|^2 = t$. Since $\gamma^{\mathfrak{M}}$ takes the value 1 exactly at the elements Q with $\mu(Q) \in D$,

$$D = \{t \in [0, 1] : \gamma^{\mathfrak{M}}(\xi(t)) > 0\} = \xi^{-1}\big[\{\gamma^{\mathfrak{M}} > 0\}\big],$$

a continuous preimage of a σ-projective set, hence σ-projective.

The consequence follows as in the proof of Theorem B, with γ in place of β_B: in a standard model, some element a has $\mu(a) \in D$, and $\gamma(a)$ takes the value 1 there. ☺

The *quantifier depth* $\mathrm{qd}(\varphi)$ of a formula φ is a countable ordinal, defined by induction. Atomic formulas have depth 0, and the depth of $f(\varphi_1, \ldots, \varphi_\ell)$, of $s \cdot \varphi$ and of $\varphi \vee \psi$ is the largest depth of the formulas to which the connective is applied. The depth of a countable join is given by $\mathrm{qd}(\bigvee_n \varphi_n) = \sup_n \mathrm{qd}(\varphi_n)$, and that of a supremum by $\mathrm{qd}(\sup_x \varphi) = \mathrm{qd}(\varphi)+1$. In the model $\mathfrak{M}$ of the proof, the induction on formulas given there shows that, for every $k < \omega$ and every formula φ with $\mathrm{qd}(\varphi) \leq k$, each set $\{\varphi^{\mathfrak{M}} > r\}$ is $\mathbf{\Delta}^1_{k+1}$. For $k = 0$, the formula φ contains no supremum over a variable, and these sets are Borel. For $k \geq 1$, if

$\mathrm{qd}(\sup_y \varphi) \leq k$, then the formula φ has depth at most $k-1$, and the set $\{(\sup_y \varphi)^{\mathfrak{M}} > r\}$ is the projection of a $\mathbf{\Delta}^1_k$ set. This projection is therefore $\mathbf{\Sigma}^1_k$, and hence $\mathbf{\Delta}^1_{k+1}$. The other cases use only countable unions, countable intersections and complements, under which $\mathbf{\Delta}^1_{k+1}$ is closed. If a formula γ with $\mathrm{qd}(\gamma) \leq k$ defines D, then the set $D = \xi^{-1}[\{\gamma^{\mathfrak{M}} > 0\}]$ is therefore $\mathbf{\Delta}^1_{k+1}$. Since the set C above is not projective, every formula that defines it has quantifier depth at least ω. As the subformula relation is well founded, any formula that defines C has a subformula of depth at least ω whose proper subformulas all have finite depth. That subformula is a countable join whose members have unbounded finite depths.

Uniformity across standard models. The definition asks for one formula that defines D in every standard model, whatever orthonormal basis interprets the constants e_n, whatever density operator gives the measure and whatever unit vectors interpret rep. The requirement cannot be weakened to a single standard model. Write $\langle\varphi\rangle$ for the two-valued formula $2\lceil\varphi\rceil - \mathbf{1}$, which takes the value 1 where $\varphi > 0$ and -1 elsewhere. We write $z_i(b)$ for the pair of predicates $\langle \mathrm{rep}(b) \mid e_i\rangle$, and we let $p(b)$ and $q(b)$ be the real and imaginary parts of $z_1(b)\overline{z_0(b)}$, written with the sum and product connectives. Let $\delta(a,b)$ be the formula $\beta_B(a,b)$ of the proof for the diagonal $B = \{(u,u) : u \in \mathbb{R}\}$. Let $\gamma_0(a)$ be the formula

$$\sup_b\Big(- e(b,0) \wedge -\mathrm{nonatom}(b) \wedge e\big(b \wedge (\mathrm{at}(e_0) \vee \mathrm{at}(e_1)), b\big) \wedge \delta(a,b) \\ \wedge \langle p(b)\rangle \wedge -\langle |q(b)|\rangle \wedge \langle \mathrm{Re}\langle \mathrm{rep}(b) \mid e_0\rangle\rangle\Big).$$

Given $X \subseteq (0,1)$, take the model of the proof of Theorem A(2) on $\ell^2(\mathbb{N})$, with the standard unit vectors as the constants e_n and with ρ the projection onto $\mathbb{C}e_0$. In this model, let rep send the atom spanned by $\sqrt{t}\,e_0 + \sqrt{1-t}\,e_1$, for $0 < t < 1$, to the vector $\sqrt{t}\,e_0 + \sqrt{1-t}\,e_1$ itself if $t \in X$ and to its negative otherwise. For $0 < t < 1$, the only atom b below $\mathrm{at}(e_0) \vee \mathrm{at}(e_1)$ with $\mu(b) = t$, $p(b) > 0$ and $q(b) = 0$ is this one, and for $t \in \{0,1\}$ there is none. Hence the formula $\gamma_0(a)$ takes the value 1 in that model exactly when $\mu(a) \in X$, and -1 otherwise. Consider now the model $\mathfrak{M}$ of the proof, whose symbols are Borel. Let D be a set such that some formula takes, at every element Q of the Solèr sort of $\mathfrak{M}$, the value 1 if $\mu(Q) \in D$ and -1 otherwise. For this model, the second half of the proof shows that the set D is σ-projective, and hence definable over T. Since the vocabulary and the set of connectives are countable, there are at most $2^{\aleph_0}$ formulas, and hence at most $2^{\aleph_0}$ sets definable over T. The interval $(0,1)$, by contrast, has $2^{2^{\aleph_0}}$ subsets, and some sets X that γ_0 defines in a single standard model are therefore not definable over T.

In a model that is not standard, a formula that defines a set D over T need not take the value 1 exactly when $\mu(a) \in D$. By Remark 11.2, the theory T has a countable model $\mathfrak{N}$; the range of μ in $\mathfrak{N}$ is countable, and some $c \in [0,1]$ is not a value of μ in $\mathfrak{N}$. By Theorem B, the formula $\sup_b \beta_{\mathbb{R}\times\{c\}}(a,b)$, with $\beta_{\mathbb{R}\times\{c\}}$ as in the proof of Theorem 8.6, defines $[0,1]$ over T. In $\mathfrak{N}$, however, this formula takes the value -1 everywhere. The formulas $\beta_B(a)$ constructed in the proof of Lemma 5.1 have no quantifiers, and in every structure they take the value 1 exactly when $\mu(a) \in B$. The cases of the remark on standard models in Section 10, which rest on facts about all models of T, depend on these properties of the formulas $\beta_B(a)$. For this reason that remark is stated for these formulas, whereas Theorem 8.6 is a theorem about the standard models, in which the measure takes every value in $[0,1]$.

9. The proof of Theorem C

In this section we prove Theorem C: in no model of T, standard or not, does the measure μ take only the values 0 and 1. The theorem concerns every model of T, and its proof accordingly draws on none of the conclusions of Theorem A, which concern the standard models. The constants e_n are orthonormal by (H2), and the entries of the constant ε are the vectors $2^{-n-1}e_n$ by (K4). Besides these constants, the proof uses only the eighteen vectors of Table 1, which are built from four of the vectors e_n. The Hilbert-space envelope of Lemma 6.3 supplies the geometry, while the rational terms built from the named basis vectors supply the vectors needed inside the model. The argument therefore requires neither the whole unit ball nor the whole lattice of closed subspaces.

We argue by contradiction, and fix throughout this section a model $\mathfrak{M}$ of T in which μ takes only the values 0 and 1. Let $\bar{E}$ be a complex Hilbert space as in Lemma 6.3: the Hilbert sort of $\mathfrak{M}$ is identified with a subset Y of the closed unit ball of $\bar{E}$. The linear span of Y with coefficients in $\mathbb{Q}(i)$, the field of complex numbers with rational real and imaginary parts, is dense in $\bar{E}$. For each element a of the Solèr sort, let $\bar{P}_a$ be the orthogonal projection of $\bar{E}$ that extends $\pi(a,\cdot)$. By (H2), the vectors e_n form an orthonormal basis of $\bar{E}$: Parseval's identity puts every vector of Y, and hence every vector of $\bar{E}$, in their closed span. The proof rests on three facts that hold in every model of T.

Lemma 9.1. *Let $\mathfrak{M}$ be a model of T, standard or not, and let $\bar{E}$ and $\bar{P}_a$ be as in Lemma 6.3. Then the following hold:*

1. *For every nonzero x in the Hilbert sort and every $\lambda \in \mathbb{Q}(i)$ with $0 < |\lambda| \leq 1$, we have $\mathrm{at}(\lambda x) = \mathrm{at}(x)$, and the range of $\bar{P}_{\mathrm{at}(x)}$ is $\mathbb{C}x$.*
2. *Let s be an element of the sequence sort, and let F be the closed span of its entries in $\bar{E}$. If every e_n lies in F or in $F^\perp$, then the range of $\bar{P}_{\mathrm{span}(s)}$ is F.*
3. *The measure satisfies $\mu(0) = 0$, and $\mathrm{span}(\varepsilon) = 1$.*

Proof. (1) By (At1) and continuity, the range of $\bar{P}_{\mathrm{at}(x)}$ contains x, and any two of its vectors are linearly dependent. Hence this range is $\mathbb{C}x$, and the range of $\bar{P}_{\mathrm{at}(\lambda x)}$ is likewise $\mathbb{C}\lambda x$. The two projections coincide, and axiom (P4) gives $\mathrm{at}(\lambda x) = \mathrm{at}(x)$.

(2) By (Sp1) and linearity, the projection $\bar{P}_{\mathrm{span}(s)}$ fixes every entry of s; hence its range contains F. If $e_n \in F^\perp$, then the vector e_n is orthogonal to every entry, and axiom (Sp2) gives $\bar{P}_{\mathrm{span}(s)}e_n = 0$. The vectors e_n that lie in $F^\perp$ span a dense subspace of $F^\perp$, because the e_n form an orthonormal basis and each of them lies in F or in $F^\perp$. By continuity, the projection $\bar{P}_{\mathrm{span}(s)}$ vanishes on $F^\perp$, and its range is F.

(3) For $s = 0$, whose entries are 0 by (K2), every vector of the Hilbert sort is orthogonal to the entries. Axiom (Sp2) gives $\bar{P}_{\mathrm{span}(0)} = 0$, which is $\bar{P}_0$ by (Sub0), and axiom (P4) then gives $\mathrm{span}(0) = 0$. Since $\mathrm{od}(0) = 0$, the remainder of Step 1 in the proof of Theorem A gives $\mu(0) = 0$; that part of the proof uses only (At5), (M3) and (M1). For $s = \varepsilon$, the entries are $2^{-n-1}e_n$ by (K4), and the space F of part (2) is then $\bar{E}$. Part (2) gives $\bar{P}_{\mathrm{span}(\varepsilon)} = \mathrm{Id}$, which is $\bar{P}_1$ by (Sub0), whence $\mathrm{span}(\varepsilon) = 1$ by (P4). ☻

Proof of Theorem C. We first show that the basis $(e_n)_{n\in\mathbb{N}}$ contains exactly one vector whose atom has measure 1. We then apply a parity argument to the bases of Table 1 within the block of four basis vectors that contains this vector. The entries of ε are mutually orthogonal, whence $\mathrm{od}(\varepsilon) = 0$, and part (1) of Lemma 9.1 gives $\mathrm{at}(2^{-n-1}e_n) = \mathrm{at}(e_n)$. By (M2), (K4),

basis	vectors (coordinates with respect to an orthonormal basis g_0, g_1, g_2, g_3)			
1	$(0,0,0,1)$	$(0,0,1,0)$	$(1,1,0,0)$	$(1,-1,0,0)$
2	$(0,0,0,1)$	$(0,1,0,0)$	$(1,0,1,0)$	$(1,0,-1,0)$
3	$(1,-1,1,-1)$	$(1,-1,-1,1)$	$(1,1,0,0)$	$(0,0,1,1)$
4	$(1,-1,1,-1)$	$(1,1,1,1)$	$(1,0,-1,0)$	$(0,1,0,-1)$
5	$(0,0,1,0)$	$(0,1,0,0)$	$(1,0,0,1)$	$(1,0,0,-1)$
6	$(1,-1,-1,1)$	$(1,1,1,1)$	$(1,0,0,-1)$	$(0,1,-1,0)$
7	$(1,1,-1,1)$	$(1,1,1,-1)$	$(1,-1,0,0)$	$(0,0,1,1)$
8	$(1,1,-1,1)$	$(-1,1,1,1)$	$(1,0,1,0)$	$(0,1,0,-1)$
9	$(1,1,1,-1)$	$(-1,1,1,1)$	$(1,0,0,1)$	$(0,1,-1,0)$

TABLE 1. The configuration of Cabello, Estebaranz and García-Alcaine [CEGA96]: nine orthogonal bases of a four-dimensional space, formed by eighteen vectors, each of which occurs in exactly two bases. In the proof of Theorem C, the vectors $g_0, \dots, g_3$ are four consecutive vectors $e_{4j}, \dots, e_{4j+3}$, for some $j \in \mathbb{N}$, of the orthonormal basis $(e_n)_{n\in\mathbb{N}}$.

(M3) and Lemma 9.1(3), we obtain

$$1 = \mu(1) = \mu(\mathrm{span}(\varepsilon)) = \sum_n \mu(\mathrm{at}(e_n)).$$

As μ takes only the values 0 and 1, there is exactly one $n_0 \in \mathbb{N}$ with $\mu(\mathrm{at}(e_{n_0})) = 1$. Let $j \in \mathbb{N}$ satisfy $4j \le n_0 < 4j+4$, put $g_i = e_{4j+i}$ for $i = 0,1,2,3$, and let F be the span of $g_0, \dots, g_3$.

The basis $g_0, \dots, g_3$ of F contains exactly one vector, e_{n_0}, whose atom has measure 1. We show that every orthogonal basis of F whose vectors have coordinates in $\{0,1,-1\}$ with respect to $g_0, \dots, g_3$ has the same property. Let $\mathcal{B}$ be such a basis, consisting of vectors $v_0, \dots, v_3$. Each vector $\frac{1}{4}v_i$ has coordinates whose absolute values sum to at most 1, and it is therefore denoted by a term built from the constants e_n by the functions $\lambda x + \lambda' y$. Hence the term $\frac{1}{2}\big(\frac{1}{2}\iota_0(\frac{1}{4}v_0) + \frac{1}{2}\iota_1(\frac{1}{4}v_1)\big) + \frac{1}{2}\big(\frac{1}{2}\iota_2(\frac{1}{4}v_2) + \frac{1}{2}\iota_3(\frac{1}{4}v_3)\big)$ denotes an element $s_{\mathcal{B}}$ of the sequence sort. By (K2), which gives $\kappa_m \iota_n x = \delta_{mn} x$, the entries $\kappa_n s_{\mathcal{B}}$ are $\frac{1}{16}v_n$ for $n < 4$ and 0 for $n \ge 4$. These entries are mutually orthogonal, whence $\mathrm{od}(s_{\mathcal{B}}) = 0$. Each e_n lies in F or in $F^\perp$; by Lemma 9.1(2), the projection $\bar{P}_{\mathrm{span}(s_{\mathcal{B}})}$ is therefore the projection onto F. For a vector v with coordinates in $\{0,1,-1\}$, we write $\mathrm{at}(v)$ for $\mathrm{at}(\frac{1}{16}v)$; by Lemma 9.1(1), this element does not change when the sign of v is changed. By (P4), all the elements $\mathrm{span}(s_{\mathcal{B}})$ are equal to one element a_F of the Solèr sort. Since $\mathrm{at}(\kappa_n s_{\mathcal{B}}) = \mathrm{at}(0) = 0$ for $n \ge 4$ by (At5) and $\mu(0) = 0$, axiom (M3) gives

$$\mu(a_F) = \sum_{v\in\mathcal{B}} \mu(\mathrm{at}(v)).$$

For the basis $g_0, \dots, g_3$ this sum contains $\mu(\mathrm{at}(e_{n_0})) = 1$ and three zeros; hence $\mu(a_F) = 1$, and every such basis $\mathcal{B}$ contains exactly one vector v with $\mu(\mathrm{at}(v)) = 1$.

The nine bases of Table 1 are orthogonal bases of F of this kind, and their eighteen vectors span pairwise distinct rays. We conclude by counting in two ways the pairs $(\mathcal{B}, v)$ that consist of one of the nine bases $\mathcal{B}$ and a vector v of $\mathcal{B}$ with $\mu(\mathrm{at}(v)) = 1$. By the preceding paragraph,

each basis contains exactly one such vector, and there are therefore exactly nine pairs. On the other hand, each of the eighteen vectors occurs in exactly two of the bases. Each vector v with $\mu(\mathrm{at}(v)) = 1$ thus belongs to exactly two pairs, and the number of pairs is even. This contradiction proves the theorem. ☺

In physical terms, the Bell–Kochen–Specker theorem, Theorem C, excludes the noncontextual hidden-variable theories that Kochen and Specker considered. Such a theory assigns, for each value of the hidden variable, a definite truth value to every experimentally verifiable proposition about a quantum system. This truth value, true or false, does not depend on the other propositions tested with that proposition. A dispersion-free measure, one with only the values 0 and 1, would record the values true and false as 1 and 0, respectively. Contextual hidden-variable theories, such as that of Bohm [Boh52], in which the value revealed by a measurement depends on the whole experimental arrangement, lie outside the scope of the theorem. In a model of T, the elements of the Solèr sort play the role of the closed subspaces, to which the experimentally verifiable propositions correspond [BvN36]. A dispersion-free measure cannot serve as the interpretation of μ: by Theorem C, the axioms of T, σ-additivity among them, exclude such a measure. Together with axiom (M1), the theorem yields the following corollary.

Corollary 9.2 (No dispersion-free measures). *In every model of T, standard or not, some element a of the Solèr sort satisfies $0 < \mu(a) < 1$.*

Proof. By (M1), the measure takes values in $[0, 1]$, and by Theorem C it does not take only the values 0 and 1. ☺

⧗ **Historical note.** Specker announced in 1960 that the closed subspaces of a three-dimensional Hilbert space cannot be mapped into a Boolean algebra in a way that respects the operations on orthogonal subspaces. Of the proof he said only that it rests on an elementary geometric argument [Spe60]. Bell derived from Gleason's theorem that no hidden-variable theory of the kind described above is possible, in a paper written in 1964 and published in 1966 [Bel66]. In 1967 Kochen and Specker gave a finite proof of this impossibility with 117 vectors in $\mathbb{R}^3$ [KS67]. The configurations then shrank: Peres found configurations of 33 vectors in $\mathbb{R}^3$ and of 24 in $\mathbb{R}^4$ in 1991 [Per91], and Kernaghan one of 20 vectors in $\mathbb{R}^4$ in 1994 [Ker94]. In 1996 Cabello, Estebaranz and García-Alcaine found the eighteen vectors of Table 1 [CEGA96]. Peres conjectured that no smaller configuration exists, and Xu, Chen and Gühne proved in 2020 that no Kochen–Specker set in any dimension has fewer than eighteen vectors [XCG20, BCG+22]. In dimension three the minimum remains unknown at the time of writing. Exhaustive computer searches raised the lower bound for this minimum to 22 vectors in 2016 [UW16] and to 24 in 2023. The bound of 24 was reached independently by Kirchweger, Peitl and Szeider and by Li, Bright and Ganesh [KPS23, LBG24]. The titles of the papers of Kernaghan and of Cabello, Estebaranz and García-Alcaine call the result the Bell–Kochen–Specker theorem, a name that records Bell's share and that Theorem C keeps.

10. The model existence game and finite obstructions

By Theorem C, the measure μ takes a value other than 0 and 1 in every model of the theory T of Theorem A, standard or not. In other words, the theory obtained from T by adjoining the axiom of Theorem C has no models. In first-order logic, compactness would turn this fact into a finite obstruction, a finite set of axioms that already has no models. Finitely many axioms would then suffice, just as finitely many vectors suffice in the Kochen–Specker theorem [KS67]. The countable suprema and infima of real-valued $L_{\omega_1\omega}$, through

which σ-additivity is expressed, carry the logic beyond the reach of compactness. In place of compactness, the logic has a model existence theorem [ADI26], which concerns Henkin types, namely the assignments of values to sentences that satisfy the Henkin conditions recalled in Section 5.2. By that theorem, every Henkin type is the type of a canonical structure for the enlarged vocabulary, one in which every element is the value of a closed term. In this section we state that theorem as a game, from which the finite obstructions arise. In the same work, the omitting types theorem is proved by an application of the Baire category theorem in the space of Henkin types. That proof, too, has a counterpart in the game (see Section A).

The game follows Väänänen's model existence game for two-valued logic [Vää11]. It is played by Olivia, who claims that a given countable theory has no models, and Josh, who claims that it has one. At each round Olivia makes a challenge, and Josh answers it by adding to a growing description of a structure (see Figure 5). Olivia wins as soon as Josh cannot answer without contradicting himself, and Josh wins if that never happens. The growing descriptions, which we call positions below, are the real-valued counterparts of the finite sets of sentences that serve as conditions in Keisler's forcing [Kei73, p. 100, Example 1.2], and the rules for infima, suprema and countable joins follow the pattern of his $\forall\exists$ formulas [Kei73, p. 107] (see Section A).

We show that the theory has no models exactly when Olivia has a winning strategy. Every winning strategy of Olivia, we show further, determines a well-founded tree of finite plays, at each leaf of which the commitments made along the play form a finite obstruction. A finite obstruction, in this sense, is a finite set of statements about the values of sentences that no structure satisfies. The tree branches infinitely only where Olivia challenges a countable supremum or infimum; once Josh's answers to those challenges are fixed, every play ends within a bounded number of rounds. The first three subsections depend only on Section 5. In the last, we return to the theory of Theorem C and describe a winning strategy of Olivia modeled on the proof of that theorem. Under this strategy, every play in which Josh makes every commitment required by the parity argument of that proof ends at a position whose commitments concern a configuration of the Kochen–Specker type.

10.1. **The setting.** Until the last subsection, we work with a countable many-sorted vocabulary $\mathcal{L}$ for the logic of Section 5 and with a nonempty countable theory T in $\mathcal{L}$. In the last subsection, the vocabulary and the theory are again those of Section 5.3 and of Theorem A. As usual, every sort of a structure is nonempty.

In the game, Josh commits the values of sentences to intervals, and the rules require for each sentence an interval fixed in advance that contains every possible value of that sentence. By the convention of Section 5, the a priori bound r_R of each predicate symbol R is a positive integer. Likewise, each countable supremum $\bigvee_n \varphi_n$ or infimum $\bigwedge_n \varphi_n$ comes, as part of its syntax, with a positive rational number r such that every φ_n takes its values in $[-r, r]$ in every structure. By induction on formulas, each formula φ receives a nonnegative rational number r_φ such that φ takes its values in the closed interval $I_\varphi = [-r_\varphi, r_\varphi]$ in every structure. The number r_φ does not change when terms are substituted for the free variables of φ. The bounds of $R(\bar{t})$ and $s \cdot \varphi$ are those of Aubone, Dueñez and Iovino [ADI26], namely r_R and $|s| r_\varphi$. From the same source come the bound $\max(r_\varphi, r_\psi)$ of $\varphi \vee \psi$ and the bound r_φ of $\sup_x \varphi$. The bound of $\bigvee_n \varphi_n$ is the number r of its family, as in Section 5; moreover, the bounds of the infima are $r_{\inf_x \varphi} = r_\varphi$ and $r_{\bigwedge_n \varphi_n} = r$. For any other continuous connective f, the number $r_{f(\varphi_1,\dots,\varphi_\ell)}$ is the least positive integer that is at least the maximum of $|f|$ on the box $I_{\varphi_1} \times \cdots \times I_{\varphi_\ell}$. In the game, the infima $\inf_x$ over a sort and the countable infima $\bigwedge_n$ are treated as primitive, with rules of their own. The formulas $s \cdot \varphi$ and $\varphi \vee \psi$, by contrast, are

handled by the rule (Cn) of Definition 10.2 as the connectives $t \mapsto st$ and $(t,u) \mapsto \max(t,u)$. Finally, the logical unity constant $\mathbf{1}$, a nullary predicate symbol, takes only the value 1. For the definition of infeasible positions below, we write $\mathcal{V}_{\mathbf{1}} = \{1\}$ and $\mathcal{V}_R = [-r_R, r_R]$ for every other predicate symbol R.

The axioms enter the game through one of its rules, which is simplest to state when all axioms have the same form. As everywhere in the paper, an axiom of T is a closed condition required of all values of its free variables. If $\bar{x}$ lists the free variables of a formula φ, then, by the conventions of Section 5, an axiom $\varphi(\bar{x}) \leq r$ is the closed condition $\sup_{\bar{x}} \varphi \leq r$ and an axiom $\varphi(\bar{x}) \geq r$ is $\inf_{\bar{x}} \varphi \geq r$. An axiom $\varphi(\bar{x}) = r$ is the union of these two closed conditions. Every axiom of T may therefore be regarded as a closed condition $\psi \leq r$ or $\psi \geq r$ for a sentence ψ and a rational number r, an axiom $\varphi(\bar{x}) = r$ counting as two. Theories that contain strict conditions are discussed in Remark 10.6.

To name the elements of a structure under construction, we add to $\mathcal{L}$ for each sort $\mathbf{S}$ a set $\mathcal{W}_{\mathbf{S}} = \{w_{\mathbf{S},i} : i \in \mathbb{N}\}$ of new constants of sort $\mathbf{S}$, the *witnesses*. We write $\mathcal{W}$ for the union of the sets $\mathcal{W}_{\mathbf{S}}$, and we denote by $\mathcal{L}_{\mathcal{W}}$ the vocabulary obtained by adding the witnesses to $\mathcal{L}$. We use the letters w and w' for witnesses. Constant symbols of $\mathcal{L}$ are treated as function symbols with no arguments, so that the rules below handle them together with the other function symbols. A sentence of $\mathcal{L}_{\mathcal{W}}$ is *basic* when it has the form $R(w_1, \dots, w_q)$, where R is a predicate symbol and $w_1, \dots, w_q$ are witnesses. The number q may be 0; in particular, the sentence $\mathbf{1}$ is basic.

In the course of a play, Josh builds, in answer to Olivia's challenges, a partial description of a structure whose elements are named by witnesses. Such a description consists of two kinds of entries, commitments and records. A commitment says that the value of a sentence lies in a given interval, whereas a record says what a function symbol does to the elements named by given witnesses. Apart from the rules that bring in the axioms and make the function symbols total, the rules of the game (see Definition 10.2) follow the inductive definition of the value of a sentence. In accordance with that definition, a commitment on a sentence formed by a continuous connective or a countable supremum or infimum is challenged through the sentences from which that sentence is formed. A commitment on a supremum or an infimum over a sort is challenged through witnesses. A commitment on an atomic sentence, finally, is challenged through records, which replace the terms of that sentence by witnesses until a basic sentence remains.

Definition 10.1. A *commitment* is an expression $\varphi \in I$, where φ is a sentence of $\mathcal{L}_{\mathcal{W}}$ and $I = [\alpha, \beta]$ is an interval with rational endpoints $\alpha \leq \beta$; the interval may be a single point. A *record* is an expression $F(\bar{w}) \mapsto w'$, where F is a function symbol and $\bar{w}, w'$ are witnesses of the appropriate sorts. A *position* is a finite set $\mathfrak{p}$ of commitments and records that contains at most one record $F(\bar{w}) \mapsto w'$ for each function symbol F and each tuple $\bar{w}$. A witness *occurs* in $\mathfrak{p}$ when it occurs in a commitment or a record of $\mathfrak{p}$. In a structure for $\mathcal{L}$, a position is *satisfied* under an interpretation of the witnesses that occur in it when each of its commitments holds and, for each of its records $F(\bar{w}) \mapsto w'$, the term $F(\bar{w})$ and the witness w' have the same value. A position $\mathfrak{p}$ is *infeasible* when there is a basic sentence $R(\bar{w})$ such that the intervals I for which $\mathfrak{p}$ contains the commitment $R(\bar{w}) \in I$, together with $\mathcal{V}_R$, have empty intersection.

An infeasible position is contradictory in the most elementary way: it assigns to a basic sentence $R(\bar{w})$ finitely many rational intervals that have no common point in $\mathcal{V}_R$ (see Figure 7). Infeasibility is the elementary conflict detected by the rules; satisfiability asks whether all the commitments and records can hold together in a structure. A position may already be unsatisfiable without being infeasible, and Olivia's subsequent challenges may expose the contradiction at the level of basic sentences.

The rule for continuous connectives asks Josh not for the exact values of the arguments of a connective but for small cells of a grid that contain those values. The grid is fine enough that the connective varies by at most a prescribed amount on each cell. The statement of the rule requires three notions: dyadic grids, a modulus of uniform continuity for each connective, and widened intervals. For $m \in \mathbb{N}$, the *dyadic grid* $\mathcal{G}_m$ is the set of intervals $[i2^{-m}, (i+1)2^{-m}]$, for $i \in \mathbb{Z}$. Let ℓ be a positive integer, let $f\colon \mathbb{R}^\ell \to \mathbb{R}$ be a continuous connective, let $\varphi_1, \dots, \varphi_\ell$ be sentences, and let $k \in \mathbb{N}$ be given. The function f is uniformly continuous on the compact box $I_{\varphi_1} \times \cdots \times I_{\varphi_\ell}$; we write $m(f, \bar{\varphi}, k)$ for the least $m \in \mathbb{N}$ such that $|f(\xi) - f(\xi')| \le 2^{-k}$ for all points ξ, ξ' of the box with $|\xi_j - \xi'_j| \le 2^{-m}$ for $j = 1, \dots, \ell$. For an interval $I = [\alpha, \beta]$ and $k \in \mathbb{N}$, we write $I^{(k)} = [\alpha - 2^{-k}, \beta + 2^{-k}]$.

When Josh chooses a witness, the rules allow witnesses already in the position and one designated fresh witness. This restriction leaves finitely many choices at each such move, while permitting a previously unused witness whenever one is needed. The choices of an index for a countable supremum or infimum retain their countable branching; this distinction is the basis of the finite-obstruction analysis in the third subsection.

Definition 10.2 (The model existence game). The game $\mathrm{G}(T)$ is played by Olivia and Josh. It starts at the empty position; at each round, Olivia makes a *challenge*, and Josh *answers* it by adding to the current position $\mathfrak{p}$ commitments and records, if any, in the manner that the rule of the challenge prescribes. When Josh answers a challenge, the *first fresh witness* of a sort $\mathbf{S}$ is the witness $w_{\mathbf{S},i}$ with the least index i that occurs neither in $\mathfrak{p}$ nor in the challenge. In each rule other than (Ax) and (Tot), the commitment named by Olivia belongs to $\mathfrak{p}$, and its interval is written $I = [\alpha, \beta]$.

(Ax): Olivia names an axiom $\psi \le r$ or $\psi \ge r$ of T. Josh adds $\psi \in [\min(-r_\psi, r), r]$ or $\psi \in [r, \max(r, r_\psi)]$, respectively.

(Tot): Olivia names a function symbol F and witnesses $\bar{w}$ of the appropriate sorts. If $\mathfrak{p}$ contains no record $F(\bar{w}) \mapsto w'$, Josh adds such a record, choosing for w' a witness of the sort of the values of F that occurs in $\mathfrak{p}$ or is the first fresh witness of that sort; otherwise he adds nothing.

(Tm): Olivia names a commitment $R(\bar{t}) \in I$, where R is a predicate symbol and $\bar{t}$ is a tuple of terms, together with a term $F(\bar{w})$ that occurs in $\bar{t}$ and in which $\bar{w}$ is a tuple of witnesses (possibly empty). Josh first answers as in (Tot) for F and $\bar{w}$; then, with w' the witness such that the enlarged position contains the record $F(\bar{w}) \mapsto w'$, he adds $R(\bar{t}') \in I$, where $\bar{t}'$ is obtained from $\bar{t}$ by replacing every occurrence of $F(\bar{w})$ by w'.

(Cn): Olivia names a commitment $f(\varphi_1, \dots, \varphi_\ell) \in I$, where f is a continuous connective, and a number $k \in \mathbb{N}$. With $m = m(f, \bar{\varphi}, k)$, Josh chooses intervals $D_1, \dots, D_\ell$ of $\mathcal{G}_m$ such that each D_j meets I_{φ_j} and f maps $(D_1 \cap I_{\varphi_1}) \times \cdots \times (D_\ell \cap I_{\varphi_\ell})$ into $I^{(k)}$, and adds the commitments $\varphi_j \in D_j \cap I_{\varphi_j}$ (see Figure 6).

($\sup \le$): Olivia names a commitment $\sup_x \varphi(x) \in I$, where x is a variable of sort $\mathbf{S}$, and a witness $w \in \mathcal{W}_{\mathbf{S}}$. Josh adds $\varphi(w) \in [-r_\varphi, \beta]$.

($\sup \ge$): Olivia names a commitment $\sup_x \varphi(x) \in I$, where x is a variable of sort $\mathbf{S}$, and a number $k \in \mathbb{N}$. Josh chooses a witness $w \in \mathcal{W}_{\mathbf{S}}$ that occurs in $\mathfrak{p}$ or is the first fresh witness of sort $\mathbf{S}$, and adds $\varphi(w) \in [\alpha - 2^{-k}, r_\varphi]$.

($\inf \ge$): Olivia names a commitment $\inf_x \varphi(x) \in I$, where x is a variable of sort $\mathbf{S}$, and a witness $w \in \mathcal{W}_{\mathbf{S}}$. Josh adds $\varphi(w) \in [\alpha, r_\varphi]$.

($\inf \le$): Olivia names a commitment $\inf_x \varphi(x) \in I$, where x is a variable of sort $\mathbf{S}$, and a number $k \in \mathbb{N}$. Josh chooses a witness w as in ($\sup \ge$), and adds $\varphi(w) \in [-r_\varphi, \beta + 2^{-k}]$.

($\bigvee \leq$): Olivia names a commitment $\bigvee_n \varphi_n \in I$, where $(\varphi_n)_{n\in\mathbb{N}}$ is a family of formulas with common rational bound $r = r_{\bigvee_n \varphi_n}$, and a number $n \in \mathbb{N}$. Josh adds $\varphi_n \in [-r, \beta]$.

($\bigvee \geq$): Olivia names a commitment $\bigvee_n \varphi_n \in I$ as in the preceding rule, and a number $k \in \mathbb{N}$. Josh chooses a number $n \in \mathbb{N}$ and adds $\varphi_n \in [\alpha - 2^{-k}, r]$.

($\bigwedge \geq$): Olivia names a commitment $\bigwedge_n \varphi_n \in I$, where $(\varphi_n)_{n\in\mathbb{N}}$ is a family of formulas with common rational bound $r = r_{\bigwedge_n \varphi_n}$, and a number $n \in \mathbb{N}$. Josh adds $\varphi_n \in [\alpha, r]$.

($\bigwedge \leq$): Olivia names a commitment $\bigwedge_n \varphi_n \in I$ as in the preceding rule, and a number $k \in \mathbb{N}$. Josh chooses a number $n \in \mathbb{N}$ and adds $\varphi_n \in [-r, \beta + 2^{-k}]$.

An answer is *legal* when it obeys the rule and every interval that it adds is nonempty, that is, has its left endpoint at most its right endpoint. Josh *loses* at a round if he has no legal answer to Olivia's challenge, or if his answer makes the position infeasible. A *play* is a finite or infinite sequence of rounds played according to these rules; it ends at the round, if any, at which Josh loses, a round that consists of Olivia's challenge alone when he has no legal answer to it. Olivia wins a play if Josh loses at some round; Josh wins a play if he never loses, in which case the play is infinite. A *strategy* for a player chooses that player's next move from the moves made so far; it is *winning* when that player wins every play in which the strategy is followed, whatever the opponent does.

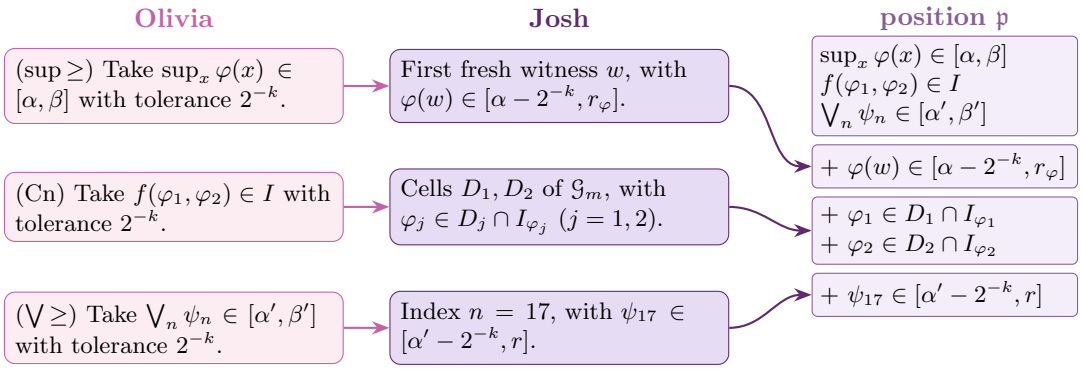


FIGURE 5. Three rounds of the game. Olivia's challenges are on the left and Josh's answers in the middle; the position, on the right, grows by one card at each answer. The first round concerns a supremum over a sort, the second a continuous connective and the third a countable supremum.

Josh's answers are of three kinds, according to the number of choices that the rules leave him. The rules (Ax), (sup $\leq$), (inf $\geq$), ($\bigvee \leq$) and ($\bigwedge \geq$) leave him no choice: the answer is determined by the challenge. Under (Tot), (Tm), (Cn), (sup $\geq$) and (inf $\leq$) he has finitely many choices because only finitely many witnesses occur in a position and only finitely many intervals of $\mathcal{G}_m$ meet a bounded interval. Only the rules ($\bigvee \geq$) and ($\bigwedge \leq$), which concern countable suprema and infima, leave him a choice among infinitely many answers. We call the challenges made under these two rules *countable challenges*, and Josh's answers to them *countable answers*.

10.2. **The model existence theorem as a game.** To extract a model from a play that Josh survives, we need Olivia to have made, sooner or later, every challenge open to her. A challenge is *available* at a round when the rules allow Olivia to make it there. Equivalently, a challenge is available at a round when the commitment that it names, if any, belongs to the position at the start of that round. Positions only grow, and a challenge that is available

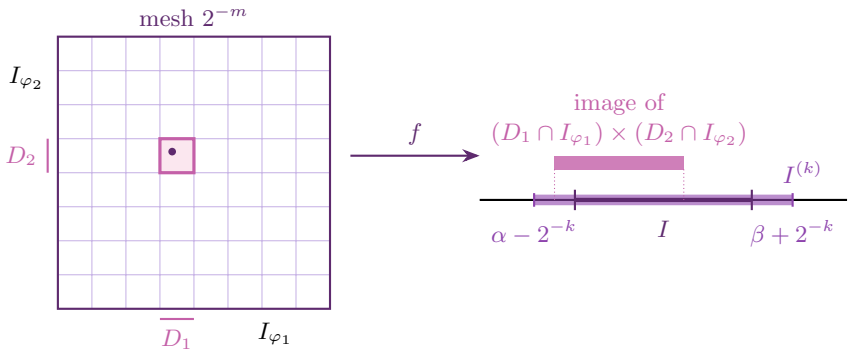


FIGURE 6. The rule (Cn) for a connective f of two variables. Josh must choose a cell $D_1 \times D_2$ of the dyadic grid of mesh 2^{-m} whose intersection with the box $I_{\varphi_1} \times I_{\varphi_2}$ is mapped by f into the widened interval $I^{(k)}$. The number k is the one named by Olivia, and the modulus of uniform continuity of f on that box, defined before Definition 10.2, is $m = m(f, \bar{\varphi}, k)$. By uniform continuity, a cell that contains the point (the dot) whose coordinates are the values of φ_1 and φ_2 in a structure satisfying the commitment $f(\varphi_1, \varphi_2) \in I$ always qualifies. The image of that cell, however, may leave I itself.

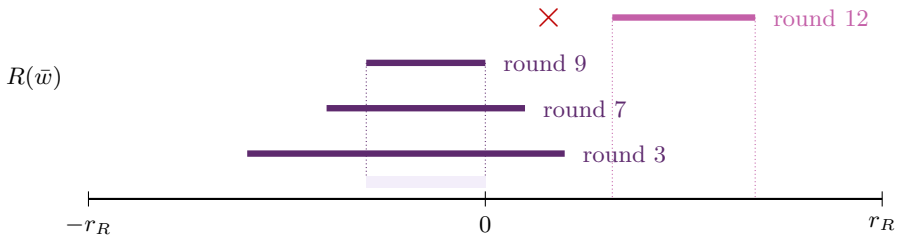


FIGURE 7. An infeasible position: at different rounds, Josh has committed the basic sentence $R(\bar{w})$ to intervals that have no common point.

at a round therefore remains available at every later round. Since a position is finite and the vocabulary $\mathcal{L}_\mathcal{W}$ and the theory T are countable, only countably many challenges are available at each round. Because T is nonempty, at least one challenge is available at each round. Moreover, every sentence occurring in a play is an atomic sentence of $\mathcal{L}_\mathcal{W}$ or is obtained from a subformula of an axiom of T by a substitution. The substitution replaces the free variables of the subformula by closed terms of $\mathcal{L}_\mathcal{W}$. The challenges that can occur in plays therefore form a countable set. A play is *fair* if every challenge that is available at some round is made by Olivia at that round or at a later one. Olivia has a *fair strategy*, one under which every infinite play is fair, whatever Josh does; we call the following one her *bookkeeping*. She fixes a bijection $i \mapsto (j(i), l(i))$ from $\mathbb{N}$ onto $\mathbb{N} \times \mathbb{N}$ with $j(i) \leq i$ for every $i \in \mathbb{N}$, and at each round she enumerates, with repetitions if necessary, the challenges available at that round. At round i she makes the $l(i)$-th challenge of her enumeration for round $j(i)$, which is still available because $j(i) \leq i$.

Theorem 10.3 (The model existence theorem as a game). *Let T be a nonempty countable theory of real-valued $L_{\omega_1\omega}$ in a countable vocabulary, whose axioms are closed conditions $\psi \le r$ or $\psi \ge r$ for sentences ψ and rational numbers r, and let $\mathrm{G}(T)$ be the game of Definition 10.2. The following conditions are equivalent:*

(1) the theory T has a model;
(2) Josh has a winning strategy in $\mathrm{G}(T)$;
(3) Olivia has no winning strategy in $\mathrm{G}(T)$.

Proof. (1) implies (2). Let $\mathfrak{N}$ be a model of T; Josh answers every challenge truthfully in $\mathfrak{N}$. To do so, he keeps along the play a map h from the witnesses occurring in the current position to $\mathfrak{N}$. The map h makes every commitment of the position hold in $\mathfrak{N}$ when each witness w is interpreted as $h(w)$. Moreover, it satisfies the equation $F^{\mathfrak{N}}(h(\bar{w})) = h(w')$ for every record $F(\bar{w}) \mapsto w'$ of the position. When a challenge names a witness that does not yet occur, Josh first extends h to it arbitrarily, which is possible because the sorts of $\mathfrak{N}$ are nonempty. The commitment that he adds under (Ax) holds because $\mathfrak{N}$ is a model of T. Under (Tot) and (Tm), if the position contains no record $F(\bar{w}) \mapsto w'$, Josh adds one with w' the first fresh witness and extends h to w' by letting $h(w') = F^{\mathfrak{N}}(h(\bar{w}))$. Under (Tm), whether the record is new or not, the value of $R(\bar{t}')$ in $\mathfrak{N}$ equals that of $R(\bar{t})$. To a challenge (Cn), Josh answers with cells $D_j \in \mathcal{G}_m$ that contain the values ξ_j of the sentences φ_j in $\mathfrak{N}$. Every point of $(D_1 \cap I_{\varphi_1}) \times \cdots \times (D_\ell \cap I_{\varphi_\ell})$ then has coordinates within 2^{-m} of those of $(\xi_1, \dots, \xi_\ell)$. By the choice of m, the connective f maps every such point to within 2^{-k} of $f(\xi_1, \dots, \xi_\ell)$, which lies in I. The commitment added under $(\sup \le)$ holds because the value of $\varphi(w)$ is at most the supremum, hence at most β. Under $(\sup \ge)$, Josh takes the first fresh witness w. Since the supremum is at least α, he lets $h(w)$ be an element of $\mathfrak{N}$ at which φ takes a value at least $\alpha - 2^{-k}$. The remaining rules are handled in the same way; under $(\bigvee \ge)$, for instance, Josh names a number n for which the value of φ_n is at least $\alpha - 2^{-k}$. Every answer is legal, since the true values lie in the added intervals. Moreover, no position becomes infeasible, since the value in $\mathfrak{N}$ of a basic sentence $R(\bar{w})$ lies in every interval committed for that sentence and in $\mathcal{V}_R$. Since both of these facts hold at every round, Josh never loses.

(2) implies (1). Let Josh play a winning strategy against a fair strategy of Olivia. The resulting play is infinite and fair; let $\mathfrak{p}_0 \subseteq \mathfrak{p}_1 \subseteq \cdots$ be its positions, indexed by $i \in \mathbb{N}$, and let $\mathfrak{p}_\infty$ be their union. We define a structure $\mathfrak{M}$ for $\mathcal{L}_{\mathcal{W}}$ whose elements are the witnesses: the universe of sort $\mathbf{S}$ is $\mathcal{W}_{\mathbf{S}}$, and each witness is interpreted as itself. For a function symbol F and witnesses $\bar{w}$, Olivia makes the challenge (Tot) for F and $\bar{w}$ at some round, because the play is fair. The union $\mathfrak{p}_\infty$ therefore contains a record $F(\bar{w}) \mapsto w'$, which is unique because every $\mathfrak{p}_i$ is a position, and we define $F^{\mathfrak{M}}(\bar{w})$ to be w'. For a basic sentence $R(\bar{w})$, the intervals I for which $\mathfrak{p}_\infty$ contains $R(\bar{w}) \in I$ are compact, and so is $\mathcal{V}_R$. Every finite subfamily of the family formed by these intervals and $\mathcal{V}_R$ has a common point, because no position $\mathfrak{p}_i$ is infeasible. By the finite intersection property, the whole family has a common point, and we let $R^{\mathfrak{M}}(\bar{w})$ be one. As a point of $\mathcal{V}_R$, this value respects the a priori bound of R, and it equals 1 when R is $\mathbf{1}$. The logic requires of the interpretations $F^{\mathfrak{M}}$ and $R^{\mathfrak{M}}$ only that they have the appropriate sorts and, for $R^{\mathfrak{M}}$, that it respect the a priori bound of R (see Section 5). The interpretations chosen above therefore complete the definition of $\mathfrak{M}$.

We claim that every commitment $\varphi \in I$ of $\mathfrak{p}_\infty$ holds in $\mathfrak{M}$. The claim is proved by induction on the construction of φ and, for atomic sentences, on the number of occurrences of function symbols. For a basic sentence, the claim holds by the choice of $R^{\mathfrak{M}}$. A non-basic atomic sentence $R(\bar{t})$ contains a term $F(\bar{w})$ whose arguments $\bar{w}$ are witnesses (possibly

none, when F is a constant symbol). By fairness, Olivia made the challenge (Tm) for the commitment $R(\bar{t}) \in I$ and this term, and Josh added $R(\bar{t}') \in I$, which holds by the induction hypothesis. The sentences $R(\bar{t})$ and $R(\bar{t}')$ have the same value in $\mathfrak{M}$, because the witness w' that replaces $F(\bar{w})$ is $F^{\mathfrak{M}}(\bar{w})$. For $f(\varphi_1, \dots, \varphi_\ell) \in I$, fairness gives, for every $k \in \mathbb{N}$, intervals $D_1, \dots, D_\ell$ such that $\mathfrak{p}_\infty$ contains the commitments $\varphi_j \in D_j \cap I_{\varphi_j}$ and the connective f maps $(D_1 \cap I_{\varphi_1}) \times \cdots \times (D_\ell \cap I_{\varphi_\ell})$ into $I^{(k)}$. By the induction hypothesis, the value of $f(\varphi_1, \dots, \varphi_\ell)$ in $\mathfrak{M}$ lies in $I^{(k)}$ for every k, hence in I. For $\sup_x \varphi(x) \in [\alpha, \beta]$, fairness puts in $\mathfrak{p}_\infty$ a commitment $\varphi(w) \in [-r_\varphi, \beta]$ for every witness w of the sort of x, that is, for every element of $\mathfrak{M}$ of that sort. Fairness also puts in $\mathfrak{p}_\infty$, for every $k \in \mathbb{N}$, a commitment $\varphi(w) \in [\alpha - 2^{-k}, r_\varphi]$ for some witness w. By the induction hypothesis, the supremum lies in $[\alpha, \beta]$. The remaining cases of the induction are similar. Finally, every axiom $\psi \leq r$ of T was named in a challenge (Ax), to which Josh answered with the commitment $\psi \in [\min(-r_\psi, r), r]$. By the claim, the structure $\mathfrak{M}$ satisfies $\psi \leq r$; the axioms $\psi \geq r$ are treated in the same way.

It remains to pass to a reduced structure. For witnesses w_1, w_2 of the same sort, let $w_1 \sim w_2$ when $\varphi(w_1, \bar{w})$ and $\varphi(w_2, \bar{w})$ have the same value in $\mathfrak{M}$ for every atomic formula $\varphi(x, \bar{y})$ of $\mathcal{L}$ and every tuple $\bar{w}$ of witnesses. If $w_1 \sim w_2$, then the witnesses $F^{\mathfrak{M}}(w_1, \bar{w}')$ and $F^{\mathfrak{M}}(w_2, \bar{w}')$ are equivalent for every function symbol F and every tuple $\bar{w}'$ of witnesses, since $\varphi(F(x, \bar{y}'), \bar{y})$ is again an atomic formula. The same holds at every argument place of F, and by transitivity the relation $\sim$ is a congruence for every function symbol. The quotient $\mathfrak{M}/\!\sim$ is reduced, and the quotient map is surjective and preserves the value of every atomic formula; by induction on formulas, it preserves the value of every formula. Hence the reduct of $\mathfrak{M}/\!\sim$ to $\mathcal{L}$ is a model of T.

(2) and (3) are equivalent. Regard $\mathrm{G}(T)$ as an infinite game in which, after Josh has lost, the play continues with arbitrary moves. Olivia wins exactly when Josh loses at some finite round. In the language of descriptive set theory, the set of her wins is *open* in the space of infinite sequences of moves. The basic open sets of that space consist of the sequences that extend a given finite one. An infinite game of perfect information is *determined* when one of the two players has a winning strategy. The Gale–Stewart theorem [GS53] says that such a game is determined whenever the set of wins of one player is open. Since the two players cannot both have winning strategies, (2) and (3) are equivalent.

Remark 10.4 (Baire category and the Banach–Mazur game). Olivia's bookkeeping can be read as a Baire category argument in the space of plays, and as a winning strategy in a Banach–Mazur game on that space. We set out both readings in Section A, beside the Baire category argument of Aubone, Dueñez and Iovino [ADI26] in the space of Henkin types.

10.3. **Finite obstructions.** A winning strategy for Olivia is a certificate that a theory has no models, and its shape shows where finite obstructions suffice. Let σ be a strategy for Olivia. The *tree of* σ, denoted $\mathcal{P}_\sigma$, is the set of finite plays in which Olivia follows σ and Josh has not lost before the last round, ordered by extension. A finite play of $\mathcal{P}_\sigma$ in which Josh has not lost has as children its extensions by Olivia's next challenge and a legal answer of Josh. When Josh has no legal answer to that challenge, the extension of the play by the challenge alone is also a child of the play. A *leaf* of $\mathcal{P}_\sigma$ is a finite play with no children, that is, one at which Josh has lost. The strategy σ is winning exactly when $\mathcal{P}_\sigma$ has no infinite branch, that is, when $\mathcal{P}_\sigma$ is *well-founded*. The *rank* of a well-founded tree is defined by recursion: a leaf has rank 0, and every other node has as rank the least ordinal exceeding the ranks of all its children. The rank of the tree is the rank of its root. The tree $\mathcal{P}_\sigma$ is countably branching,

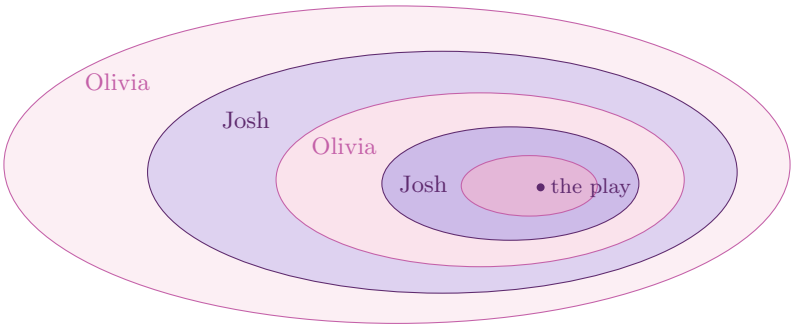


FIGURE 8. A play of the game as a sequence of nested basic open sets. Olivia (pink) and Josh (purple) move alternately, and each move, like a move of the Banach–Mazur game, shrinks the set of infinite plays that extend the current finite play. For an infinite play, the intersection of the nested sets consists of that play alone.

hence countable; when it is well-founded, its rank is therefore a countable ordinal. That rank is finite exactly when the plays in which Olivia follows σ have bounded length.

In Corollary 10.5 below, we show that $\mathcal{P}_\sigma$ branches infinitely only at nodes where σ makes a countable challenge; the following notions keep track of Josh's answers to such challenges. The *countable record* of a node of $\mathcal{P}_\sigma$ is the finite sequence of the numbers n that Josh names in his countable answers along the branch from the root to the node. The numbers appear in that sequence in the order in which Josh names them. The *skeleton* of σ is the set of countable records of nodes of $\mathcal{P}_\sigma$, ordered by extension; it is a tree of finite sequences of natural numbers.

Corollary 10.5 (Finite obstructions). *Let T be a nonempty countable theory of real-valued $L_{\omega_1\omega}$ in a countable vocabulary, whose axioms are closed conditions $\psi \le r$ or $\psi \ge r$ for sentences ψ and rational numbers r, and suppose that T has no models. Then Olivia has a winning strategy in $\mathrm{G}(T)$, and every winning strategy σ of Olivia has the following properties:*

(1) The tree $\mathcal{P}_\sigma$ is well-founded: every play in which Olivia follows σ ends after finitely many rounds, at a leaf.

(2) At every leaf, the last position $\mathfrak{p}$ is a finite set of commitments and records, involving finitely many axioms of T and finitely many witnesses, that is satisfied in no structure under any interpretation of the witnesses. Either $\mathfrak{p}$ is infeasible, in which case a basic sentence $R(\bar{w})$ receives finitely many rational intervals with no common point in $\mathcal{V}_R$, or Josh has no legal answer to Olivia's last challenge, which names a commitment of $\mathfrak{p}$ that holds in no structure.

(3) A node of $\mathcal{P}_\sigma$ has infinitely many children only if σ makes a countable challenge at it; every other node has finitely many children.

(4) If every node of $\mathcal{P}_\sigma$ has finitely many children, as is the case when σ makes no countable challenge, and in particular when no axiom of T contains a countable supremum or infimum, then the tree $\mathcal{P}_\sigma$ is finite. In that case there is a number N_σ such that every play in which Olivia follows σ ends within N_σ rounds, and some finite set of axioms of T has no models.

(5) The skeleton of σ is well-founded, and for every element $\vec{n}$ of the skeleton there is a number $N_\sigma(\vec{n})$ such that every play in which Olivia follows σ and whose countable record is $\vec{n}$ ends within $N_\sigma(\vec{n})$ rounds.

Proof. By Theorem 10.3, Olivia has a winning strategy; let σ be any winning strategy of Olivia. Part (1) holds because σ is winning. For part (2), recall that a position contains finitely many commitments and records, and consider one that is satisfied in some structure under some interpretation of the witnesses. The argument for "(1) implies (2)" in the proof of Theorem 10.3 shows that such a position is not infeasible and allows Josh a legal answer to every challenge. The same argument shows that Josh has a legal answer to a challenge whenever the commitment that the challenge names holds in some structure under some interpretation of its witnesses. Under (Ax), (Tot) and (Tm), Josh always has a legal answer. Part (3) follows from the count of Josh's possible answers in the paragraph after Definition 10.2.

Parts (4) and (5) rest on Kőnig's lemma: an infinite tree in which every node has finitely many children has an infinite branch. For part (4), the tree $\mathcal{P}_\sigma$ is well-founded and by hypothesis finitely branching; hence it is finite, and the length of its longest branch gives N_σ. By part (3), the hypothesis holds when σ makes no countable challenge. In particular, the hypothesis holds when no axiom of T contains a countable supremum or infimum. Indeed, every sentence that occurs in a play is atomic or is an instance of a subformula of an axiom. It remains to be shown that some finite set of axioms of T has no models. Let T_σ be the set of axioms named in challenges (Ax) at nodes of $\mathcal{P}_\sigma$, together with one axiom of T if there are none. In $\mathrm{G}(T_\sigma)$, every challenge that σ makes at a node of $\mathcal{P}_\sigma$ is allowed by the rules, and Josh has the same answers as in $\mathrm{G}(T)$. The strategy σ therefore wins $\mathrm{G}(T_\sigma)$, and the finite theory T_σ has no models by Theorem 10.3.

For part (5), let $\vec{n}$ be a finite sequence of natural numbers, and let $\mathcal{P}_\sigma^{\vec{n}}$ be the set of nodes of $\mathcal{P}_\sigma$ whose countable record is an initial segment of $\vec{n}$. The set $\mathcal{P}_\sigma^{\vec{n}}$ is a subtree of $\mathcal{P}_\sigma$, and every play in which Olivia follows σ and whose countable record is $\vec{n}$ runs inside that subtree. In $\mathcal{P}_\sigma^{\vec{n}}$, a node at which σ makes a countable challenge has at most one child. If Josh has no legal answer to the challenge, its only child is the extension by the challenge alone. Otherwise, if the countable record of the node is $\vec{n}$ itself, the node has no child in $\mathcal{P}_\sigma^{\vec{n}}$. In the remaining case, the only child of the node is the extension by the countable answer whose number is the next entry of $\vec{n}$. By part (3), every other node of $\mathcal{P}_\sigma^{\vec{n}}$ has finitely many children. Since $\mathcal{P}_\sigma^{\vec{n}}$ is well-founded, it is finite by Kőnig's lemma, and the length of its longest branch gives $N_\sigma(\vec{n})$. Finally, suppose, towards a contradiction, that the skeleton has an infinite branch $\vec{n}_1 \subsetneq \vec{n}_2 \subsetneq \cdots$. The set of nodes of $\mathcal{P}_\sigma$ whose countable record is an initial segment of some $\vec{n}_i$ then forms a finitely branching subtree of $\mathcal{P}_\sigma$. This subtree contains nodes of every countable record $\vec{n}_i$ and is therefore infinite. By Kőnig's lemma, it has an infinite branch, contradicting part (1); the skeleton is therefore well-founded. ☺

For countable theories of the finitary part of the logic, part (4) of Corollary 10.5 yields the compactness theorem. That theorem asserts that if a countable theory of closed conditions without countable suprema and infima has no models, then some finite set of its axioms has no models. Part (3) locates the infinitary character of the obstruction: only at the countable challenges, where Josh chooses an index n among infinitely many, can the tree branch infinitely. By part (5), the skeleton (a well-founded countably branching tree) records the whole infinitary shape of the obstruction: once Josh's countable answers are fixed, the plays have bounded length (see Figures 9 and 10). For a theory with countable suprema and infima, the single finite obstruction of the compact case thus gives way to a well-founded tree of finite obstructions, one at each leaf.

Remark 10.6 (Strict conditions). The conditions of the logic are strict inequalities, and a closed condition is a countable theory of strict ones (see Section 5). For a theory that contains strict conditions, the game has one more rule.

(Str): Olivia names a condition $\varphi < r$ of T, for a sentence φ. Josh chooses a rational number $\varrho < r$ and adds $\varphi \in [-r_\varphi, \varrho]$. For a condition $\varphi > r$ of T, Josh chooses a rational number $\varrho > r$ and adds $\varphi \in [\varrho, r_\varphi]$.

With one more case for this rule, the proof of Theorem 10.3 applies to theories that consist of closed conditions as above and of strict conditions on sentences. In the additional case, a truthful Josh chooses ϱ between the value of φ and r, and in the structure built from a fair play the commitment $\varphi \in [-r_\varphi, \varrho]$ gives $\varphi \le \varrho < r$. Under (Str), however, Josh chooses among infinitely many answers, and the challenges (Str) are accordingly counted as countable challenges in Corollary 10.5. In the countable record, Josh's rational answers are coded by natural numbers; parts (1), (3) and (5) of the corollary then hold. Part (2) holds once the condition named in the last challenge is added to $\mathfrak{p}$. Indeed, Josh has no legal answer to (Str) for $\varphi < r$ exactly when $r \le -r_\varphi$, and none for $\varphi > r$ exactly when $r \ge r_\varphi$. In either case, the condition named is itself unsatisfiable. With the same counting, part (4) holds, except for its clause on theories without countable suprema and infima. The exception arises because a challenge (Str) with a legal answer has infinitely many children and because such theories may have no models although each finite set of their axioms has one. For a nullary predicate symbol R with $r_R > 0$, the theory $\{R > 0\} \cup \{R < \frac{1}{n} : n \ge 1\}$ is such a theory. Olivia wins by naming first $R > 0$, to which Josh answers with some rational $\varrho > 0$ and the commitment $R \in [\varrho, r_R]$, and then a condition $R < \frac{1}{n}$ with $\frac{1}{n} < \varrho$. The tree of this strategy branches infinitely at its root. The contrast between the two kinds of conditions has a topological counterpart in the space of Henkin types of Aubone, Dueñez and Iovino [ADI26], which we describe in Section A.

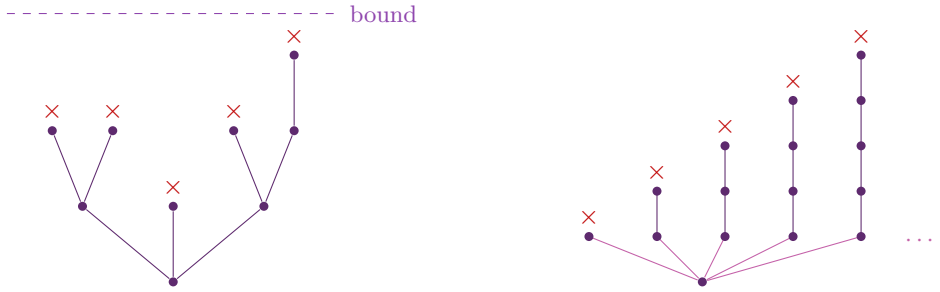


FIGURE 9. Trees grow upward from their roots. Left: a finitely branching tree without infinite branches is finite, by Kőnig's lemma, and its branches have bounded length. Right: a countably branching tree without infinite branches may have branches of every finite length, as in this fan with one branch of each length.

10.4. **The Bell–Kochen–Specker obstruction.** We now return to the theory T of Theorem A, with its measure μ, and to the axiom of Theorem C, which says that μ takes only the values 0 and 1. In the normal form described in the first subsection, that axiom becomes $\sup_a\big(\mu(a) \wedge (\mathbf{1} - \mu(a))\big) \le 0$. By Theorem C, the theory obtained by adding the resulting

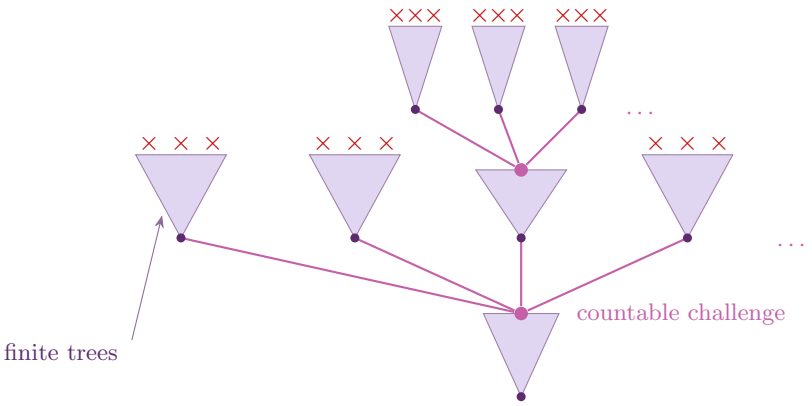


FIGURE 10. The tree of a winning strategy for Olivia, growing upward from its root. At each countable challenge (pink dot) to which Josh has a legal answer, the tree branches infinitely (pink fan), one branch for each number that he may name. The sequences of numbers that Josh names along the branches form the skeleton. The parts of the tree between countable challenges are finite (shaded), and at each of their leaves, marked $\times$, the commitments made along the play that ends there form a finite obstruction.

axiom to T has no models, and the conclusions of Corollary 10.5 hold for it. We show that the tree of every winning strategy of Olivia for this theory branches infinitely, and we describe a winning strategy, modeled on the proof of Theorem C. Under that strategy, every play in which Josh makes all the commitments required by the parity argument of Theorem C ends at a position whose commitments concern a configuration of the Kochen–Specker type.

Proposition 10.7. *Let T^{01} be the theory T of Theorem A together with the axiom*

$$\sup_a \big(\mu(a) \wedge (\mathbf{1} - \mu(a))\big) \leq 0.$$

(1) Olivia has a winning strategy in $\mathrm{G}(T^{01})$; every play in which she follows it ends after finitely many rounds, at a position that is satisfied in no structure.
(2) Every finite set of axioms of T^{01} has a model.
(3) The skeleton of every winning strategy of Olivia in $\mathrm{G}(T^{01})$ has more than one element: in some play in which Olivia follows such a strategy, Josh answers a countable challenge.

Proof. Part (1) follows from parts (1) and (2) of Corollary 10.5, since T^{01} has no models by Theorem C.

For part (2), let a finite set of axioms of T^{01} be given; the numbers n for which this set contains the instance $\kappa_n\varepsilon \doteq 2^{-n-1}e_n$ of axiom (K4) form a finite set $Z \subseteq \mathbb{N}$. In the proof of Theorem C, it is the instances of (K4) taken together that make the entries of ε span the whole space. In that proof, the σ-additivity of μ along ε then gives some atom the measure 1. The given finite set contains only finitely many instances of (K4). In a structure whose sequences are all finitely supported, axiom (M3) involves only finite-dimensional subspaces, on which a measure that vanishes on every finite-dimensional subspace satisfies that axiom. Such a structure is the following. Its Hilbert sort is the closed unit ball of the space ℓ^2 of square-summable complex sequences, and its Solèr sort is the lattice L_{fc} of Example 7.6. The symbols e, π, at, rep and the constants e_n are interpreted as in that example. The

sequence sort consists of the finitely supported sequences in the closed unit ball of the space $\ell^2(\mathbb{N}, \ell^2)$ of square-summable sequences in ℓ^2. The interpretations of ι_n and κ_n are those of the proof of Theorem 8.3, and the function span sends each element s of the sequence sort to the finite-dimensional span of its entries. The measure μ vanishes on the finite-dimensional subspaces and equals 1 on their orthogonal complements. The constant ε is the finitely supported sequence whose entry of index n is $2^{-n-1}e_n$ for $n \in Z$ and 0 otherwise; its norm is less than 1. The axioms of T_0 hold as in Example 7.6, and the axioms (K1)–(K3) and (Sp) hold as in the proof of Theorem 8.3, since the span of finitely many vectors is closed. Of the instances of (K4), the only axiom in which ε occurs, exactly those with $n \in Z$ hold. The clause on μ in (OL2) holds because e is interpreted as equality; axioms (M1), (M2) and the axiom $\sup_a\big(\mu(a) \wedge (\mathbf{1} - \mu(a))\big) \leq 0$ hold because μ takes only the values 0 and 1, with $\mu(1) = 1$. For (M3), let s be an element of the sequence sort. If the entries of s are mutually orthogonal, the element $\mathrm{span}(s)$ and the elements $\mathrm{at}(\kappa_n s)$ are finite-dimensional subspaces, and both sides of (M3) vanish. Otherwise the right side of (M3) is 2, and the left side is at most 2. The structure is therefore a model of the given finite set of axioms.

For part (3), suppose that the skeleton of a winning strategy σ of Olivia had only one element, the empty sequence. Then Josh would give no countable answer at any node of $\mathcal{P}_\sigma$: a node at which σ makes a countable challenge would have only one child, the extension by the challenge alone. Every other node would have finitely many children by Corollary 10.5(3). By Corollary 10.5(4), the theory T^{01} would then have a finite set of axioms without models, contradicting part (2). ☺

Proof of Theorem D. In the model existence game, Olivia is the player who claims that the theory has no models, and the tree of plays of a strategy σ of hers is the tree $\mathcal{P}_\sigma$. Let T' be a countable theory that has no models, with its axioms put in the normal form of the first subsection. The theory T' is then nonempty, since the empty theory has models, and the assertions of Theorem D about T' are those of Theorem 10.3 and Corollary 10.5, with T' in place of T. The theory of Theorem C, written in this normal form, is the theory T^{01} of Proposition 10.7. By part (2) of that proposition, every finite set of axioms of T^{01} has a model, whose measure takes only the values 0 and 1. It follows from part (3) that Josh gives a countable answer at some node of the tree of every winning strategy of Olivia. The interval that Josh adds under the rules $(\bigvee \geq)$ and $(\bigwedge \leq)$ does not depend on the number he names. Every number is therefore a legal answer at that node, and the tree branches infinitely there. ☺

We describe informally a winning strategy for Olivia that follows the proof of Theorem C (see Figures 11 and 12). The strategy has two phases, the first of which finds a basis vector whose atom has measure close to 1. The second phase runs the parity argument of Theorem C in the block of four basis vectors that contains this basis vector. Recall that the constant ε names the sequence $(2^{-n-1}e_n)_{n\in\mathbb{N}}$ built from the constants e_n, which form an orthonormal basis. Recall also that the function at sends a nonzero vector x to an element of the Solèr sort whose members are multiples of x (see Lemma 9.1(1)). Olivia begins with the instance of axiom (M3) at ε and challenges that instance together with instances of the axioms on which Lemmas 6.3 and 9.1 and the proof of Theorem C rest. Among these axioms are (H1), (H2), (OL2), (P1)–(P4), (Sub0), (At1), (At5), (K2), (K4), (Sp1), (Sp2), (M1) and (M2). Through these challenges she obliges Josh to commit the sum of the values $\mu(\mathrm{at}(\kappa_n\varepsilon))$ to an interval close to 1. She makes a countable challenge $(\bigvee \geq)$ to the join over N of the truncated partial sums $\sum_{n<N} \mu(\mathrm{at}(\kappa_n\varepsilon))$ in (M3). This challenge forces Josh to name a number N for which the finite sum $\sum_{n<N} \mu(\mathrm{at}(\kappa_n\varepsilon))$ is close to 1. Challenges to the axiom of T^{01} at

witnesses that name the elements $\mathrm{at}(\kappa_n\varepsilon)$, for $n < N$, are followed by challenges (Cn) and by challenges to (M1). Together, these challenges make Josh commit each value $\mu(\mathrm{at}(\kappa_n\varepsilon))$ to an interval close to 0 or close to 1. Olivia chooses the tolerance of these challenges after Josh has named N, and she can take it finer than $1/(2N)$. Consequently, the finite sum forces at least one $n_0 < N$ for which $\mu(\mathrm{at}(\kappa_{n_0}\varepsilon))$ is close to 1. Through (K4) and the axioms behind Lemma 9.1(1), the element $\mathrm{at}(\kappa_{n_0}\varepsilon)$ is identified with $\mathrm{at}(e_{n_0})$. Further countable challenges arise, here and in the second phase, from the countable suprema in (H2) and in the formula od that precedes (M3). In both phases, the countable suprema in the uses of the formula $\lceil\varphi\rceil$ of (2) in (P4), (At5), (Sp2) and (M3) give rise to further countable challenges as well.

In the second phase, Olivia turns to the block of four basis vectors $e_{4j}, \dots, e_{4j+3}$, where the index $j \in \mathbb{N}$ satisfies $4j \le n_0 < 4j + 4$. She challenges the instances of the axioms that the proof of Theorem C uses for the nine bases of Table 1 and for the block itself. As in that proof, we write $\mathrm{at}(v)$ for $\mathrm{at}(\frac{1}{16}v)$ when v is one of the eighteen vectors of the table. Josh's answers commit each of the eighteen values $\mu(\mathrm{at}(v))$ to an interval close to 0 or close to 1. The same answers commit each of the nine sums over a basis to an interval close to 1. Whatever finitely many choices Josh makes, the parity argument in the proof of Theorem C shows that these commitments are incompatible. In that proof, each of the nine bases $\mathcal{B}$, with vectors $v_0, \dots, v_3$, gives an element $s_{\mathcal{B}}$ of the sequence sort whose entries are $\frac{1}{16}v_0, \dots, \frac{1}{16}v_3$ followed by zeros. The instances of (K2) and (At5) identify each element $\mathrm{at}(\kappa_i s_{\mathcal{B}})$, for $i < 4$, with $\mathrm{at}(v_i)$, and the instances of (P4) identify the nine elements $\mathrm{span}(s_{\mathcal{B}})$ with one another. The clause on μ in (OL2) carries the values of μ along these identifications. Through these instances of (K2), (At5), (P4) and (OL2), Olivia can then reduce the incompatibility to a basic sentence that receives intervals with no common point. Thus every play in which Josh makes all the commitments that the parity argument requires ends at a position whose commitments concern a finite configuration of the Kochen–Specker type. That configuration lies inside one block of four basis vectors, the block that Josh's answer N bounds and his finitely many choices of intervals select.

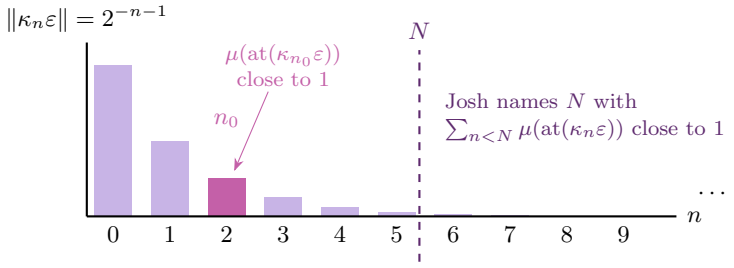


FIGURE 11. The decisive countable answer. The entries of ε have norms 2^{-n-1}; Josh names N with $\sum_{n<N} \mu(\mathrm{at}(\kappa_n\varepsilon))$ close to 1. The axiom of T^{01}, together with (M1), then forces some $n_0 < N$ with value close to 1.

Josh gives countable answers in some plays of every winning strategy of Olivia in $\mathrm{G}(T^{01})$, by Proposition 10.7(3). By Corollary 10.5(5), the plays of such a strategy have bounded length once those answers are fixed. Whether the length can be bounded independently of them remains open.

Question 10.8. Does Olivia have a winning strategy in $\mathrm{G}(T^{01})$ whose tree has finite rank, that is, such that every play in which she follows it ends within a fixed number of rounds?

Remark 10.9 (Standard models). The game $\mathrm{G}(T)$ concerns all models of T, standard or not, whereas Theorem B concerns the standard models, which are singled out by a hypothesis

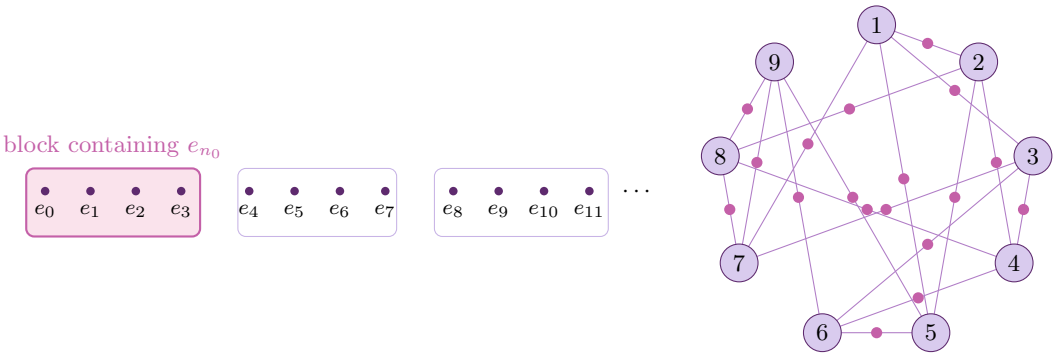


FIGURE 12. The finite obstruction at a leaf. Left: the basis vectors e_n grouped in blocks of four, with the block that contains e_{n_0} highlighted (here $n_0 = 2$, as in Figure 11). Right: the nine bases of Table 1 (numbered circles) and their eighteen rays (dots), each ray drawn on the segment that joins the two bases containing it. The element a_F of the proof of Theorem C, which stands for the span of the block, has measure 1. A $\{0,1\}$-valued measure would therefore give the value 1 to exactly one ray in each basis. The pairs formed by a basis and its ray of value 1 would then number nine when counted basis by basis. Counted ray by ray, their number would be even, since each ray lies in exactly two bases.

on the model rather than by axioms. For a Borel set B that meets $[0,1]$, let T_B be the theory T together with the axiom $\sup_a \beta_B(a) \leq -1$, the normal form of the axiom of Theorem B. Here the formula $\beta_B(a)$, given by Lemma 5.1, takes the value 1 at a when $\mu(a) \in B$ and -1 otherwise. By Theorem 10.3, Olivia has a winning strategy in $\mathrm{G}(T_B)$ exactly when T_B has no models at all, standard or not. This is the case when B contains 0 or 1, by Lemma 9.1(3) and (M2), and when B contains $(0,1)$, by Corollary 9.2. Other Borel sets B give theories with models. Indeed, the range of μ in a countable model $\mathfrak{N}$ of T (see Remark 11.2) is countable and contains 0 and 1 and, by Corollary 9.2, a point of $(0,1)$. For the complement B of that range in $[0,1]$, the structure $\mathfrak{N}$ is a model of T_B. We do not know which of the remaining Borel sets B give theories T_B with models.

Historical note. The Banach–Mazur game began as Problem 43 of the Scottish Book, posed by Mazur. On 4 August 1935 Banach wrote in the book that Mazur's conjecture is true, but Banach's proof was never published, and Oxtoby published one in 1957 [Mau81, Oxt57]. Gale and Stewart proved in 1953 that open and closed games are determined [GS53]; Dénes Kőnig had published his lemma on finitely branching trees in 1927 [Kőn27]. The witnesses of the game descend from the new constants of Henkin's completeness proof of 1949 [Hen49]. For Väänänen there are essentially three kinds of games in logic, the semantic game, the model existence game and the Ehrenfeucht–Fraïssé game; his book of 2011 is built on them [Vää11].

11. REMARKS

In these remarks we explain why the vocabulary links vectors to elements of the Solèr sort through projections, and why standardness is essential to part (1) of Theorem A and to Theorem B. We also explain why axiom (K4) is essential to Theorem C, and we then describe the changes needed for real and quaternionic scalars. Finally, we record the logical form of the axioms and the redundancies among them.

Remark 11.1 (Projections rather than membership). The vocabulary links a vector and an element of the Solèr sort through the projection π rather than through a membership relation. In terms of π, a vector x belongs to an element a of the Solèr sort exactly when $\|\frac{1}{2}x - \frac{1}{2}\pi(a,x)\| = 0$. The choice matters for joins: with a membership relation, the members of $a \vee b$ would naturally be described existentially. These members would be described as the vectors x with $x = y + z$ for some member y of a and some member z of b. In the logic, an existential statement of this kind is expressed by the vanishing of an infimum, here the infimum of a scaled norm of $x - y - z$ over such y and z. Consider a standard model whose Hilbert sort is the closed unit ball of a Hilbert space H. In that model, the infimum of the scaled norm of $x-y-z$ vanishes exactly at the sums $y+z$ of such members, since these sums form a closed set. Indeed, bounded sequences in the closed subspaces S_a and S_b spanned by the members of a and of b have weakly convergent subsequences. When $S_a + S_b$ is not closed, these sums therefore miss the vectors of the unit ball that lie in the join $\overline{S_a + S_b}$ of S_a and S_b in $L(H)$ but not in $S_a + S_b$. Axiom (Sub3) reaches the join through the projections instead, by von Neumann's alternating projection theorem. The functions at and rep provide the bijection between the rays of H, which correspond to the unit vectors up to a phase, and the atoms of the Solèr sort (see Proposition 7.4). Completeness of the Solèr sort is a theorem rather than an axiom: in standard models it follows from the axioms of T_1 (see Theorem 8.3).

Remark 11.2 (Nonstandard models). Standardness is essential to Theorem B and to part (1) of Theorem A: by the downward Löwenheim–Skolem theorem, the theory T has countable models. In each of these models, the range of μ is countable and the Solèr sort, being countable, is not isomorphic to the lattice of closed subspaces of an infinite-dimensional Hilbert space.

In a nonstandard model of T_0, even the conclusion of Gleason's theorem for T_0 (see Theorem 7.7) can fail. Let E_0 be the space of finitely supported complex sequences with the inner product of ℓ^2. We denote by Λ_0 the lattice of finite-dimensional subspaces of E_0 and their orthogonal complements in E_0 (see Example 4.3). Take the closed unit ball of E_0 as Hilbert sort, the standard unit vectors as the constants e_n for $n \in \mathbb{N}$, and the lattice Λ_0 as Solèr sort, with e interpreted as equality. The function π sends (a, x) to the orthogonal projection of x onto a, and the function at sends x to $\mathbb{C}x$. Finally, the function rep assigns to each $a \neq \{0\}$ a unit vector $\mathrm{rep}(a)$ chosen in a, and to $\{0\}$ the vector 0. The resulting structure is a model of T_0 that is not standard. Every maximal orthonormal family of E_0 is countable and has join 1 in Λ_0; a σ-additive state on Λ_0 would therefore yield a frame function on E_0. Here a frame function on E_0 is a function on the unit vectors subject to two conditions. Its sum over every maximal orthonormal family is 1, and its restriction to the unit vectors of every finite-dimensional subspace has the same sum over all the orthonormal bases of that subspace. By a theorem of Dvurečenskij [Dvu95, Theorem 3.1] (compare Hamhalter and Pták [HP87]), an inner product space that carries such a function is complete, which E_0 is not. The lattice Λ_0 therefore has no σ-additive states. On the other hand, every density operator ρ on ℓ^2 defines a map $a \mapsto \mathrm{Tr}(\rho P_a)$ on Λ_0, with P_a the projection of ℓ^2 onto the closure of a. The conclusion of Theorem 7.7, which identifies the σ-additive states with these maps, therefore fails for this nonstandard model.

In standard models, the axioms of Theorem B exclude every range other than $[0, 1]$, one Borel set at a time, or more generally one σ-projective set at a time (see Theorem 8.6). No single sentence, however, says that the range of μ is disconnected. Indeed, suppose that a sentence ψ of real-valued $L_{\omega_1\omega}$ in the vocabulary of T took the value 1 exactly in the models of T in which the range of μ is disconnected. Let $\mathfrak{M}$ be a standard model of T, which exists by Theorem A(2). By Theorem B, the range of μ in $\mathfrak{M}$ is $[0, 1]$, and so the value of ψ in

$\mathfrak{M}$ is not 1. By the downward Löwenheim–Skolem theorem, the model $\mathfrak{M}$ has a countable elementary substructure $\mathfrak{N}$ for a countable fragment containing T and ψ. The value of ψ in $\mathfrak{N}$ is also not 1, yet the range of μ in $\mathfrak{N}$ is countable and contains $\mu(0) = 0$ and $\mu(1) = 1$, whence it is disconnected, a contradiction.

We leave open whether the measure μ of a nonstandard model of T can differ from every map $a \mapsto \mathrm{Tr}(\rho \bar{P}_a)$ with ρ a density operator on a complex Hilbert space $\bar{E}$ as in Lemma 6.3. Here $\bar{P}_a$ is the orthogonal projection of $\bar{E}$ that extends $\pi(a, \cdot)$.

Axiom (K4) is essential to Theorem C. Indeed, with Z empty, the structure described in the proof of Proposition 10.7(2) satisfies every axiom of T except (K4), and the measure of that structure takes only the values 0 and 1.

Remark 11.3 (Scalars). The vocabulary fixes the complex numbers as scalars. For real Hilbert spaces, replace $\mathbb{Q}(i)$ by $\mathbb{Q}$ and omit the predicate $\mathrm{Im}\langle x \mid y\rangle$. For quaternionic spaces, use the quaternions with rational coordinates, acting on the left, and four predicates for the real components of the form. In the reading of (At4), "complex number of modulus one" then becomes "real number of modulus one" or "quaternion of modulus one". Each choice yields the analogue of Theorem 8.3 for the corresponding case of Theorem 3.6, and the proof of Theorem C applies in all three cases, since the vectors of Table 1 have real coordinates.

Remark 11.4 (The logical form of the axioms). The axioms (H1), (OL), (P1)–(P3), (Sub0), (Sub4), (At1), (At2), (K1), (K2), (K4), (Sp1), (M1) and (M2) are universal closed conditions built from finitary formulas. Axioms (H2), (Sub2), (Sub3), (K3), (Sp2), (M3) and the formula od use countable suprema or infima of uniformly bounded families. Axioms (P4), (Sub1), (At4), (At5), (Sp2), (Sp3) and (M3) use the formula $\lceil\varphi\rceil$ of (2), a countable join of a family with common bound 1 that records whether φ is positive. The functions at and span take values in the Solèr sort, on which the two-valued predicate e defines the discrete topology. Neither function is continuous from the norm topology of its vector sort to the discrete topology. The logic of Section 5 accommodates this discontinuity, since that logic places no condition on the interpretation of a function symbol beyond its sorts. Axiom (At3) uses the formula $\mathrm{nonatom}(a)$, which recognizes atoms and contains the connective $\sup_b$, a supremum over the Solèr sort.

Remark 11.5 (Redundancy). In standard models several axioms follow from others: axiom (Sub1) follows from (Sub2) and (P4), and axiom (Sub3) from (Sub2), (Sub4), (OL) and (P4). In standard models, axiom (At3) likewise follows from (At1), (At2) and Proposition 7.3, and axiom (At4) from (At1), (P4) and Proposition 7.3. These four axioms are kept nonetheless, so that each property has an axiom of its own.

Historical note. Löwenheim proved in 1915 that a first-order sentence with a model has a countable model [Lö15]. Skolem extended the result to countable sets of sentences in 1920, and in 1922 he observed that axiomatic set theory, in which uncountable sets exist, therefore has countable models [Sko70]. The countable models of Remark 11.2 are of the same kind: they satisfy every axiom of a theory whose standard models are uncountable. That an incomplete inner product space can be recognized by the absence of states goes back to Hamhalter and Pták. In 1987 they proved that a separable inner product space is complete exactly when the lattice of its orthogonally closed subspaces admits a σ-additive state [HP87]. The theorem of Dvurečenskij used in Remark 11.2 belongs to the same line of work [Dvu95].

12. Finite obstructions and Kochen–Specker configurations

Kochen and Specker certified, by a finite configuration of rays in a finite-dimensional space, that no quantum probability measure on the lattice of closed subspaces of that space takes only the values 0 and 1. Such a configuration is a finite set of rays in a space of finite dimension, read together with the orthonormal bases that it contains. An assignment of the values 0 and 1 to its rays is *admissible* if it gives the value 1 to exactly one ray of each of these bases and to no two orthogonal rays. The configurations of Kochen and Specker have no admissible assignment. The model existence game certifies that the theory of Theorem C, whose constants e_n name an infinite orthonormal family, has no models. The certificates of the game are finite sets of commitments that stand at the leaves of a well-founded tree of plays. A finite obstruction of the game is the last position of a play won by Olivia (see Corollary 10.5). In this section we compare the two kinds of certificate: their material, their demands on the measure, their location, and the topological property of the logic behind their finiteness.

A configuration of Kochen and Specker builds the hypothesis of noncontextuality into its vertex set, since each ray is a single vertex in all the bases that contain it. In a finite obstruction of the game, a ray that lies in two bases gives two vertices, one for each basis. The finite obstructions share this representation with contextuality-by-default, the approach of Dzhafarov and Kujala. In that approach, a quantity measured in two contexts gives two random variables with a common content (see Kujala, Dzhafarov and Larsson [KDL15] and Dzhafarov, Kujala and Larsson [DKL15]). In the sheaf-theoretic framework of Abramsky and Brandenburger, likewise, a measurement shared by two contexts enters the probability distributions on both, and these distributions are required to agree on it [AB11]. In both frameworks, that the two contexts share one quantity is given with the measurement scenario. In the finite obstructions, the identity of the two vertices of a ray is derived from axioms of T that do not mention the measure. The measure then carries the value across: by the clause on μ in (OL2), it takes equal values on equal elements of the lattice. The two vertices of a ray thus receive the same probability. There are eighteen identifications of rays, and the parity argument needs each of them. The commitments about the measure are intervals with rational endpoints, one for the probability of each ray and one for the sum over each basis. The tolerances that bound the widths of these intervals are part of the certificate. Finally, the obstruction is located by the hypothetical measure. Through σ-additivity, the play commits the measure to give probability close to 1 to the span of a block of four consecutive vectors of a fixed orthonormal basis. The rays of the obstruction lie in that span. In the first two respects, the finite obstructions give finer information than the configurations of Kochen and Specker. A configuration of Kochen and Specker, in turn, is a single certificate valid against every assignment, and it can be realized by measurements. The third respect, the location, is the form that the infinite dimension takes for obstructions built from rays: projections of infinite rank need no location (see the paragraph on σ-additivity below).

Throughout the section, we write T^{01} for the theory of Proposition 10.7, which adds to the theory T of Theorem A the axiom that μ takes only the values 0 and 1. We consider the finite obstructions reached under the strategy of Olivia described in Section 10.4, in the plays in which Josh makes every commitment that the parity argument requires. The comparison of the two kinds of certificate is summarized in Table 2.

In a configuration of Kochen and Specker, a vertex is a ray, a single vertex in all the bases that contain it. That a ray receives one value in all its bases is thus part of the data. In a finite obstruction of the game, a vertex is a ray as it occurs in a particular basis. As in

the proof of Theorem C, each of the nine bases $\mathcal{B}$ of Table 1, with vectors $v_0, \dots, v_3$, gives an element $s_{\mathcal{B}}$ of the sequence sort with entries $\frac{1}{16}v_0, \dots, \frac{1}{16}v_3, 0, 0, \dots$. The vertices of $\mathcal{B}$ are the elements $\mathrm{at}(\kappa_i s_{\mathcal{B}})$ of the Solèr sort, for $i < 4$, where $\kappa_i s_{\mathcal{B}}$ is the entry of index i of $s_{\mathcal{B}}$. Here at sends a nonzero vector x to the element whose members are the multiples of x. Every ray of the table, lying in two bases, is thus represented by two vertices. The instances of (K2) and (At5) identify each vertex $\mathrm{at}(\kappa_i s_{\mathcal{B}})$ with the element $\mathrm{at}(\frac{1}{16}v_i)$, which, as in the proof of Theorem C, we write $\mathrm{at}(v_i)$. Hence these instances identify the two vertices of a ray with each other, while the instances of (P4) identify the nine elements $\mathrm{span}(s_{\mathcal{B}})$ with a single element a_F. The element a_F stands for the span F of the block. These identifications follow from axioms that do not mention μ, and they record that the two vertices of a ray are one projection and the nine spans one subspace. The clause $e(a,b) \wedge |\mu(a) - \mu(b)| \leq 0$ of (OL2), which every quantum probability measure satisfies, says that μ respects the equality predicate e. This clause carries the measure across each identification, and thus requires the two vertices of a ray to have the same probability.

Each of the eighteen identifications of rays is indispensable to the parity argument. Indeed, the nine bases can be arranged in the cycle $1, 2, 4, 3, 6, 5, 9, 8, 7$ of their numbers in Table 1, in which any two consecutive bases (the last and the first included) share a ray. If the identification attached to a ray of a basis $\mathcal{B}$ is omitted, give the value 1 to the vertex of that ray in $\mathcal{B}$. Next, pair off the other eight bases, which follow one another along the cycle, by four of the rays that consecutive bases share. Finally, give the value 1 to both vertices of each of these four rays and the value 0 to all the other vertices. Each basis then has exactly one vertex of value 1. Without identifications, a vertex of value 1 could be chosen in each basis independently. An independent choice is possible in the dense families of orthonormal bases that Clifton and Kent constructed in each finite dimension. In each of these families, any two bases generate Boolean algebras of projections with only 0 and the identity in common [CK00].

A configuration of Kochen and Specker prescribes exact values. In a finite obstruction, the axiom of T^{01} still requires the measure to take only the values 0 and 1, but Josh's commitments about the measure are intervals with rational endpoints. The widths of these intervals are governed by the tolerances 2^{-k} that Olivia names. Each of the eighteen probabilities $\mu(\mathrm{at}(v))$ receives an interval within δ of 0 or of 1, and each of the nine sums over a basis an interval within η of 1. The constraint on a basis is derived. Each basis of Table 1, completed by $F^{\perp}$, decomposes the identity into five pairwise orthogonal projections, and the first phase of the strategy has given a_F a probability close to 1. These commitments are incompatible as soon as $4\delta + \eta < 1$. Indeed, the probabilities lie in $[0, 1]$. Hence a basis with no ray whose probability lies within δ of 1 would have a sum at most $4\delta < 1 - \eta$, and a basis with two such rays a sum at least $2 - 2\delta$. The bound $2 - 2\delta$ exceeds $1 + \eta$ because $2\delta + \eta < 1$. Each basis therefore has exactly one such ray; by the identifications, such a ray has the same probability in both of its bases, and the parity argument applies. The threshold depends only on the dimension: for bases of d rays, the same argument applies whenever $d\delta + \eta < 1$. The threshold thus comes from the rays and their bases, not from the game. It holds equally for approximate values assigned to the rays themselves, each a single vertex in both of its bases. The validity of the threshold for such approximate values is in the spirit of the forms of the Kochen–Specker theorem for imperfect measurements [SBZ01, Lar02]. What belongs to the game is that the tolerances are part of the certificate itself; they concern the values of the measure, while the rays remain exact.

A configuration of Kochen and Specker is fixed in advance, and one configuration serves against every assignment. Such a configuration can be realized in the laboratory, each of

its bases as a measurement; D'Ambrosio and his coauthors implemented the nine bases of Table 1 with single photons in 2013 [DHA+13]. The finite obstruction of a play, by contrast, is a set of commitments about a hypothetical measure, and it is located by Josh's answers. Consider the first phase of the strategy, where the constants e_n of T name an orthonormal basis $(e_n)_{n\in\mathbb{N}}$ and the constant ε names the sequence $(2^{-n-1}e_n)_{n\in\mathbb{N}}$. In this phase, a countable challenge to (M3) at the constant ε makes Josh name a number N for which the probabilities of the rays of $e_0, \dots, e_{N-1}$ add up nearly to 1. His later answers single out some $n_0 < N$ for which the probability of the ray of e_{n_0} is close to 1. The eighteen rays then lie in the span F of the block $e_{4j}, \dots, e_{4j+3}$ with $4j \leq n_0 < 4j + 4$. Since Olivia's challenges are dictated by her strategy, a play is determined by Josh's answers. The dependence on Josh's answers is shared by every winning strategy of Olivia. Indeed, since every finite set of axioms of T^{01} has a model, the tree of every winning strategy is well-founded and branches infinitely at some node. At that node, every natural number is a legal answer for Josh (see Proposition 10.7 and Theorem D). Whether the length of the plays can be bounded independently of Josh's answers is the subject of Question 10.8.

The need to locate the obstruction comes from σ-additivity. In the models of the finite sets of axioms of T^{01} constructed in the proof of Proposition 10.7(2), the measure vanishes on every finite-dimensional subspace. In these models the parity argument finds no ray of value 1 to count. It is σ-additivity over all the vectors e_n, through infinitely many instances of (K4), that excludes a measure vanishing on every finite-dimensional subspace. In finite dimension this step is automatic, since a configuration of Kochen and Specker spans the space. This role of σ-additivity belongs to configurations of rays. Write a separable infinite-dimensional Hilbert space H as $\mathbb{C}^4 \otimes \ell^2$, and, for each vector v of Table 1, let P_v be the projection of $\mathbb{C}^4$ onto the ray of v. For each basis $\mathcal{B}$ of the table, the four projections $P_v \otimes 1$ with v in $\mathcal{B}$ are pairwise orthogonal with sum 1. The eighteen projections $P_v \otimes 1$, which have infinite rank, therefore form a configuration that is fixed in advance and refutes every finitely additive $\{0, 1\}$-valued probability measure on $L(H)$ by the same parity argument. More generally, Döring proved the Kochen–Specker theorem for von Neumann algebras without summands of types I_1 and I_2 [Dör05]. His argument passes through the finitely additive form of Gleason's theorem due to Christensen, Yeadon, and Bunce and Wright [Chr82, Yea83, BW92] (see Hamhalter [Ham03]). The Solèr sort of the models of finite sets of axioms of T^{01} constructed in the proof of Proposition 10.7(2) is the lattice L_{fc} of Example 7.6. Since this lattice contains no projection of infinite rank and infinite corank, the finite obstructions of the game are built from rays. The dependence of these obstructions on the hypothetical measure is the form that the infinite dimension takes for them.

A configuration of Kochen and Specker certifies the impossibility by the absence of an admissible assignment, a finite obstruction by a finite set of commitments that no structure satisfies. Behind each kind of certificate stands a topological property of the logic. The assignments of 0 and 1 to the rays form a compact space. In that space, the condition that a finite basis contain exactly one ray of value 1 involves finitely many rays and defines a closed set. It is through these closed sets that compactness, the topological form of finiteness, turns the theorem of J. S. Bell into a finite configuration [Bel96, Pit98]. The condition that the probabilities of the rays of the vectors e_n add up to 1 involves infinitely many rays at once and defines no closed set. Indeed, as n grows, the assignments that give the value 1 to the ray of e_n alone converge to the assignment that vanishes everywhere, which violates that condition. Such conditions are expressed by the countable suprema and infima of real-valued $L_{\omega_1\omega}$, whose space of Henkin types is Čech-complete, whence the Baire category theorem holds in it (see Section A). The finite certificates come from the game, whose analysis rests on the

determinacy theorem of Gale and Stewart. Olivia's bookkeeping is a Baire category argument on the space of plays, the analogue of the one in the space of Henkin types. The passage from compactness to Čech-completeness is thus the passage from what finite configurations of rays can witness to what σ-additivity demands. The game returns what σ-additivity demands as finite certificates, one at each leaf of a well-founded tree.

Historical note. In 1990 Peres gave a proof of the Kochen–Specker theorem by observables on two qubits, valid for a particular state. Building on that proof, Mermin found a finite obstruction of another kind, formed by nine such observables arranged in three rows and three columns of commuting observables. No assignment of the values ± 1 to these nine observables respects the multiplicative relations along the rows and the columns [Per90, Mer90]. The array thus gives a state-independent proof of the theorem. In 1999 Meyer observed that a coloring of Godsil and Zaks [GZ88] assigns 0 and 1 to the unit vectors of $\mathbb{R}^3$ with rational coordinates, which are dense in the unit sphere. In this assignment, each orthogonal triple contains exactly one vector of value 1, and no two orthogonal vectors have the value 1 [Mey99]. Kent extended the result in the same year to dense sets of resolutions of the identity in every finite dimension [Ken99], and the families of Clifton and Kent date from 2000 [CK00].

Appendix A. The space of Henkin types and the model existence game

The model existence theorem of Section 10 and the omitting types theorem of Aubone, Dueñez and Iovino [ADI26] are two treatments of the Henkin construction. In the first, the description of a structure whose elements are named by witnesses is built move by move in the plays of a game. In the second, such descriptions are points of the space of Henkin types recalled in Section 5, and structures are obtained from the Baire category theorem. In this appendix we recall the facts about the space of Henkin types which bear on the game, referring to the paper of Aubone, Dueñez and Iovino for the proofs. We also compare the positions and the rules of the game with the conditions and the $\forall\exists$ formulas of Keisler's model-theoretic forcing [Kei73]. Throughout, we distinguish the correspondences that rest on proof from those that we present as analogies.

The space of Henkin types. The vocabulary $\mathcal{L}$ and the theory T are as in the setting of Section 10. The continuous connectives come from a countable set, as in Section 5, whence the fragment $\mathcal{F}$ generated by the sentences of the axioms of T is countable (see [ADI26, Section 6]). For the set C of new constants of Section 5 we take the set $\mathcal{W}$ of witnesses of Section 10. The fragment $\mathcal{F}_C$ that $\mathcal{F}$ generates in $\mathcal{L}_\mathcal{W}$ then has a countable set $S_C^\mathcal{F}$ of sentences. The set $S_C^\mathcal{F}$ contains every sentence that occurs in a play of the game $\mathrm{G}(T)$ of Definition 10.2. Indeed, a sentence that occurs in a play is an atomic sentence of $\mathcal{L}_\mathcal{W}$ or is obtained from a subformula of an axiom by substituting closed terms. We write $\prod S_C^\mathcal{F}$ for the product of the closed intervals $I_\varphi = [-r_\varphi, r_\varphi]$, for $\varphi \in S_C^\mathcal{F}$, and $\varphi^\mathfrak{m}$ for the coordinate at φ of a point $\mathfrak{m}$ of the product. By Tychonov's theorem, the product is a compact Hausdorff space; it contains the space $\mathcal{H}_C^\mathcal{F}$ of Henkin types as a subspace. For a set T' of closed conditions on sentences of $S_C^\mathcal{F}$, we write $\langle\langle T' \rangle\rangle$ for the set of points that satisfy every condition of T'. The set $\langle\langle \emptyset \rangle\rangle$ is thus the whole product.

The correspondence rests on three results of [ADI26]:

(i) A structure for $\mathcal{L}_\mathcal{W}$ is *canonical* when every element is the value of a closed term. The family of the values of the sentences of $S_C^\mathcal{F}$ in a canonical structure, its $S_C^\mathcal{F}$*-type*, belongs to $\mathcal{H}_C^\mathcal{F}$. Conversely, every member of $\mathcal{H}_C^\mathcal{F}$ is the $S_C^\mathcal{F}$-type of a canonical structure, the term structure of its restriction to the atomic sentences [ADI26, Lemma 2].

(ii) For a countable vocabulary and a countable fragment, every structure has a substructure whose sorts are countable and in which every sentence of the fragment keeps its value [ADI26, Lemma 3 and its proof]. A structure for $\mathcal{L}$ whose sorts are countable has an expansion to $\mathcal{L}_W$ that names every element by a witness, and this expansion is canonical [ADI26, proof of Proposition 4].

(iii) The space $\mathcal{H}_C^{\mathcal{F}}$ is a G_δ subset of $\prod S_C^{\mathcal{F}}$ [ADI26, Proposition 5]. Every Henkin condition defines a closed set except the upper bounds on suprema, each of which defines a countable intersection of open sets. For a supremum $\sup_x \varphi(x)$, the upper bound says that its value does not exceed the supremum of the values of the sentences $\varphi(t)$. For a countable join, the upper bound says that the value of the join does not exceed the supremum of the values of its members.

These facts, together with Henkin Realizability and the Čech-completeness of the space of Henkin types, are proved in the paper of Aubone, Dueñez and Iovino for a many-sorted logic without continuous connectives. In that logic, the a priori bound of a join is the supremum of the bounds of its members. The proofs of Aubone, Dueñez and Iovino apply unchanged to the logic of this paper once the Henkin conditions include a further condition for each continuous connective f. The condition for f says that the value of $f(\varphi_1, \ldots, \varphi_\ell)$ is the value of f at the values of $\varphi_1, \ldots, \varphi_\ell$. Indeed, this condition, an equation in finitely many coordinates, defines a closed set. Moreover, each induction on formulas acquires a case for f that follows from the definitions, as the case of $\varphi \vee \psi$ does. Finally, the bound r that accompanies a join in Section 5 serves only to fix the interval of its values.

By (iii), the space $\mathcal{H}_C^{\mathcal{F}}$ is a G_δ subset of the product. The set $\langle\!\langle T \rangle\!\rangle \cap \mathcal{H}_C^{\mathcal{F}}$ is likewise a G_δ subset of the product, since each axiom $\psi \leq r$ or $\psi \geq r$ of T defines the intersection of the open sets $\{\mathfrak{m} : \psi^{\mathfrak{m}} < r + \frac{1}{n}\}$ or $\{\mathfrak{m} : \psi^{\mathfrak{m}} > r - \frac{1}{n}\}$, for $n \geq 1$. Let K be either of these two sets. The closure of K in $\prod S_C^{\mathcal{F}}$ is compact (see Engelking [Eng89, Theorem 3.1.2]), hence a compactification of K. The remainder of this compactification, the trace of the complement of a G_δ set, is a countable union of closed sets. By Theorem 3.9.1 of Engelking [Eng89] and the definition that follows it, the space K is Čech-complete. Aubone, Dueñez and Iovino define Čech-completeness by the equivalent internal characterization of Theorem 3.9.2 in the same book. In K, the union of countably many nowhere dense subsets has dense complement, by the Baire category theorem [Eng89, Theorem 3.9.3].

Positions and plays. Let $\mathfrak{p}$ be a position of $\mathrm{G}(T)$ whose sentences belong to $S_C^{\mathcal{F}}$, as do those of every position reached in a play. The position $\mathfrak{p}$ defines the set $\langle\!\langle \mathfrak{p} \rangle\!\rangle$ of the points $\mathfrak{m} \in \prod S_C^{\mathcal{F}}$ that satisfy two requirements. The first is that $\varphi^{\mathfrak{m}} \in I$ for every commitment $\varphi \in I$ of $\mathfrak{p}$. The second is that $\theta(F(\bar{w}))^{\mathfrak{m}} = \theta(w')^{\mathfrak{m}}$ for every record $F(\bar{w}) \mapsto w'$ of $\mathfrak{p}$ and every atomic formula $\theta(x)$ of $\mathcal{L}_W$ whose only free variable is x. These equations say that $F(\bar{w})$ and w' are indiscernible by atomic formulas with parameters, as equal elements are. Since the projections are continuous, the sets $\langle\!\langle \mathfrak{p} \rangle\!\rangle$ and $\langle\!\langle T' \rangle\!\rangle$ are closed. A position, which commits sentences to closed intervals, thus defines a closed set and not, as finitely many strict conditions would, a basic open set of the product. Recall from Definition 10.1 the satisfaction of a position in a structure under an interpretation of the witnesses that occur in it. A position is satisfied under such an interpretation when its commitments hold and, for each of its records $F(\bar{w}) \mapsto w'$, the term $F(\bar{w})$ and the witness w' have the same value.

Proposition A.1. *Let $\mathfrak{p}$ be a position whose sentences belong to $S_C^{\mathcal{F}}$, and let T' be a subset of T. The set $\langle\!\langle \mathfrak{p} \rangle\!\rangle \cap \langle\!\langle T' \rangle\!\rangle \cap \mathcal{H}_C^{\mathcal{F}}$ is nonempty if and only if $\mathfrak{p}$ is satisfied in some model of T' under some interpretation of the witnesses. In particular, the theory T has a model if and only if $\langle\!\langle T \rangle\!\rangle \cap \mathcal{H}_C^{\mathcal{F}}$ is nonempty.*

Proof. Let $\mathfrak{m}$ be a point of the set; by (i), it is the $S_C^{\mathcal{F}}$-type of a canonical structure $\mathfrak{N}$. As at the end of the proof of Theorem 10.3, the quotient of $\mathfrak{N}$ by indiscernibility in atomic formulas with parameters is reduced, and the quotient map preserves the value of every formula. The reduct of this quotient to $\mathcal{L}$ is therefore a model of T', which satisfies $\mathfrak{p}$ when each witness is interpreted as the class of its value. Indeed, the equations defining $\langle\!\langle \mathfrak{p} \rangle\!\rangle$ place the values of $F(\bar{w})$ and w' in one class. Conversely, let a model $\mathfrak{M}$ of T' satisfy $\mathfrak{p}$ under an interpretation h. We apply (ii) to the expansion of $\mathfrak{M}$ by constants for the elements $h(w)$ and to the fragment generated by $\mathcal{F}$ and the sentences of $\mathfrak{p}$. The structure $\mathfrak{M}$ thus has a substructure $\mathfrak{M}_0$ with countable sorts that contains these elements and satisfies T' and $\mathfrak{p}$ under h, the records included. The infinitely many witnesses of each sort that do not occur in $\mathfrak{p}$ can name all the elements of $\mathfrak{M}_0$. The resulting expansion is canonical, and by (i) its $S_C^{\mathcal{F}}$-type belongs to $\langle\!\langle \mathfrak{p} \rangle\!\rangle \cap \langle\!\langle T' \rangle\!\rangle \cap \mathcal{H}_C^{\mathcal{F}}$. ☺

In these terms, Josh's truthful answers in a model of T, in the proof that (1) implies (2) in Theorem 10.3, lead to positions that the same model satisfies. These answers therefore keep the set $\langle\!\langle \mathfrak{p} \rangle\!\rangle \cap \langle\!\langle T \rangle\!\rangle \cap \mathcal{H}_C^{\mathcal{F}}$ nonempty. At a leaf of the tree of a winning strategy of Olivia, on the other hand, the last position $\mathfrak{p}$ is satisfied in no structure (see Corollary 10.5(2)). By the proposition, applied with T' empty, the closed set $\langle\!\langle \mathfrak{p} \rangle\!\rangle$ does not meet $\mathcal{H}_C^{\mathcal{F}}$.

Through the sets $\langle\!\langle \mathfrak{p} \rangle\!\rangle$, the game can also be compared with Keisler's model-theoretic forcing [Kei73]. Keisler adjoins a countable set C of new constants and takes as forcing conditions the finite sets of sentences with constants from C that are satisfiable in a given class of models, ordered by inclusion [Kei73, pp. 98 and 100, Example 1.2]. The witnesses play the role of his constants, and a position, which grows along a play, is the real-valued counterpart of such a finite set. By Proposition A.1, a position $\mathfrak{p}$ is satisfied in some model of T under some interpretation of the witnesses exactly when $\langle\!\langle \mathfrak{p} \rangle\!\rangle \cap \langle\!\langle T \rangle\!\rangle \cap \mathcal{H}_C^{\mathcal{F}}$ is nonempty; such positions are the counterparts of his conditions for the class of models of T. A record $F(\bar{w}) \mapsto w'$ is the counterpart, in a logic without equality, of an atomic sentence $c = \tau$. For every closed term τ, Keisler's definition of a forcing property requires each condition to have an extension that contains such a sentence for some $c \in C$ [Kei73, p. 99, clause (iv)], and the rule (Tot) meets the corresponding requirement in the game. A fair play that Josh survives corresponds to the increasing sequence of conditions by which Keisler obtains a generic set [Kei73, p. 101, Lemma 1.4], and the structure built from it is canonical, as his generic models are [Kei73, p. 101].

The alternation of challenges and answers follows the pattern of Keisler's $\forall\exists$ formulas [Kei73, p. 107]. His formulas have the form $\forall \bar{x} \bigvee_n \exists \bar{y} (\varphi_{n1} \wedge \cdots \wedge \varphi_{nj_n})$. Consider an axiom $\inf_{\bar{x}} \bigvee_n \sup_{\bar{y}} \theta_n \geq r$ of the same shape, in which each formula θ_n is a minimum of finitely many formulas. Once Josh has committed its value under (Ax), the rule ($\inf \geq$) lets Olivia choose the witnesses for $\bar{x}$, whereas ($\bigvee \geq$) and ($\sup \geq$) require Josh to choose the index n and the witnesses for $\bar{y}$, up to the tolerances that Olivia names. The universal choices are thus Olivia's and the existential choices Josh's. Keisler's general omitting types theorem asks that every condition remain satisfiable together with every instance $\psi_n(\bar{c})$ of the given $\forall\exists$ sentences [Kei73, p. 108]. In the game, the corresponding requirement is that Josh answer each challenge at a position satisfied in a model of T by a position with the same property, and the proof that (1) implies (2) in Theorem 10.3 shows that he can do so whenever T has a model. When T has no model, no position is satisfied in a model of T, and a winning strategy of Olivia makes this fact finite. At each leaf of its tree, the last position is a finite set of commitments and records satisfied in no structure, and hence contained in no position that some structure satisfies, that is, in no counterpart of a forcing condition. The refined

Kochen–Specker graphs of Section 12 picture such positions, since their vertices, identifications and intervals are given by commitments and records of the last position of a play. The correspondence is not literal in three respects: commitments confine the values of sentences to closed intervals instead of asserting sentences, the axioms of T need not have the $\forall\exists$ shape, and Keisler's logic has equality [Kei73, p. 97], whose role in the game is taken by the records.

A fair play that Josh survives determines a nonempty closed subset of $\langle\!\langle T\rangle\!\rangle \cap \mathcal{H}_C^{\mathcal{F}}$. Let $\mathfrak{p}_0 \subseteq \mathfrak{p}_1 \subseteq \cdots$ be its positions, and let $\mathfrak{p}_\infty$ be their union. The set $\bigcap_i \langle\!\langle \mathfrak{p}_i\rangle\!\rangle \cap \mathcal{H}_C^{\mathcal{F}}$, which is closed in $\mathcal{H}_C^{\mathcal{F}}$, is contained in $\langle\!\langle T\rangle\!\rangle$, since every axiom of T is named in a challenge (Ax). The structure $\mathfrak{M}$ built in the proof that (2) implies (1) in Theorem 10.3 is canonical, since its elements are the witnesses, each named by itself. It satisfies every commitment and every record of $\mathfrak{p}_\infty$; by (i), its $S_C^{\mathcal{F}}$-type lies in the set, which is therefore nonempty. In general the set has more than one point, since a play leaves freedom to the structure that it describes. Suppose, for instance, that a predicate symbol R other than $\mathbf{1}$ occurs in no axiom of T; then no sentence that occurs in the play contains R. The construction leaves the value of $R^{\mathfrak{M}}$ at each tuple of witnesses free in $[-r_R, r_R]$, and different choices yield canonical structures whose $S_C^{\mathcal{F}}$-types differ at the sentences $R(\bar{w})$.

Closed conditions, strict conditions and compactness. The contrast between closed and strict conditions drawn in Remark 10.6 has a topological counterpart. The Henkin types that satisfy a closed condition form a closed subset of $\mathcal{H}_C^{\mathcal{F}}$, and those that satisfy a strict condition an open subset. In the compact space $\prod S_C^{\mathcal{F}}$, a family of closed sets with empty intersection has a finite subfamily with empty intersection, whereas a family of open sets need not. For the theory $\{R > 0\} \cup \{R < \frac{1}{n} : n \geq 1\}$ of Remark 10.6, each finite part has a model, whence the open set that it defines meets the space of Henkin types by (i) and (ii). The whole theory, on the other hand, defines the empty set. The tree of Olivia's strategy for that theory branches infinitely at its root. If $\mathcal{H}_C^{\mathcal{F}}$ were closed, the compactness of the product would yield, for a theory T of closed conditions without models, a finite subset T' of T with $\langle\!\langle T'\rangle\!\rangle \cap \mathcal{H}_C^{\mathcal{F}} = \emptyset$. By Proposition A.1, the set T' would be a finite set of axioms without models. The space $\mathcal{H}_C^{\mathcal{F}}$ is a G_δ subset of the product, and what it inherits from the compact product is not compactness but Čech-completeness. On Čech-completeness rest the Baire category theorem in that space and, with it, the omitting types theorem recalled below. By (iii), the Henkin conditions that do not define closed sets are the upper bounds on suprema, which in the game Josh certifies by naming an instance. Under $(\sup \geq)$ and $(\inf \leq)$ the instance is a witness, chosen among the finitely many that the freshness of witnesses leaves him. Under $(\bigvee \geq)$ and $(\bigwedge \leq)$ it is an index, chosen among infinitely many. Parts (3) and (4) of Corollary 10.5 record these two cases. This correspondence concerns the rules of the game, from which Corollary 10.5 is proved directly, without recourse to the topology of $\mathcal{H}_C^{\mathcal{F}}$.

Baire category. Olivia's bookkeeping can also be read as a Baire category argument. Let σ' be a winning strategy for Josh, and let $X_{\sigma'}$ be the set of infinite plays in which Josh follows σ'. We equip $X_{\sigma'}$ with the topology whose basic open sets consist of the plays that extend a given finite play. Since the moves range over countable sets, the space $X_{\sigma'}$ is homeomorphic to a closed subspace of the Baire space $\mathbb{N}^{\mathbb{N}}$. It is therefore a Polish space, in which the Baire category theorem holds. For each challenge and each round, the set of plays in which the challenge, if available at that round, is made at that round or later is open and dense in $X_{\sigma'}$. The fair plays therefore form a countable intersection of dense open subsets of $X_{\sigma'}$, which by the Baire category theorem is dense, hence nonempty. The bookkeeping can be

read, too, through the *Banach–Mazur game*. In that game, two players alternately choose nonempty open sets of a space, each contained in the one before. The second player wins if the intersection of the chosen sets is contained in a given set (see Figure 8). Consider the Banach–Mazur game on $X_{\sigma'}$ in which the players choose basic open sets and the second player wins if the resulting play is fair. In this game the second player has a winning strategy modeled on Olivia's bookkeeping (compare Oxtoby [Oxt57]). At the move of index k of the second player, for each $k \in \mathbb{N}$, this strategy extends the current finite play by one round. In that round, Olivia makes the $l(k)$-th challenge of her enumeration for round $j(k)$, and Josh answers by σ'. The challenge is available because at least k rounds precede the new one and because $j(k) \leq k$. In the proof of Theorem 10.3, the category argument reduces to the bookkeeping, because only countably many challenges are available at each round.

The Baire category argument on plays is the analogue of the one by which Eagle [Eag14] proved the omitting types theorem for $[0,1]$-valued $L_{\omega_1\omega}$, applying Čech's form of the Baire category theorem to a space of countable metric structures. Aubone, Dueñez and Iovino prove the omitting types theorem for their logic [ADI26, Theorem 8] by Morley's Baire category method, applied in the Čech-complete space of Henkin types. The points of that space are the types of structures built from closed terms, as Keisler and Miller build models in logic without equality [KM01, Section 4]. In the terms of this appendix, for a sequence $(\Xi_n(\bar{x}_n))_{n\in\mathbb{N}}$ of types, sets of closed conditions $\varphi(\bar{x}_n) \geq r$ with $\varphi \in \mathcal{F}$ and r rational, their theorem can be stated as follows. Suppose that T has a model and that the instances $\Xi_n(\bar{t})$ of these types at tuples $\bar{t}$ of closed terms of $\mathcal{L}_\mathcal{W}$ meet $\langle\langle T\rangle\rangle \cap \mathcal{H}^\mathcal{F}_C$ in sets with empty interior in that space. Then T has a model whose sorts are countable and in which no tuple satisfies any $\Xi_n(\bar{x}_n)$. Indeed, the complements of these sets are countably many dense open subsets of the nonempty space $\langle\langle T\rangle\rangle \cap \mathcal{H}^\mathcal{F}_C$, in which the Baire category theorem holds. By (i), a point of the intersection of these complements is the $S^\mathcal{F}_C$-type of a canonical structure, whose tuples consist of values of closed terms. The reduced quotient of this structure changes no value, and in the reduct of that quotient to $\mathcal{L}$ no tuple satisfies any $\Xi_n(\bar{x}_n)$. The hypothesis of empty interior at tuples of closed terms explicates, as a statement about interiors in a Čech-complete space, the hypothesis of Keisler's general omitting types theorem [Kei73, p. 108] in the case in which his $\forall\exists$ formulas are taken over the whole fragment (see [ADI26, Introduction]). The two applications of the Baire category theorem take place in different spaces, the Polish space of plays and the Čech-complete space of Henkin types. They correspond as analogues rather than through a map. Indeed, as observed above, a fair play determines a closed set of Henkin types, not a single point. Nor do we know of any continuous map from the one space to the other that carries one argument onto the other.

Historical note. The omitting types theorem goes back to Henkin, in 1954 [Hen54], and to Orey, in 1956 [Ore56]. Robinson introduced forcing into model theory in an article published in 1971 [Rob71], adapting the method that Cohen had created for set theory in 1963 [Coh63]. Barwise extended Robinson's forcing to infinitary logic and used it to give a new proof of the omitting types theorem (see Keisler [Kei73, p. 96]). In 1973 Keisler built countable models by forcing with finite sets of sentences satisfiable in a given class of models [Kei73, p. 100, Example 1.2]. He described his treatment of forcing as a reformulation, in terms of ordinary models, of the Boolean-valued models of Rasiowa and Sikorski [RS63], and he regarded its basic result, the Generic Model Theorem, as essentially due to them [Kei73, p. 97]. From generic models he derived a general omitting types theorem, in which $\forall\exists$ formulas take the place of principal types [Kei73, pp. 107–108]. In 2009 Ben Yaacov and Iovino developed model-theoretic forcing for an infinitary continuous logic [BYI09].

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	Kochen–Specker configurations	finite obstructions of the model existence game for T^{01}
framework	partial Boolean algebras: in dimension at least three, the projections admit no homomorphism into the two-element Boolean algebra	real-valued $L_{\omega_1\omega}$: the theory T^{01} has no models, although each of its finite sets of axioms has one
topological property of the logic	compactness, the topological form of finiteness: each condition on a basis involves finitely many rays and defines a closed set, whence the theorem of J. S. Bell yields a finite configuration (J. L. Bell, Pitowsky); Kochen and Specker exhibited one	the game, whose analysis rests on determinacy, gives the finite certificates: σ-additivity, indispensable for configurations of rays, involves infinitely many rays at once, and the inconsistency of T^{01} is genuinely infinitary; Olivia's bookkeeping is a Baire category argument on the space of plays, by analogy with the one on the Čech-complete space of Henkin types (see Section A)
		for every winning strategy of Olivia, who claims that T^{01} has no models
tree of plays	none: one configuration serves against every assignment	well-founded, and branching infinitely at some node
certificate of impossibility	no assignment of 0 and 1 to the rays gives 1 to exactly one ray of each basis and to no two orthogonal rays	at each leaf, a finite set of commitments satisfied in no structure
		under the strategy of Section 10.4, in a play in which Josh, who undertakes to build a model, makes every commitment that the parity argument requires
bases	orthonormal bases of the space	the nine bases of Table 1 in the span F of a block, each completed by $F^\perp$
vertex	a ray	a ray as it occurs in one of the nine bases
identifications	none needed: a ray is one vertex in all the bases that contain it (the hypothesis of noncontextuality)	eighteen identifications of rays and nine of the spans of the bases with F, all derived from axioms of T that do not mention μ, along which the clause on μ in (OL2) carries probabilities; the parity argument needs all eighteen identifications of rays
values	0 or 1 at each ray	an interval with rational endpoints, within δ of 0 or of 1, for the probability of each ray
constraint on a basis	exactly one ray of value 1	an interval with rational endpoints, within η of 1, for the sum of the probabilities of its rays; any tolerances with $4\delta + \eta < 1$ suffice
location of the rays	fixed in advance	the eighteen rays of Table 1, in the block determined by Josh's answers
space	finite dimension, spanned by the rays	infinite dimension; the rays lie in the span of a block $e_{4j}, \dots, e_{4j+3}$ of the orthonormal basis $(e_n)_{n\in\mathbb{N}}$ named by constants of T
role of σ-additivity	none: every basis is finite; in infinite dimension, the eighteen projections $P_v \otimes 1$ on $\mathbb{C}^4 \otimes \ell^2$ need only finite additivity	indispensable for configurations of rays, which it locates: Josh names a number N such that the probabilities of the rays of $e_0, \dots, e_{N-1}$ add up nearly to 1, and the rays lie in the block that contains some e_{n_0} with $n_0 < N$

TABLE 2. The configurations of Kochen and Specker and the finite obstructions of the model existence game for the theory T^{01}. The theory T^{01} adds to T the axiom that μ takes only the values 0 and 1.